Linear Algebra

Chapter 3: Finite Dimensional Vector Spaces

From the very basics to advanced – every example, every exercise

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Part 0 Before We Start: The Tools You Need

Read this first – how this document is built

Every topic in Chapter 3 follows the *same four-step rhythm*:

1. **Picture it** (teal box) – the idea in everyday words, before any symbols.
2. **Definition / Theorem** (blue / purple box) – the exact statement from your notes, followed by a **Plain English** box (green) that re-says it in simple words.
3. **Worked Examples** – first the ones from your notes (grey boxes, fully solved), then extra ones I added (gold boxes).
4. **Questions** (orange boxes) – every exercise from your notes and textbook. The question and its short step-by-step solution are in the *same* box.

If a word or symbol feels strange, it is explained in Part 0 or in the box right before it. Nothing is assumed.

0.1 What is a vector? (just a list of numbers)

Picture it

A **vector** is an *ordered list of numbers*. "Ordered" means the position matters: $(2, 5)$ and $(5, 2)$ are different vectors.

- $(2, 5)$ has **2 entries**, so it lives in $\mathbb{R}^2$ (the "plane").
- $(1, 0, -3)$ has **3 entries**, so it lives in $\mathbb{R}^3$ ("space").
- $(x_1, x_2, \ldots, x_n)$ has n entries, so it lives in $\mathbb{R}^n$.

$\mathbb{R}^n$ simply means *"the set of all lists of n real numbers"*. The small letter n is only a counter: $\mathbb{R}^2$ is the plane, $\mathbb{R}^3$ is 3-d space, and $\mathbb{R}^5$ is a space with 5 coordinates (you cannot draw it, but the rules are the same).

Two vectors are **equal** only if they agree *in every position*: $(1, 2, 3) = (1, 2, 3)$, but $(1, 2, 3) \neq (1, 3, 2)$.

Definition – The two basic operations on vectors in $\mathbb{R}^n$

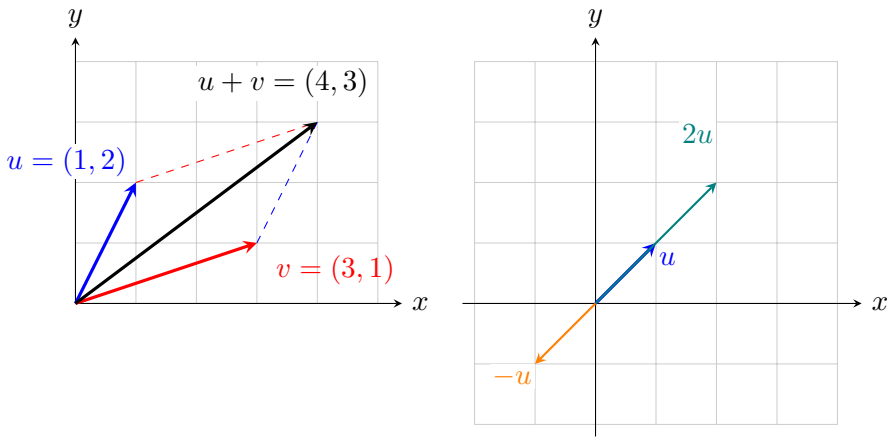
Addition – add position by position. **Scalar multiplication** – multiply every position by the same number.

A **scalar** is just "an ordinary number" (here, a real number). The word is used to tell numbers apart from vectors.

Example 0.1 – adding and scaling

- $(1, 2) + (3, -1) = (1 + 3,\ 2 + (-1)) = (4, 1)$.
- $3 \cdot (1, 2) = (3 \cdot 1,\ 3 \cdot 2) = (3, 6)$.
- $-1 \cdot (1, 2) = (-1, -2)$. We call this $-(1, 2)$, the *negative* of $(1, 2)$.
- $(1, 2) + (-1, -2) = (0, 0)$. The vector $(0, 0)$ (all entries 0) is called the **zero vector**, written $\mathbf{0}$.

- A **mix** of vectors (called a **linear combination**): $2(1,0)+3(0,1)=(2,0)+(0,3)=(2,3)$.



Left: adding vectors = completing the parallelogram. Right: scaling u moves you along the same line through the origin.

Row or column?

We sometimes write $(1,2,3)$ across and sometimes down as $\begin{pmatrix}1\\2\\3\end{pmatrix}$. It is the same vector. When a matrix multiplies a vector we need the column form, so we write $x=(x_1,\dots,x_n)^t$ (the small t means "turn the row into a column").

0.2 Which numbers may we use? Real and complex scalars

Picture it

In Chapter 3 the numbers you are *allowed to multiply by* are not always the same. Sometimes only real numbers $\mathbb{R}$ are allowed; sometimes complex numbers $\mathbb{C}$ are allowed. The set of allowed numbers is called the **field of scalars**, written $\mathbb{F}$ (so $\mathbb{F}=\mathbb{R}$ or $\mathbb{F}=\mathbb{C}$).

Complex numbers in 6 lines. A complex number is $z=a+ib$ with a,b real, and i is a symbol with $i^2=-1$.

1. **Equal:** $a+ib=c+id \iff a=c$ and $b=d$ (compare real parts and imaginary parts separately). In particular $a+ib=0$ with a,b real forces $a=0$ and $b=0$. **This one fact is used again and again in this chapter.**
2. **Add:** $(2+3i)+(1-4i)=3-i$ (add real parts, add imaginary parts).
3. **Multiply:** expand like brackets and use $i^2=-1$: $(2+3i)(1-4i)=2-8i+3i-12i^2=2-5i+12=14-5i$.
4. **Conjugate:** $\overline{a+ib}=a-ib$. For example $\overline{2+3i}=2-3i$.
5. **Size (modulus):** $|a+ib|=\sqrt{a^2+b^2}$. For example $|2+3i|=\sqrt{13}$. Also $z\cdot\overline{z}=|z|^2$, e.g. $(2+3i)(2-3i)=4+9=13$.

6. **Divide:** multiply top and bottom by the conjugate: $\dfrac{1}{2+3i} = \dfrac{2-3i}{(2+3i)(2-3i)} = \dfrac{2-3i}{13}$.

Every real number is also complex ($5 = 5 + 0i$), so $\mathbb{R} \subseteq \mathbb{C}$. But $i \notin \mathbb{R}$.

Definition – The notation $V(\mathbb{F})$, $\mathbb{R}^n(\mathbb{R})$, $\mathbb{C}^n(\mathbb{C})$, $\mathbb{C}^n(\mathbb{R})$

$V(\mathbb{F})$ means "the set V of vectors, where the allowed scalars are the numbers in $\mathbb{F}$". The letter in the bracket tells you **which numbers you may multiply by**.

Name	Vectors look like	Allowed scalars – example
$\mathbb{R}^2(\mathbb{R})$	$(1,2)$, $(\pi,0)$	real only; $\sqrt{2}\cdot(1,2) = (\sqrt{2}, 2\sqrt{2})$ is fine.
$\mathbb{C}^2(\mathbb{C})$	$(1+i,\ 3)$	complex allowed; $i\cdot(1,0) = (i,0)$ is fine.
$\mathbb{C}^2(\mathbb{R})$	$(1+i,\ 3)$ (same vectors!)	real only; $2\cdot(1+i,3) = (2+2i,6)$ is fine, but **you may not multiply by** i.

Same set, different scalars, different space

$\mathbb{C}^2(\mathbb{C})$ and $\mathbb{C}^2(\mathbb{R})$ have exactly the same vectors but they are *different vector spaces*, because the allowed scalars differ. Later you will see that this changes the dimension (2 versus 4). Always read the letter in the bracket.

0.3 Three machines from Chapter 2 (you will use them in every question)

Picture it

Almost every question in Chapter 3 secretly asks: *"does this system of linear equations have a solution, and what is it?"* So we recall three tools. If you master these three, about 80% of Chapter 3 becomes mechanical.

Machine 1: row reduction (solving linear systems)

Row operations (they never change the solutions of the system):

1. swap two rows; 2. multiply a row by a non-zero number; 3. add a multiple of one row to another.

Row-reduced echelon form (RREF): each non-zero row starts with a 1 (a **pivot**); each pivot is the only non-zero entry in its column; pivots move to the right as you go down; zero rows are at the bottom. Columns with a pivot are **pivot columns**; the other columns are **free columns**.

Example 0.2 – a system with exactly one solution

Solve $x + y + z = 6$, $2x - y + z = 3$, $x + 2y - z = 2$.

Step 1 – write it as a matrix. The augmented matrix is $\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & -1 & 1 & 3 \\ 1 & 2 & -1 & 2 \end{array}\right)$.

Step 2 – row-reduce. Use $R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 - R_1$ to clear column 1, then clear the other columns the same way. The final form is $\left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array}\right)$.

Step 3 – read the answer. The rows say $x = 1$, $y = 2$, $z = 3$. **Check:** $1 + 2 + 3 = 6$ ✓, $2 - 2 + 3 = 3$ ✓, $1 + 4 - 3 = 2$ ✓.

Example 0.3 – free variables (infinitely many solutions)

Solve $A\mathbf{x} = \mathbf{0}$ for $A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \end{pmatrix}$, i.e. $x + 2y + z = 0$ and $2x + 4y + 3z = 0$.

Step 1 – row-reduce. $R_2 \to R_2 - 2R_1$ gives $\begin{pmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \end{pmatrix}$; then $R_1 \to R_1 - R_2$ gives $\begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$.

Step 2 – find pivot and free columns. Pivots are in columns 1 and 3, so x and z are *pivot variables*. Column 2 has no pivot, so y is a *free variable*: it can be any number; call it t.

Step 3 – solve for the pivot variables. Row 1: $x + 2y = 0 \Rightarrow x = -2t$. Row 2: $z = 0$.

Step 4 – write the answer as a vector. $(x, y, z) = (-2t, t, 0) = t\,(-2, 1, 0)$. So *every* solution is a multiple of the single vector $(-2, 1, 0)$.

Pattern to remember: number of free variables = number of unknowns − number of pivots. Here $3 - 2 = 1$ free variable, and the solution set is built from 1 vector.

Machine 2: the determinant test

For a *square* matrix: 2×2: $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$. For 3×3 expand along a row with signs $+, -, +$:

$$\det \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{pmatrix} = 1 \cdot (1 \cdot 0 - 4 \cdot 6) - 2 \cdot (0 \cdot 0 - 4 \cdot 5) + 3 \cdot (0 \cdot 6 - 1 \cdot 5) = -24 + 40 - 15 = 1.$$

Key Fact K1 (from Chapter 2) – the "one test, many meanings" fact

For an $n \times n$ matrix A the following all say the same thing:

$$\det A \neq 0 \iff A\mathbf{x} = \mathbf{0} \text{ has } \textit{only} \text{ the solution } \mathbf{x} = \mathbf{0} \iff \text{the columns of } A \text{ are independent}$$
$$\iff \text{the rows of } A \text{ are independent} \iff A \text{ is invertible} \iff \operatorname{rank} A = n.$$

("Independent" is explained in Section 3.2; for now just remember: *determinant* $\neq 0$ *is the good case*.)

Machine 3: rank

Definition – Rank

The **rank** of a matrix is the number of non-zero rows left after row reduction (equivalently, the number of pivots). Example: $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 0 & 1 \end{pmatrix} \to \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$ has **rank 2** (two non-zero rows).

Key Fact K2 – "more unknowns than equations"

A homogeneous system $Ax = \mathbf{0}$ with *more unknowns than equations* always has a non-zero solution (there is at least one free variable, so there are infinitely many solutions). This is exactly why, later, "too many vectors must be dependent".

0.4 The language of proofs (symbols and habits)

Symbol	Meaning, with an example
$x \in S$	"x is an element of S": $2 \in \mathbb{Z}$, $(1,0) \in \mathbb{R}^2$.
$S \subseteq T$	"every element of S is also in T" (S is a subset of T): $\mathbb{Z} \subseteq \mathbb{R}$.
$S \cap T$	elements in *both* S and T (intersection).
$S \cup T$	elements in S *or* T (or both) (union).
$\{0\}$ vs 0 vs $\varnothing$	0 is a number (or the zero vector), $\{0\}$ is a *set with one element*, $\varnothing$ is the *empty set* (no elements).
$\forall$, "for every"	$\forall\, \alpha \in \mathbb{R}$ means "whichever real number α you pick".
$\exists$, "there exists"	"you can find at least one".
$\iff$ ("iff")	two-way implication: you must prove *both* directions.
$\Rightarrow$	"implies": if the left is true, the right is true.
$\sum_{i=1}^{n} a_i$	$a_1 + a_2 + \cdots + a_n$ (a sum with n terms).
$L(S)$	the "span" of S (all mixes of vectors in S); defined in Section 3.1.

Habit 1 – how to prove a statement "for every" element

Start with "*Let u, v be any two elements of S*" (arbitrary, not special ones). Then show the required property using only what is given. If you used a particular example, you have proved nothing in general.

Habit 2 – how to prove two sets are equal ($A = B$)

Prove **two inclusions**: (i) $A \subseteq B$ (take any $x \in A$ and show $x \in B$) and (ii) $B \subseteq A$.

Habit 3 – how to prove something is false

One **counterexample** is enough. To show "S is not closed under addition" find two elements of S whose sum is outside S.

Habit 4 – proof by contradiction

Assume the opposite of what you want, and keep deducing until you reach something impossible (like $1 = 0$ or a statement contradicting the hypothesis). Then the opposite must be false, so the original claim is true.

0.5 Roadmap of the chapter

Section	The question it answers	Key words
3.1	Which sets behave like $\mathbb{R}^n$ (can be added and scaled safely)? Which parts of them are again of that kind? What can you reach by mixing a few vectors?	vector space, subspace, span, row/column/null space
3.2	When is a list of vectors *wasteful* (some vector is a mix of the others)?	linear independence
3.3	What is the *smallest* list that still reaches everything, and how long is it?	basis, dimension, $\dim(M+N)$, rank
3.4	How do we give every vector a numerical "address", and convert between address systems?	ordered basis, coordinates, change of basis

3.1 Vector Spaces

3.1.1 Why do we need this idea?

Picture it – the motivating example from your notes

Take the two planes $2x+3y+z+u=0$ and $3x+y+2z+u=0$ in $\mathbb{R}^4$ and let V be the set of points lying on *both*. Let us test V:

1. $(0,0,0,0)\in V$ (every equation is $0=0$).
2. $(-1,0,1,1)\in V$ and $(-5,1,7,0)\in V$. **Check the first:** $2(-1)+3\cdot 0+1+1=0$ ✓ and $3(-1)+0+2+1=0$ ✓. Their sum $(-6,1,8,1)$ is also in V: $2(-6)+3+8+1=0$ ✓, $3(-6)+1+16+1=0$ ✓.
3. A multiple $\alpha(-1,0,1,1)=(-\alpha,0,\alpha,\alpha)$ is again in V for every real α.

The *same three behaviours* (contains $\mathbf{0}$; adding two members stays inside; scaling a member stays inside) are shown by the solution set of *any* system $Ax=\mathbf{0}$. And they are also shown by sets of polynomials, matrices, functions, ... So mathematicians collect all such sets under one name: a **vector space**. Whatever we prove once for "vector spaces" then works for all of them at once.

3.1.2 The definition, axiom by axiom

Definition – Vector Space (Definition 3.1.1 in your notes)

A **vector space** over $\mathbb{F}$, written $V(\mathbb{F})$, is a non-empty set V together with two operations

$$\oplus:\ V\times V\to V\ \text{ (vector addition)},\qquad \odot:\ \mathbb{F}\times V\to V\ \text{ (scalar multiplication)},$$

such that for all $u,v,w\in V$ and all scalars $\alpha,\beta\in\mathbb{F}$ the following 8 rules hold.
Vector addition: (1a) $u\oplus v=v\oplus u$. (1b) $(u\oplus v)\oplus w=u\oplus(v\oplus w)$. (1c) There is a (unique) element $\mathbf{0}\in V$ with $u\oplus\mathbf{0}=u$ for all u. (1d) For every u there is a (unique) element $-u\in V$ with $u\oplus(-u)=\mathbf{0}$.
Scalar multiplication: (2a) $\alpha\odot(\beta\odot u)=(\alpha\beta)\odot u$. (2b) $1\odot u=u$.
Distributive laws: (3a) $\alpha\odot(u\oplus v)=(\alpha\odot u)\oplus(\alpha\odot v)$. (3b) $(\alpha+\beta)\odot u=(\alpha\odot u)\oplus(\beta\odot u)$.

Plain English – break the definition into small pieces

- **"a non-empty set V"** – a collection of objects called *vectors*. There must be at least one (in fact the zero vector).
- **"over $\mathbb{F}$"** – the scalars come from $\mathbb{F}$ ($\mathbb{R}$ or $\mathbb{C}$).
- "$\oplus: V\times V\to V$" – if you take *any two vectors* and add them, *the answer is again in* V. You never leave the set by adding. (This is called **closure under addition**.)
- "$\odot:\mathbb{F}\times V\to V$" – if you multiply *any vector* by *any scalar*, the answer is again in V. (**Closure under scalar multiplication**.)
- The circle in $\oplus,\odot$ is just a reminder that these operations might be *strange* (not the usual $+$ and $\times$). When the operations are the usual ones, we write plainly $u+v$ and αu.

- **The 8 rules** simply say: "these two operations obey the same sensible rules that $+$ and $\times$ obey for ordinary vectors in $\mathbb{R}^n$".
- **Zero vector vs zero number.** Bold $\mathbf{0}$ is the zero *vector* (an element of V); plain 0 is the zero *number* (an element of $\mathbb{F}$). They are different objects.

The 8 rules, each checked with real numbers in $\mathbb{R}^2$

Take $u = (1, 2)$, $v = (3, -1)$, $w = (0, 4)$ and scalars $\alpha = 2$, $\beta = 3$ (ordinary operations).

Rule	In words	Numerical check in $\mathbb{R}^2$
1a	order of adding does not matter	$u + v = (4, 1)$ and $v + u = (4, 1)$.
1b	brackets in a triple sum do not matter	$(u + v) + w = (4, 1) + (0, 4) = (4, 5)$; $u + (v + w) = (1, 2) + (3, 3) = (4, 5)$.
1c	a "do nothing" vector $\mathbf{0}$ exists	$u + (0, 0) = (1, 2) = u$.
1d	every vector has an "opposite"	$u + (-1, -2) = (0, 0)$, so $-u = (-1, -2)$.
2a	scaling twice = scaling once by the product	$2 \cdot (3 \cdot u) = 2 \cdot (3, 6) = (6, 12)$; $(2 \cdot 3)u = 6u = (6, 12)$.
2b	multiplying by 1 changes nothing	$1 \cdot u = (1, 2)$.
3a	scalar spreads over a vector sum	$2(u + v) = 2(4, 1) = (8, 2)$; $2u + 2v = (2, 4) + (6, -2) = (8, 2)$.
3b	a vector spreads over a scalar sum	$(2 + 3)u = 5u = (5, 10)$; $2u + 3u = (2, 4) + (3, 6) = (5, 10)$.

Why bother with 8 rules if $\mathbb{R}^n$ obeys them automatically?

Because the chapter's real point is that *strange sets with strange operations* can also be vector spaces (example 4 and 5 below) – and that *reasonable-looking* sets can fail (the non-examples below). The 8 rules are the referee.

Recipe – how to decide whether $(V, \oplus, \odot)$ is a vector space

1. **Closure.** Add two general elements and scale one: does the answer stay in V? (If not, stop: not a vector space.)
2. **Zero vector.** Solve $u \oplus e = u$ for e. Is e in V and the same for every u?
3. **Opposite.** For a given u solve $u \oplus w = e$ for w. Is w in V?
4. **Rule 2b.** Compute $1 \odot u$. Is it u?
5. **Remaining rules** (1a, 1b, 2a, 3a, 3b): expand both sides and compare.

To *disprove*, one failing rule with numbers is enough (a counterexample).

3.1.3 Three consequences of the rules (Theorem 3.1.3)

Theorem 3.1.3

Let V be a vector space over $\mathbb{F}$. Then
1. $u \oplus v = u$ implies $v = \mathbf{0}$.
2. $\alpha \odot u = \mathbf{0}$ if and only if $\alpha = 0$ or $u = \mathbf{0}$.
3. $(-1) \odot u = -u$ for every $u \in V$.

Plain English

(1) The only vector that can be added to u without changing it is the zero vector. (2) Multiplying by the number 0 always gives $\mathbf{0}$, multiplying the vector $\mathbf{0}$ by anything gives $\mathbf{0}$, and nothing else gives $\mathbf{0}$. (3) "Multiply by -1" is the same as "take the opposite".
These look obvious for $\mathbb{R}^n$. The point is that they follow from the 8 rules *alone*, so they are true in *every* vector space, even strange ones.

Proof in small steps – Part 1: $u \oplus v = u \Rightarrow v = \mathbf{0}$

Step 1 – get an opposite. By rule 1d there is $-u$ with $u \oplus (-u) = \mathbf{0}$ (and by 1a also $(-u) \oplus u = \mathbf{0}$).

Step 2 – add $-u$ to both sides. From $u \oplus v = u$ we get $(-u) \oplus (u \oplus v) = (-u) \oplus u$.

Step 3 – re-bracket (rule 1b). $\big((-u) \oplus u\big) \oplus v = (-u) \oplus u$.

Step 4 – use $(-u) \oplus u = \mathbf{0}$. $\mathbf{0} \oplus v = \mathbf{0}$.

Step 5 – use rule 1c. $\mathbf{0} \oplus v = v$. Hence $v = \mathbf{0}$. ■

Tiny example: in $\mathbb{R}^2$, $(1, 2) + v = (1, 2)$ forces $v = (0, 0)$.

Proof in small steps – Part 2: $\alpha \odot u = \mathbf{0} \iff \alpha = 0$ or $u = \mathbf{0}$

(a) $\alpha \odot \mathbf{0} = \mathbf{0}$.

Step 1 – split zero. $\mathbf{0} = \mathbf{0} \oplus \mathbf{0}$, so $\alpha \odot \mathbf{0} = \alpha \odot (\mathbf{0} \oplus \mathbf{0})$.

Step 2 – rule 3a. $\alpha \odot (\mathbf{0} \oplus \mathbf{0}) = (\alpha \odot \mathbf{0}) \oplus (\alpha \odot \mathbf{0})$.

Step 3 – use Part 1. We now have $x = x \oplus x$ where $x = \alpha \odot \mathbf{0}$. Part 1 (with $u = x$, $v = x$) says $x = \mathbf{0}$. So $\alpha \odot \mathbf{0} = \mathbf{0}$.

(b) $0 \odot u = \mathbf{0}$.

Step 1 – split the number zero. $0 = 0 + 0$, so $0 \odot u = (0 + 0) \odot u$.

Step 2 – rule 3b. $= (0 \odot u) \oplus (0 \odot u)$. Again $x = x \oplus x$ with $x = 0 \odot u$, so by Part 1, $0 \odot u = \mathbf{0}$.

(c) If $\alpha \odot u = \mathbf{0}$ and $\alpha \neq 0$, then $u = \mathbf{0}$.

Step 1 – use the reciprocal. Since $\alpha \neq 0$, the number $1/\alpha$ exists in $\mathbb{F}$ (true for $\mathbb{R}$ and $\mathbb{C}$).

Step 2 – chain of equalities. $\mathbf{0} = \frac{1}{\alpha} \odot \mathbf{0}$ (by (a)) $= \frac{1}{\alpha} \odot (\alpha \odot u)$ (given) $= \big(\frac{1}{\alpha} \cdot \alpha\big) \odot u$ (rule 2a) $= 1 \odot u = u$ (rule 2b). So $u = \mathbf{0}$. ■

Proof in small steps – Part 3: $(-1)\odot u = -u$

Step 1 – start from 0. $\mathbf{0} = 0\odot u$ by Part 2(b).

Step 2 – write $0 = 1+(-1)$. $0\odot u = (1+(-1))\odot u = (1\odot u)\oplus((-1)\odot u)$ by rule 3b.

Step 3 – use rule 2b. $= u\oplus((-1)\odot u)$.

Step 4 – conclude. So $u\oplus((-1)\odot u) = \mathbf{0}$: the vector $(-1)\odot u$ is an opposite of u. Rule 1d says the opposite is unique, so $(-1)\odot u = -u$. ■

Tiny example: $(-1)(1,2) = (-1,-2) = -(1,2)$.

3.1.4 Examples of vector spaces (Example 3.1.4 of your notes, items 1–12)

How to read the list

For each example ask four questions: **(i)** what do the vectors look like? **(ii)** what are $\oplus$ and $\odot$? **(iii)** what is the zero vector $\mathbf{0}$? **(iv)** what are the scalars? Items 1, 2, 3, 8, 9, 10, 11, 12 are "obviously fine" (the operations are done position by position). Items 4, 5, 6, 7 are the ones to study slowly.

#	Space	Operations	Zero vector; scalars
1	$\mathbb{R}$	usual $+$ and $\times$	0; $\mathbb{R}$
2	$\mathbb{R}^2$	$(x_1,x_2)\oplus(y_1,y_2) = (x_1+y_1, x_2+y_2)$, $\alpha\odot(x_1,x_2) = (\alpha x_1, \alpha x_2)$	$(0,0)$; $\mathbb{R}$
3	$\mathbb{R}^n$	the same, with n positions	$(0,\dots,0)$; $\mathbb{R}$
4	$\mathbb{R}^+$ (positive reals)	$v_1\oplus v_2 = v_1\cdot v_2$, $\alpha\odot v = v^\alpha$	1 (!); $\mathbb{R}$
5	$\mathbb{R}^2$ (twisted)	$(x_1,x_2)\oplus(y_1,y_2) = (x_1{+}y_1{+}1,\ x_2{+}y_2{-}3)$, $\alpha\odot(x_1,x_2) = (\alpha x_1{+}\alpha{-}1,\ \alpha x_2{-}3\alpha{+}3)$	$(-1,3)$ (!); $\mathbb{R}$
6a	$\mathbb{C} = \{x+iy\}$	usual addition, $\alpha\odot(x+iy) = \alpha x + i\alpha y$ with $\alpha\in\mathbb{R}$	0; $\mathbb{R}$
6b	$\mathbb{C}$	usual addition, $(\alpha+i\beta)\odot(x+iy) = (\alpha x-\beta y)+i(\alpha y+\beta x)$	0; $\mathbb{C}$
7a, 7b	$\mathbb{C}^n$	position by position	$(0,\dots,0)$; 7a: $\mathbb{C}$, 7b: $\mathbb{R}$
8	$M_n(\mathbb{R})$: all $n\times n$ real matrices	matrix addition; scalar $\times$ matrix	zero matrix; $\mathbb{R}$
9	$P_n(\mathbb{R})$: polynomials of degree $\le n$	add polynomials; multiply by a number	zero polynomial; $\mathbb{R}$
10	$P(\mathbb{R})$: all real polynomials	the same	zero polynomial; $\mathbb{R}$

#	Space	Operations	Zero vector; scalars
11	$C[-1,1]$: continuous functions on $[-1,1]$	$(f \oplus g)(x) = f(x) + g(x)$, $(\alpha \odot f)(x) = \alpha f(x)$	the function 0 (zero at every x); $\mathbb{R}$
12	$V \times W$ (pairs)	see below	$(\mathbf{0}_V, \mathbf{0}_W)$; $\mathbb{R}$

Example 3.1.4 – item 4: $\mathbb{R}^+$ with $v_1 \oplus v_2 = v_1 v_2$ and $\alpha \odot v = v^\alpha$

Idea. Under the *usual* + and ×, $\mathbb{R}^+$ fails (no zero, no negatives). But if we decide that "adding" means *multiplying* and "scaling by α" means *raising to the power* α, then everything works. Here the vectors are positive numbers, and the scalars are real.
First, feel the strange operations: $2 \oplus 3 = 2 \cdot 3 = 6$; $3 \odot 2 = 2^3 = 8$; $(-1) \odot 4 = 4^{-1} = \frac{1}{4}$.
Closure: a product of positive numbers is positive, and $v^\alpha > 0$ for $v > 0$. So $\oplus, \odot$ really output elements of $\mathbb{R}^+$. ✓
Now the 8 rules. Let $v_1, v_2, v_3, v \in \mathbb{R}^+$ and $\alpha, \beta \in \mathbb{R}$.

- **1a:** $v_1 \oplus v_2 = v_1 v_2 = v_2 v_1 = v_2 \oplus v_1$. **1b:** $(v_1 \oplus v_2) \oplus v_3 = (v_1 v_2) v_3 = v_1 (v_2 v_3) = v_1 \oplus (v_2 \oplus v_3)$.
- **1c (zero vector):** we need e with $v \oplus e = v$, i.e. $v \cdot e = v$, so $e = 1$. The "zero vector" of this space is the number 1.
- **1d (opposite):** need w with $v \oplus w = 1$, i.e. $vw = 1$, so $w = 1/v \in \mathbb{R}^+$. Example: the opposite of 5 is $\frac{1}{5}$, since $5 \oplus \frac{1}{5} = 1$.
- **2b:** $1 \odot v = v^1 = v$. **2a:** $\alpha \odot (\beta \odot v) = (v^\beta)^\alpha = v^{\alpha\beta} = (\alpha\beta) \odot v$.
- **3a:** $\alpha \odot (v_1 \oplus v_2) = (v_1 v_2)^\alpha = v_1^\alpha v_2^\alpha = (\alpha \odot v_1) \oplus (\alpha \odot v_2)$. *Numbers:* $3 \odot (2 \oplus 5) = 10^3 = 1000$ and $(3 \odot 2) \oplus (3 \odot 5) = 8 \oplus 125 = 1000$. ✓
- **3b:** $(\alpha + \beta) \odot v = v^{\alpha+\beta} = v^\alpha v^\beta = (\alpha \odot v) \oplus (\beta \odot v)$. *Numbers:* $(2+3) \odot 4 = 4^5 = 1024$ and $(2 \odot 4) \oplus (3 \odot 4) = 16 \oplus 64 = 1024$. ✓

All 8 rules hold, so $\mathbb{R}^+$ is a real vector space. **Moral:** "zero vector" means the element that *does nothing under* $\oplus$; it need not be the number 0.

Example 3.1.4 – item 5: the twisted plane

$V = \mathbb{R}^2$ with $(x_1, x_2) \oplus (y_1, y_2) = (x_1 + y_1 + 1,\ x_2 + y_2 - 3)$ and $\alpha \odot (x_1, x_2) = (\alpha x_1 + \alpha - 1,\ \alpha x_2 - 3\alpha + 3)$. Scalars: $\mathbb{R}$.
Feel the operations: $(1,1) \oplus (2,5) = (1+2+1,\ 1+5-3) = (4,3)$. $2 \odot (1,1) = (2+2-1,\ 2-6+3) = (3,-1)$.
The idea – a "change of glasses". Define $\varphi(x_1, x_2) = (x_1 + 1,\ x_2 - 3)$. Each point is shifted by the same amount. Then:

$$\varphi\big(p \oplus q\big) = \varphi(p) + \varphi(q), \qquad \varphi\big(\alpha \odot p\big) = \alpha\, \varphi(p),$$

where on the right the *ordinary* + and scalar multiplication are used. Let us check with numbers $p = (1,1)$, $q = (2,5)$: $\varphi(p) = (2,-2)$, $\varphi(q) = (3,2)$, so $\varphi(p) + \varphi(q) = (5,0)$; and $p \oplus q = (4,3)$ has $\varphi(4,3) = (5,0)$. ✓ Also $\varphi(2 \odot p) = \varphi(3,-1) = (4,-4) = 2 \cdot (2,-2) = 2\varphi(p)$. ✓
Symbolic check: $\varphi(p \oplus q) = (x_1 + y_1 + 2,\ x_2 + y_2 - 6) = (x_1 + 1, x_2 - 3) + (y_1 + 1, y_2 - 3)$ ✓; $\varphi(\alpha \odot p) = (\alpha x_1 + \alpha,\ \alpha x_2 - 3\alpha) = \alpha(x_1 + 1,\ x_2 - 3)$ ✓.
Why this proves the 8 rules. φ is a perfect re-labelling (it has an inverse: shift back). So any rule for $(\mathbb{R}^2, \oplus, \odot)$ becomes, after applying φ, the same rule for ordinary $\mathbb{R}^2$, which we know is true. Hence every rule holds.
Zero vector: the point that φ sends to $(0,0)$, i.e. $(-1,3)$. Check: $(x_1, x_2) \oplus (-1,3) = (x_1 - 1 + 1,\ x_2 + 3 - 3) = (x_1, x_2)$. ✓
Opposite of (x_1, x_2): $(-x_1 - 2,\ -x_2 + 6)$. Check: $(x_1, x_2) \oplus (-x_1 - 2, -x_2 + 6) = (x_1 - x_1 - 2 + 1,\ x_2 - x_2 + 6 - 3) = (-1, 3) = \mathbf{0}$. ✓
Rule 2b: $1 \odot (x_1, x_2) = (x_1 + 1 - 1,\ x_2 - 3 + 3) = (x_1, x_2)$. ✓

Example 3.1.4 – items 6 and 7: the same set, but real versus complex scalars

Item 6(a): $\mathbb{C}$ over $\mathbb{R}$. Vectors: complex numbers $x+iy$. Addition: $(x_1+iy_1)\oplus(x_2+iy_2) = (x_1+x_2)+i(y_1+y_2)$. Scalars are *real*: $\alpha\odot(x+iy)=\alpha x+i\alpha y$. Example: $3\odot(1+3i)=3+9i$. This is just $\mathbb{R}^2$ in disguise (write $x+iy\leftrightarrow(x,y)$), so it is a real vector space. *Here you cannot multiply by i as a scalar, because $i\notin\mathbb{R}$.*
Item 6(b): $\mathbb{C}$ over $\mathbb{C}$. Now the scalars are complex: $(\alpha+i\beta)\odot(x+iy)=(\alpha x-\beta y)+i(\alpha y+\beta x)$. **This is exactly ordinary complex multiplication:** $(\alpha+i\beta)(x+iy)=\alpha x+i\alpha y+i\beta x+i^2\beta y=(\alpha x-\beta y)+i(\alpha y+\beta x)$. *Example:* $(2+i)\odot(1+3i)=(2\cdot1-1\cdot3)+i(2\cdot3+1\cdot1)=-1+7i$. Check by expanding $(2+i)(1+3i)=2+6i+i+3i^2=-1+7i$. ✓ Since ordinary multiplication of complex numbers is associative and distributes over addition, rules 2a, 3a, 3b hold; $1\odot z=z$ is rule 2b. So $\mathbb{C}$ is a complex vector space.
Item 7: $\mathbb{C}^n$. Vectors are lists $(z_1,\dots,z_n)$ of complex numbers, with position-by-position operations $(z_1,\dots,z_n)\oplus(w_1,\dots,w_n)=(z_1+w_1,\dots,z_n+w_n)$ and $\alpha\odot(z_1,\dots,z_n)=(\alpha z_1,\dots,\alpha z_n)$.

- **7(a):** scalars $\alpha\in\mathbb{C}$. This is $\mathbb{C}^n(\mathbb{C})$. Allowed: $i\odot(1,0)=(i,0)$ *(Remark 3.1.5 in your notes).*
- **7(b):** scalars $\alpha\in\mathbb{R}$. This is $\mathbb{C}^n(\mathbb{R})$. Allowed: $2\odot(1+i,\,3)=(2+2i,\,6)$. *Not* allowed: "$i\odot(1,0)$", because i is not a real number; the vector $(i,0)$ still exists in the set, but you cannot reach it by scaling $(1,0)$.

Example 3.1.4 – items 8, 9, 10, 11, 12 (spaces of matrices, polynomials, functions, pairs)

Item 8: $M_n(\mathbb{R})$. Example with $n=2$: $\begin{pmatrix}1&2\\3&4\end{pmatrix}\oplus\begin{pmatrix}0&1\\-1&5\end{pmatrix}=\begin{pmatrix}1&3\\2&9\end{pmatrix}$ and $3\odot\begin{pmatrix}1&2\\3&4\end{pmatrix}=\begin{pmatrix}3&6\\9&12\end{pmatrix}$. A matrix is just a list of n^2 numbers written in a grid, so everything works exactly as in $\mathbb{R}^{n^2}$. Zero vector: the zero matrix.
Item 9: $P_n(\mathbb{R})$ = polynomials of degree $\le n$. Example with $n=2$: $(1+2x+3x^2)\oplus(4-x)=5+x+3x^2$ and $2\odot(1+2x+3x^2)=2+4x+6x^2$. A polynomial $a_0+a_1x+\cdots+a_nx^n$ is determined by its list of coefficients $(a_0,a_1,\dots,a_n)$, and adding/scaling polynomials just adds/scales the coefficients. So $P_n(\mathbb{R})$ behaves exactly like $\mathbb{R}^{n+1}$. *Why "degree $\le n$"?* Because the sum of two polynomials of degree $\le n$ again has degree $\le n$ (it can drop, e.g. $x^2+(-x^2+x)=x$, but never rise). Zero vector: the zero polynomial.
Item 10: $P(\mathbb{R})$ = all polynomials. Any two polynomials have finite degree; pad the shorter coefficient list with zeros and add. So closure holds. It is a vector space with "infinitely many coordinates" (no highest degree).
Item 11: $C[-1,1]$. Vectors are *functions*. Example: $f(x)=x$ and $g(x)=x^2$. Then $(f\oplus g)(x)=x+x^2$, $(3\odot f)(x)=3x$, and the opposite of f is $(-f)(x)=-x$. The zero vector is the function $\mathbf{0}(x)=0$ for all x. *Closure:* the sum of continuous functions is continuous, and a constant times a continuous function is continuous (calculus facts). Each rule holds because it holds for numbers at every single point x.
Item 12: $V\times W$. Let V,W be real vector spaces. Vectors are pairs (x,y) with $x\in V$, $y\in W$. Define $(x_1,y_1)\oplus(x_2,y_2)=(x_1+x_2,\ y_1\oplus_W y_2)$ and $\alpha\odot(x_1,y_1)=(\alpha x_1,\ \alpha\odot_W y_1)$. *Example:* $V=\mathbb{R}$, $W=\mathbb{R}^2$: $(2,(1,1))\oplus(3,(0,5))=(5,(1,6))$ and $2\odot(2,(1,1))=(4,(2,2))$. The zero vector is $(\mathbf{0}_V,\mathbf{0}_W)$. Each rule for $V\times W$ splits into the same rule for V in the first slot and for W in the second slot, and both are known to hold. So $V\times W$ is a vector space.

3.1.5 Non-examples: what goes wrong (extra examples)

Extra Example 3.1-A – six sets that look reasonable but fail

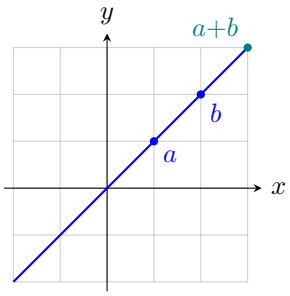
1. $\mathbb{R}^+$ with the usual $+$ and $\times$ (real scalars). Not closed under scalar multiplication: $(-1)\cdot2=-2\notin\mathbb{R}^+$. Also there is no zero vector ($0\notin\mathbb{R}^+$). *(But with the strange operations of item 4 it works!)*
2. The first quadrant $Q=\{(x,y):x\ge0,\,y\ge0\}$ with the usual operations. Closed under

adding, but not under scaling: $(-1)\cdot(1,2)=(-1,-2)\notin Q$. Rule 1d fails: $(1,2)$ has no opposite inside Q.
3. $\mathbb{Z}^2=\{(a,b):a,b$ **integers**$\}$ **over** $\mathbb{R}$**.** Not closed under scaling: $\frac{1}{2}\cdot(1,0)=(\frac{1}{2},0)\notin\mathbb{Z}^2$.
4. $\mathbb{R}^2$ **with the usual** $+$ **but** $\alpha\odot(x,y)=(\alpha x,\,y)$**.** Rule 3b fails: $(1+1)\odot(1,1)=(2,1)$, but $1\odot(1,1)\oplus 1\odot(1,1)=(1,1)+(1,1)=(2,2)$. Different answers, so the set is not a vector space.
5. $\mathbb{R}^2$ **with the usual** $+$ **but** $\alpha\odot(x,y)=(0,0)$ **for every** α**.** Rule 2b fails: $1\odot(1,2)=(0,0)\neq(1,2)$. (All other rules hold! One failing rule is enough.)
6. $\{$**polynomials of degree *exactly* 2**$\}$**.** Not closed under addition: $x^2+(-x^2+1)=1$ has degree 0. It also does not contain the zero polynomial.

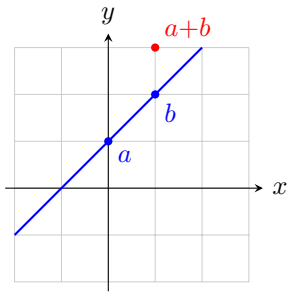
3.1.6 Subspaces

Picture it

A vector space is a big "room". A **subspace** is a smaller "room inside it" that is *closed*: whatever you do inside (add two things, or scale one thing) you never walk out. In $\mathbb{R}^2$ a line *through the origin* is such a room; a line that misses the origin is not (it does not even contain $\mathbf{0}$).



line $y=x$: stays on the line



line $y=x+1$: sum falls off the line

Definition – Subspace (Definition 3.1.6)

Let S be a non-empty subset of V. Then $S(\mathbb{F})$ is a **subspace** of $V(\mathbb{F})$ if

$$\alpha u+\beta v\in S \qquad \text{whenever } \alpha,\beta\in\mathbb{F} \text{ and } u,v\in S.$$

Remark 3.1.7. Any subspace is a vector space in its own right, with the same $\oplus$ and $\odot$.

Plain English – unpacking the one-line test

- "S *non-empty subset of* V" – S is a part of V and contains at least one vector.
- "$\alpha u+\beta v\in S$" – take *any* two vectors u,v of S and *any* two scalars; mix them. The result must still be in S. ("A mix of members is a member.")
- **One test replaces two.** Put $\alpha=\beta=1$: you get $u+v\in S$ (closed under addition). Put $\beta=0$: you get $\alpha u\in S$ (closed under scaling), because $0\cdot v=\mathbf{0}$ (Theorem 3.1.3). Conversely, if S is closed under both, then αu and βv are in S, so their sum is too. So "closed under both" $\iff$ "the one-line test".
- **Zero is forced.** Put $\alpha=\beta=0$ to get $0u+0v=\mathbf{0}\in S$. So *every subspace contains* $\mathbf{0}$. Also $-u=(-1)u\in S$.
- **Why Remark 3.1.7 is true.** The 8 rules hold for *all* vectors in V, hence for those in S. Closure

holds by the test. $\mathbf{0}$ and opposites lie in S (previous bullets). So S is a vector space. This is why proving "subspace" is much cheaper than proving "vector space": *you skip the 8 rules.*

Recipe – how to prove that S is a subspace (three steps)

Step 0 (zero first). Is $\mathbf{0} \in S$? If not, *stop: S is not a subspace.* (If yes, this does *not* yet prove it is a subspace; continue.)
Step 1 (take arbitrary members). Let $u, v \in S$ and let α, β be scalars. Write down the condition each of u, v satisfies.
Step 2 (mix and check). Compute $\alpha u + \beta v$ and show it satisfies the same condition.
To disprove: give one explicit counterexample (specific u, v, α, β with $\alpha u + \beta v \notin S$).

Typical condition	**Subspace? (rule of thumb)**
$a_1x_1 + \cdots + a_nx_n = 0$ (linear, right side 0), e.g. $x + y - z = 0$, $x = y$, $x_1 = 0$	**Yes.** Linear and homogeneous conditions survive mixing.
Same with right side $\neq 0$, e.g. $x + y + z = 3$	**No.** $\mathbf{0}$ fails the equation.
Squares, products, absolute values: $z = x^2$, $xy = 0$, $\lvert x\rvert \leq 1$	**No** (counterexample needed).
Inequalities: $x \geq 0$	**No.** Scaling by -1 breaks it.
Special number types: x integer / rational	**No.** Scaling by $\sqrt{2}$ or $\frac{1}{2}$ breaks it.
"$f(a) = 0$" for functions, "$p(1) = 0$" for polynomials, "trace $= 0$", "$A = A^t$"	**Yes** (linear in the object).

Example 3.1.8 of your notes – five basic subspaces and non-subspaces

(1) $S = \{\mathbf{0}\}$ and $S = V$ are subspaces (the *trivial* ones).
(2) $S = \{(x, y, z) \in \mathbb{R}^3 : \ x + y - z = 0\}$ (a plane through the origin) is a subspace.
(3) $S = \{(x, y, z) \in \mathbb{R}^3 : \ x + y + z = 3\}$ (a plane missing the origin) is *not.*
(4) $S = \{(x, y, z) \in \mathbb{R}^3 : \ z = x\}$ is a subspace.
(5) $P_n(\mathbb{R})$ is a subspace of $P(\mathbb{R})$.
Solution.

Step 1 – trivial subspaces. $\{\mathbf{0}\}$: it is non-empty and $\alpha\mathbf{0} + \beta\mathbf{0} = \mathbf{0} \in \{\mathbf{0}\}$. V: any mix of vectors of V lies in V because V is a vector space (closure).

Step 2 – plane $x + y - z = 0$. *Number sanity check first:* $u = (1, 2, 3)$ and $w = (2, -1, 1)$ are in S ($1 + 2 - 3 = 0$, $2 - 1 - 1 = 0$). Then $u + w = (3, 1, 4)$ and $3 + 1 - 4 = 0$ ✓; $5u = (5, 10, 15)$ and $5 + 10 - 15 = 0$ ✓. *General proof:* let $u = (x, y, z)$, $w = (x', y', z') \in S$, so $x + y - z = 0$ and $x' + y' - z' = 0$. For scalars α, β: $\alpha u + \beta w = (\alpha x + \beta x',\ \alpha y + \beta y',\ \alpha z + \beta z')$ and

$$(\alpha x + \beta x') + (\alpha y + \beta y') - (\alpha z + \beta z') = \alpha(x + y - z) + \beta(x' + y' - z') = \alpha \cdot 0 + \beta \cdot 0 = 0.$$

So $\alpha u + \beta w \in S$. Subspace. ✓

Step 3 – plane $x + y + z = 3$. $\mathbf{0} = (0, 0, 0)$ gives $0 + 0 + 0 = 0 \neq 3$, so $\mathbf{0} \notin S$. Every subspace contains $\mathbf{0}$. **Not a subspace.**

Step 4 – $z = x$. Let $(x, y, z), (x', y', z') \in S$, so $z = x$ and $z' = x'$. The mix has third coordinate $\alpha z + \beta z' = \alpha x + \beta x'$, which equals its first coordinate. So the mix is in S. Subspace. ✓

Step 5 – $P_n(\mathbb{R}) \subseteq P(\mathbb{R})$. Every polynomial of degree $\leq n$ is a real polynomial, and $P_n(\mathbb{R}) \neq \varnothing$. If p, q have degree $\leq n$, then $\alpha p + \beta q$ has degree $\leq n$ (degree never goes up when you add or scale). So it is in $P_n(\mathbb{R})$. Subspace. ✓

Extra Example 3.1-B – more subspace tests (do these on paper first!)

(1) $S = \{(x, y) \in \mathbb{R}^2 : y = 3x\}$. *Zero:* $0 = 3 \cdot 0$ ✓. Let $(x, 3x), (x', 3x') \in S$. Then $\alpha(x, 3x) + \beta(x', 3x') = (\alpha x + \beta x',\ 3(\alpha x + \beta x'))$, again of the form $(t, 3t)$. **Subspace** (a line through the origin).
(2) $S = \{(x, y) \in \mathbb{R}^2 : y = 3x + 1\}$. Zero: $0 = 3 \cdot 0 + 1$? No. **Not a subspace.**
(3) $S = \{(x, y) \in \mathbb{R}^2 : xy = 0\}$ (the two axes together). Zero is in S ✓, but $(1, 0) \in S$ and $(0, 1) \in S$ while $(1, 0) + (0, 1) = (1, 1)$ has $1 \cdot 1 = 1 \neq 0$. **Not a subspace.**
(4) $S = \{(x, y, z) : x + y + z = 0$ **and** $x - z = 0\}$. Two homogeneous linear equations: the mix satisfies both (same computation as (2) in Example 3.1.8). **Subspace.** Solve it: $x = z$ and $2x + y = 0$ give $(x, -2x, x) = x(1, -2, 1)$, a line through the origin. Check: $1 - 2 + 1 = 0$ ✓.
(5) Polynomials. $S = \{p \in P_2(\mathbb{R}) : p(0) = 0\}$ is a subspace: if $p(0) = 0 = q(0)$ then $(\alpha p + \beta q)(0) = \alpha p(0) + \beta q(0) = 0$. But $\{p : p(0) = 1\}$ is *not*: the zero polynomial has value 0 at 0.
(6) Matrices. Upper triangular 2×2 matrices $\begin{pmatrix} a & b \\ 0 & d \end{pmatrix}$: a mix of two such matrices again has 0 in the lower-left corner: **subspace**. The set of *invertible* 2×2 matrices is **not**: I and $-I$ are invertible but $I + (-I)$ is the zero matrix (no zero matrix inside!). The set of matrices with $\det = 0$ is also **not**: $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = I$ has $\det = 1$.

Extra Example 3.1-C – the null space is always a subspace (the notes' motivating example, proved)

Let A be an $m \times n$ real matrix and $N(A) = \{x \in \mathbb{R}^n : Ax = \mathbf{0}\}$.
Step 1 – zero. $A\mathbf{0} = \mathbf{0}$, so $\mathbf{0} \in N(A)$.
Step 2 – two members. Let $x, y \in N(A)$, i.e. $Ax = \mathbf{0}$ and $Ay = \mathbf{0}$. Let $\alpha, \beta \in \mathbb{R}$.
Step 3 – use linearity of matrix multiplication. $A(\alpha x + \beta y) = \alpha Ax + \beta Ay = \alpha\mathbf{0} + \beta\mathbf{0} = \mathbf{0}$.
Step 4 – conclude. So $\alpha x + \beta y \in N(A)$: $N(A)$ is a subspace of $\mathbb{R}^n$. ✓
Number check: with $A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \end{pmatrix}$ we found $N(A) = \{t(-2, 1, 0)\}$ in Example 0.3; the multiples $t(-2, 1, 0)$ are closed under mixing, just like the line $y = 3x$ above.

Which subspaces exist? (preview – you will prove the $\mathbb{R}^2$ case in Question 1.5)

- Subspaces of $\mathbb{R}^2$: only $\{\mathbf{0}\}$, the *lines through the origin*, and $\mathbb{R}^2$ itself.
- Subspaces of $\mathbb{R}^3$: only $\{\mathbf{0}\}$, lines through the origin, planes through the origin, and $\mathbb{R}^3$.

Nothing curved, nothing that misses the origin, nothing bounded.

Extra Example 3.1-D – intersection, union and sum of two subspaces (the three operations you will meet in the questions)

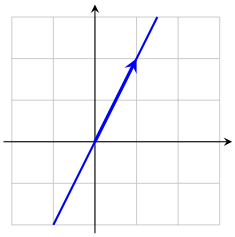
Let $P = \{(x, 0)\}$ (the x-axis) and $Q = \{(0, y)\}$ (the y-axis), both subspaces of $\mathbb{R}^2$.
- $P \cap Q$ (in *both*) $= \{\mathbf{0}\}$. A subspace. (Question 1.6 proves: the intersection of subspaces is always a subspace.)
- $P \cup Q$ (in *at least one*) is the two axes. *Not* a subspace: $(1, 0) + (0, 1) = (1, 1)$ is on neither axis. (Question 1.6: the union is a subspace only if one of P, Q contains the other.)

- $P + Q = \{u + v : u \in P, v \in Q\} = \{(x, 0) + (0, y)\} = \{(x, y)\} = \mathbb{R}^2$. A subspace. (Question 1.7 proves: $P + Q$ is always a subspace.)

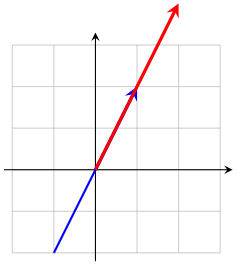
3.1.7 Linear combinations and span

Picture it

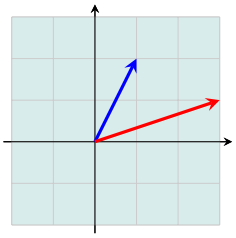
You have a few "ingredient" vectors. A **linear combination** is any *recipe*: "3 spoons of the first, -2 of the second, ...". The **span** is *everything you can cook*, i.e. every point you can reach. One non-zero vector reaches a line; two vectors pointing in different directions reach a whole plane; two vectors pointing the same way still reach only a line.



$L\{(1, 2)\}$: a line



$L\{(1, 2), (2, 4)\}$: same line



$L\{(1, 2), (3, 1)\}$: whole plane

Definition – Linear span (Definition 3.1.10)

Let $S = \{u_1, \ldots, u_n\}$ be a non-empty subset of $V(\mathbb{F})$. The **linear span** of S is

$$L(S) = \{\alpha_1 u_1 + \alpha_2 u_2 + \cdots + \alpha_n u_n : \ \alpha_i \in \mathbb{F}\}.$$

Each such expression is a **linear combination** of $u_1, \ldots, u_n$. By convention $L(\varnothing) = \{\mathbf{0}\}$.

Plain English

$L(S)$ is the set of *all* the mixes $\alpha_1 u_1 + \cdots + \alpha_n u_n$ you get by letting the scalars α_i run through $\mathbb{F}$. "Is b in $L(S)$?" means "can I find scalars α_i so that the mix equals b?" That is a *system of linear equations* in the unknowns $\alpha_1, \ldots, \alpha_n$.

Recipe 1 – is b a linear combination of $v_1, \ldots, v_k$?

1. Write the equation $\alpha_1 v_1 + \cdots + \alpha_k v_k = b$ coordinate by coordinate. This is a linear system in $\alpha_1, \ldots, \alpha_k$.
2. Row-reduce the augmented matrix $[\, v_1 \ \cdots \ v_k \,|\, b \,]$ (vectors as columns).
3. If the system is consistent, $b \in L(v_1, \ldots, v_k)$ and the solution gives the coefficients. If inconsistent, $b \notin L$.
4. The coefficients are unique *iff* there are no free variables (that is, iff $v_1, \ldots, v_k$ are independent – Section 3.2).

Recipe 2 – describe $L(S)$ by an equation

Write the general mix with letters $a, b, \ldots$, read off x, y, z in terms of them, then eliminate a, b. (For two vectors in $\mathbb{R}^3$ one can also use: $(x, y, z) \in L(v_1, v_2) \iff \det[v_1 \ v_2 \ (x, y, z)] = 0$.)

Example 3.1.11 of your notes – fully explained

(1) Show $(4,5,5)$ is a linear combination of $(1,0,0),(1,1,0),(1,1,1)$, and that the combination is unique.
(2) Is $(4,5,5)$ a linear combination of $(1,2,3),(-1,1,4),(3,3,2)$? Is the combination unique?
(3) Show $(4,5,5)$ is *not* a linear combination of $(1,2,1),(1,1,0)$.
(4) Describe $L(S)$ for $S=\{(1,1,1),(2,1,3)\}$.

Solution (1).

Step 1 – set up. Want $\alpha(1,0,0)+\beta(1,1,0)+\gamma(1,1,1)=(4,5,5)$. In coordinates: $\alpha+\beta+\gamma=4$, $\beta+\gamma=5$, $\gamma=5$.

Step 2 – solve from the bottom. $\gamma=5$; then $\beta=5-5=0$; then $\alpha=4-0-5=-1$.

Step 3 – answer and check. $(4,5,5)=-1(1,0,0)+0(1,1,0)+5(1,1,1)$. Check: $(-1+0+5,\ 0+0+5,\ 5)=(4,5,5)$ ✓. (The notes write it as $5(1,1,1)-1(1,0,0)+0(1,1,0)$; same thing.)

Step 4 – unique?. The system had exactly one solution (no free variable): the coefficient matrix $\begin{pmatrix}1&1&1\\0&1&1\\0&0&1\end{pmatrix}$ is triangular with $\det=1\neq 0$. So the representation is **unique**.

Solution (2).

Step 1 – set up. $\alpha_1(1,2,3)+\alpha_2(-1,1,4)+\alpha_3(3,3,2)=(4,5,5)$ gives the augmented matrix $\left(\begin{array}{ccc|c}1&-1&3&4\\2&1&3&5\\3&4&2&5\end{array}\right)$.

Step 2 – row-reduce. It becomes $\left(\begin{array}{ccc|c}1&0&2&3\\0&1&-1&-1\\0&0&0&0\end{array}\right)$. There is a zero row with $0=0$ (consistent) and column 3 is free.

Step 3 – read off. $\alpha_1=3-2t$, $\alpha_2=t-1$, $\alpha_3=t$ for any real t.

Step 4 – check two members of the family. $t=0$: $3(1,2,3)-1(-1,1,4)+0=(3+1,\ 6-1,\ 9-4)=(4,5,5)$ ✓. $t=1$: $1(1,2,3)+0(-1,1,4)+1(3,3,2)=(4,5,5)$ ✓.

Step 5 – answer. Yes, it is a combination, but **not unique** (infinitely many choices of t). Reason: $\det\begin{pmatrix}1&-1&3\\2&1&3\\3&4&2\end{pmatrix}=0$, so the three vectors are dependent (Section 3.2).

Solution (3).

Step 1 – set up. $b_1(1,2,1)+b_2(1,1,0)=(4,5,5)$ gives $b_1+b_2=4$, $2b_1+b_2=5$, $b_1=5$.

Step 2 – first two equations. Subtract: $b_1=1$, then $b_2=3$.

Step 3 – contradiction. But the third equation demands $b_1=5$. Impossible. So **no** b_1,b_2 exist: $(4,5,5)\notin L\{(1,2,1),(1,1,0)\}$.

Solution (4).

Step 1 – write the general mix. $\alpha(1,1,1)+\beta(2,1,3)=(\alpha+2\beta,\ \alpha+\beta,\ \alpha+3\beta)=(x,y,z)$.

Step 2 – eliminate α,β. Note $2x-y=2(\alpha+2\beta)-(\alpha+\beta)=\alpha+3\beta=z$. So every point of $L(S)$ satisfies $z=2x-y$.

Step 3 – the other direction. Conversely, if $z=2x-y$: solve $x=\alpha+2\beta$, $y=\alpha+\beta$ to get $\beta=x-y$, $\alpha=2y-x$; then $\alpha+3\beta=(2y-x)+3(x-y)=2x-y=z$ ✓. So the mix reaches (x,y,z).

Step 4 – answer. $L(S)=\{(x,y,z)\in\mathbb{R}^3:\ 2x-y=z\}$, a plane through the origin. (Check a member: $(1,1,1)$: $2-1=1$ ✓.)

Extra Example 3.1-E – more span questions

(1) Is $(1,2,3) \in L\{(1,0,1),(0,1,1)\}$**? Is** $(1,1,1)$**?** Solve $a(1,0,1)+b(0,1,1)=(x,y,z)$: the first two coordinates force $a=x$, $b=y$; the third then requires $z=a+b=x+y$. For $(1,2,3)$: $3=1+2$ ✓, so $(1,2,3)=1\cdot(1,0,1)+2\cdot(0,1,1)$. For $(1,1,1)$: $1\neq 1+1$, so **no**.
(2) Describe $L\{(1,2,3),(0,1,1)\}$ **by an equation.** General mix: $(a,\ 2a+b,\ 3a+b)$. Then $x+y=a+2a+b=3a+b=z$. So $L=\{x+y=z\}$, the plane $x+y-z=0$. (Check: $(1,2,3)$: $1+2-3=0$ ✓; $(0,1,1)$: $0+1-1=0$ ✓.) Interesting: it is the same plane as $\{x+y-z=0\}$ in Example 3.1.8(2).
(3) $L\{(1,0),(0,1)\}=\mathbb{R}^2$ **and** $L\{(1,1),(1,-1)\}=\mathbb{R}^2$**.** For the second: solve $a(1,1)+b(1,-1)=(x,y)$: $a+b=x$, $a-b=y$, so $a=\frac{x+y}{2}$, $b=\frac{x-y}{2}$. Example: $(3,1)=2(1,1)+1(1,-1)$ (check: $(2+1,\ 2-1)=(3,1)$ ✓). So two *different* spanning sets can give the same span.
(4) Polynomials. $L\{1,x,x^2\}=P_2(\mathbb{R})$ because $a_0+a_1x+a_2x^2=a_0\cdot 1+a_1\cdot x+a_2\cdot x^2$. Also $L\{1+x,\ 1-x\}=P_1(\mathbb{R})$ since $a+bx=\frac{a+b}{2}(1+x)+\frac{a-b}{2}(1-x)$ (check: constant term $\frac{a+b}{2}+\frac{a-b}{2}=a$ ✓, coefficient of x: $\frac{a+b}{2}-\frac{a-b}{2}=b$ ✓).
(5) Matrices. $L\left\{\begin{pmatrix}1&0\\0&0\end{pmatrix},\begin{pmatrix}0&0\\0&1\end{pmatrix}\right\}=\left\{\begin{pmatrix}a&0\\0&d\end{pmatrix}\right\}$, the diagonal 2×2 matrices.

Lemma 3.1.12, Remark 3.1.13 and Theorem 3.1.14

Lemma 3.1.12. For a non-empty $S\subseteq V$, $L(S)$ is a subspace of V.
Remark 3.1.13. If $W\subseteq V$ is a subspace and $S\subseteq W$, then $L(S)\subseteq W$.
Theorem 3.1.14. $L(S)$ is the *smallest* subspace of V containing S.

Plain English

Mixing vectors never takes you out of a subspace, so $L(S)$ is itself a subspace (3.1.12). If a subspace W already holds all of S, it holds all mixes of S too (3.1.13). Therefore $L(S)$ fits inside every subspace that contains S: it is the *least* one (3.1.14).

Proof in small steps – Lemma 3.1.12: $L(S)$ **is a subspace**

Step 1 – non-empty. $S\subseteq L(S)$ (take $\alpha_i=1$ for one u_i and 0 for the others), and $S\neq\varnothing$.
Step 2 – two members. Let $u,v\in L(S)$: $u=\alpha_1w_1+\cdots+\alpha_nw_n$ and $v=\beta_1w_1+\cdots+\beta_nw_n$ (use the same list $w_1,\ldots,w_n$ for both, padding with zero coefficients if needed).
Step 3 – mix them. For scalars a,b: $au+bv=(a\alpha_1+b\beta_1)w_1+\cdots+(a\alpha_n+b\beta_n)w_n$.
Step 4 – conclude. This is again a combination of $w_1,\ldots,w_n$ with scalars $a\alpha_i+b\beta_i\in\mathbb{F}$. So $au+bv\in L(S)$. ■

Proof in small steps – Remark 3.1.13 and Theorem 3.1.14

3.1.13. Let $W\supseteq S$ be a subspace and $x=\alpha_1u_1+\cdots+\alpha_nu_n\in L(S)$ with each $u_i\in S\subseteq W$. A subspace is closed under mixes, so $x\in W$. Hence $L(S)\subseteq W$.
3.1.14. (i) $L(S)$ is a subspace (Lemma 3.1.12) and contains S. (ii) If W is any subspace containing S, then $L(S)\subseteq W$ by 3.1.13. So no subspace containing S is smaller than $L(S)$. ■

3.1.8 Row space, column space and null space of a matrix

Definition – Definition 3.1.15

Let A be an $m \times n$ real matrix with rows $a_1^t, \dots, a_m^t \in \mathbb{R}^n$ and columns $b_1, \dots, b_n \in \mathbb{R}^m$. Then:

1. $\text{Row}(A) = L(a_1, \dots, a_m)$ (the **row space**, a subspace of $\mathbb{R}^n$).
2. $\text{Col}(A) = L(b_1, \dots, b_n)$ (the **column space**, a subspace of $\mathbb{R}^m$).
3. $N(A) = \{x \in \mathbb{R}^n : Ax = \mathbf{0}\}$ (the **null space**, a subspace of $\mathbb{R}^n$).
4. $\text{Range}(A) = \{y \in \mathbb{R}^m : y = Ax \text{ for some } x \in \mathbb{R}^n\}$.

Note: $\text{Col}(A) = \text{Range}(A)$, because $Ax = x_1 b_1 + \cdots + x_n b_n$ is a mix of the columns. Also $\text{Col}(A) = \{b : Ax = b \text{ has a solution}\}$.

Extra Example 3.1-F – all four spaces for one matrix

Let $A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 3 & 6 & 4 \end{pmatrix}$. Rows: $(1,2,1), (2,4,3), (3,6,4)$. Columns: $(1,2,3), (2,4,6), (1,3,4)$.

Ax is a mix of the columns: $A \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} = 1 \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + 0 \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix} + 2 \begin{pmatrix} 1 \\ 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 3 \\ 8 \\ 11 \end{pmatrix}$. (Check by rows: $1+0+2=3$, $2+0+6=8$, $3+0+8=11$ ✓.)

Row-reduce: $A \to \begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$ (the RREF). Its rank is 2.

- **Row space:** spanned by the non-zero rows of the RREF: $(1,2,0)$ and $(0,0,1)$. (Why: Lemma 3.1.16 below.)
- **Null space:** $x + 2y = 0$, $z = 0$, so $N(A) = \{t(-2,1,0)\}$.
- **Column space:** spanned by the columns of the *original* A that are pivot columns (columns 1 and 3): $(1,2,3)$ and $(1,3,4)$. Both satisfy $x + y - z = 0$, so $\text{Col}(A)$ is the plane $x + y = z$.

Warning (the big trap). The columns of the RREF, e.g. $(1,0,0)$, are *not* in $\text{Col}(A)$: $1+0-0 = 1 \neq 0$. Row operations change the column space. Use the pivot columns of the *original* matrix.

Lemma 3.1.16 and Theorem 3.1.17

Lemma 3.1.16. If $B = EA$ for an elementary matrix E (that is, B comes from A by one row operation), then $\text{Row}(A) = \text{Row}(B)$.

Theorem 3.1.17. For an $m \times n$ real matrix A: (1) $N(A)$ is a subspace of $\mathbb{R}^n$. (2) The non-zero rows of the row-reduced form of A form a basis of $\text{Row}(A)$; hence $\dim \text{Row}(A) =$ row rank of A.

Plain English

Row operations only *recombine* the rows, so the collection of rows reachable by mixing stays the same. Row-reducing therefore does not change the row space, and in RREF the non-zero rows are exactly the leftover "useful" rows. ("Basis" is explained in Section 3.3: a spanning list with no waste.)

Proof in small steps – Lemma 3.1.16: a row operation does not change the row space

Step 1 – what a row operation does. It replaces the row list $a_1, \dots, a_m$ by a new list in which each new row is a mix of the old rows. (Swap: same rows reordered. Scale a_i by $c \neq 0$: new row ca_i. Add ca_j to a_i: new row $a_i + ca_j$.)

Step 2 – new $\subseteq$ old. So every new row lies in L(old rows). Since L(old rows) is a subspace, every mix of new rows lies in it: $\text{Row}(B) \subseteq \text{Row}(A)$.

Step 3 – every row operation can be undone. Swap again; scale by $1/c$; add $-c\,a_j$. So A is obtained from B by a row operation too.

Step 4 – old $\subseteq$ new. Apply Step 2 with the roles of A, B swapped: $\text{Row}(A) \subseteq \text{Row}(B)$. Together: $\text{Row}(A) = \text{Row}(B)$. ■

Consequence: row operations applied k times (all the way to RREF) keep the row space the same.

Proof in small steps – Theorem 3.1.17(1) and (2)

(1) Proved in Extra Example 3.1-C.
(2) By Lemma 3.1.16 repeated, $\text{Row}(A) = \text{Row}(D)$, D the RREF. The zero rows add nothing to a span, so $\text{Row}(D) = L(\text{non-zero rows } d_1, \ldots, d_r)$. They are independent: each d_i has a pivot 1 in some column where all the other rows have 0; so if $c_1 d_1 + \cdots + c_r d_r = \mathbf{0}$, looking at that pivot column gives $c_i = 0$. Hence they form a basis, and $\dim \text{Row}(A) = r =$ number of non-zero rows = row rank. ■

Extra Example 3.1-G – row operations keep the row space but change the column space

$A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ $(R_2 \to R_2 - R_1)$. Rows: both have row space $L\{(1,1)\}$. Columns: $\text{Col}(A) = L\{(1,1)\}$, but $\text{Col}(B) = L\{(1,0)\}$. These are different lines ($(1,1) \notin L\{(1,0)\}$). This is Question 1.4's second part.

3.1.9 Questions for Section 3.1 (Exercises 3.1.9 and 3.1.18 of your notes)

Before you start – the plan for every question below

For a "subspace?" question: **(0)** check $\mathbf{0} \in S$; **(1)** pick arbitrary members; **(2)** mix them and test the condition. For a "not a subspace" answer, always give *one explicit counterexample.*

Question 1.1 (Exercise 3.1.9, item 1)

Which of the following are correct statements?
(a) Let $S = \{(x, y, z) \in \mathbb{R}^3 : z = x^2\}$. Then S is a subspace of $\mathbb{R}^3$.
(b) Let $V(\mathbb{F})$ be a vector space and $x \in V$. Then $\{\alpha x : \alpha \in \mathbb{F}\}$ is a subspace of V.
(c) Let $W = \{f \in C[-1,1] : f(1/2) = 0\}$. Then W is a subspace of $C[-1,1]$.
Solution.

Step 1 – (a) is false. $(1,0,1) \in S$ since $1 = 1^2$. Multiply by 2: $2(1,0,1) = (2,0,2)$, and $2 \neq 2^2 = 4$. So $2u \notin S$. (Note: $\mathbf{0} \in S$, so the zero test alone does *not* catch this.)

Step 2 – (b) is true. Let $u = \alpha x$ and $v = \alpha' x$ be members and $a, b \in \mathbb{F}$. Then $au + bv = (a\alpha + b\alpha')x$, and $a\alpha + b\alpha' \in \mathbb{F}$. So it is again of the form (scalar)$\cdot x$. Also $\mathbf{0} = 0 \cdot x \in S$. (Example: $x = (1,2)$ gives the line $\{(t, 2t)\}$.)

Step 3 – (c) is true. The zero function has value 0 at $\frac{1}{2}$. For $f, g \in W$ and real a, b: $h = af + bg$ is continuous and $h(\frac{1}{2}) = af(\frac{1}{2}) + bg(\frac{1}{2}) = 0 + 0 = 0$. So $h \in W$. (Example members: $f(x) = 2x - 1$, $g(x) = x^2 - \frac{1}{4}$.)

One-line summary: (a) false, (b) true, (c) true.

Question 1.2 (Exercise 3.1.9, item 2)

Which of the following are subspaces of $\mathbb{R}^n$?
(a) $\{(x_1,\dots,x_n) : x_1 \geq 0\}$. (b) $\{(x_1,\dots,x_n) : x_1 + 2x_2 = 4x_3\}$. (c) $\{(x_1,\dots,x_n) : x_1 \text{ is rational}\}$.
(d) $\{(x_1,\dots,x_n) : x_1 = x_3^2\}$. (e) $\{(x_1,\dots,x_n) : \text{either } x_1 \text{ or } x_2 \text{ or both is } 0\}$. (f) $\{(x_1,\dots,x_n) : |x_1| \leq 1\}$.
Solution. (Take $n \geq 3$ so all coordinates used exist.)

Step 1 – (a) No. $u = (1,0,\dots,0)$ is in the set. $(-1)u = (-1,0,\dots,0)$ has $x_1 = -1 < 0$: not in the set.

Step 2 – (b) Yes. A linear equation with right side 0. Proof: if $x_1 + 2x_2 = 4x_3$ and $y_1 + 2y_2 = 4y_3$, then $(\alpha x_1 + \beta y_1) + 2(\alpha x_2 + \beta y_2) = \alpha(x_1 + 2x_2) + \beta(y_1 + 2y_2) = 4(\alpha x_3 + \beta y_3)$. ✓

Step 3 – (c) No. $u = (1,0,\dots,0)$ has rational x_1. Multiply by the real scalar $\sqrt{2}$: $x_1 = \sqrt{2}$ is not rational.

Step 4 – (d) No. $u = (1,0,1,0,\dots)$: $x_1 = 1 = 1^2 = x_3^2$ ✓. But $2u = (2,0,2,\dots)$ has $x_1 = 2 \neq 4 = x_3^2$.

Step 5 – (e) No. $u = (1,0,0,\dots)$ and $v = (0,1,0,\dots)$ are in the set, but $u + v = (1,1,0,\dots)$ has $x_1 = 1 \neq 0$ and $x_2 = 1 \neq 0$.

Step 6 – (f) No. $u = (1,0,\dots)$ has $|x_1| = 1 \leq 1$, but $2u$ has $|x_1| = 2 > 1$.

One-line summary: Only (b) is a subspace.

Question 1.3 (Exercise 3.1.9, item 3)

Which of the following are subspaces of (i) $\mathbb{C}^n(\mathbb{R})$ and (ii) $\mathbb{C}^n(\mathbb{C})$?
(a) $\{(z_1,\dots,z_n) : z_1 \text{ is real}\}$. (b) $\{(z_1,\dots,z_n) : z_1 + z_2 = z_3\}$. (c) $\{(z_1,\dots,z_n) : |z_1| = |z_2|\}$.
Solution. Remember: in (i) the scalars are real, in (ii) they are complex.

Step 1 – (a), scalars real – Yes. If z_1 and w_1 are real and α, β are real, then $\alpha z_1 + \beta w_1$ is real. ✓

Step 2 – (a), scalars complex – No. $u = (1,0,\dots)$ is in the set. Multiply by the complex scalar i: $iu = (i,0,\dots)$ has $z_1 = i$, not real. **So (a) is a subspace of $\mathbb{C}^n(\mathbb{R})$ but not of $\mathbb{C}^n(\mathbb{C})$.**

Step 3 – (b), both – Yes. Let $z_1 + z_2 = z_3$, $w_1 + w_2 = w_3$ and α, β be scalars (real *or* complex). Then $(\alpha z_1 + \beta w_1) + (\alpha z_2 + \beta w_2) = \alpha z_3 + \beta w_3$. The same computation works for any scalars, so (b) is a subspace of both.

Step 4 – (c), both – No. $u = (1,1,0,\dots)$ ($|1| = |1|$) and $v = (1,i,0,\dots)$ ($|1| = |i| = 1$) are in the set. $u + v = (2,1+i,0,\dots)$ has $|2| = 2$ but $|1+i| = \sqrt{2}$. Not equal. Only the scalars $\alpha = \beta = 1$ (real!) were used, so (c) fails in both.

One-line summary: (a): (i) yes, (ii) no. (b): yes, yes. (c): no, no.

Question 1.4 (Exercise 3.1.18, item 1)

Show that any two row-equivalent matrices have the same row space. Give examples to show that the column space of two row-equivalent matrices need not be the same.
Solution.

Step 1 – what "row-equivalent" means. B is obtained from A by a finite list of row operations.

Step 2 – one operation does not change the row space. This is Lemma 3.1.16, proved above (each new row is a mix of old rows, and the operation can be undone).

Step 3 – many operations. Apply Step 2 once for each operation: $\text{Row}(A) = \text{Row}(A_1) = \text{Row}(A_2) = \cdots = \text{Row}(B)$. ✓

Step 4 – example for columns. $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ (use $R_2 \to R_2 - R_1$). Then $\text{Col}(A) = L\{(1,1)\}$ but $\text{Col}(B) = L\{(1,0)\}$. They differ, since $(1,1) = (a,0)$ is impossible.

One-line summary: Row operations keep the row space; they can change the column space.

Question 1.5 (Exercise 3.1.18, item 2)

Find all the vector subspaces of $\mathbb{R}^2$.

Solution. Answer: $\{\mathbf{0}\}$, the lines through the origin $\{t\,d : t \in \mathbb{R}\}$ (with $d \neq \mathbf{0}$), and $\mathbb{R}^2$.

Step 1 – these are subspaces. $\{\mathbf{0}\}$ and $\mathbb{R}^2$ are trivial subspaces. A line $\{td\}$ is $L(d)$, a span, hence a subspace (Lemma 3.1.12).

Step 2 – nothing else. Let S be any subspace. **Case A:** $S = \{\mathbf{0}\}$. Done.

Step 3 – Case B: S has a non-zero vector d. Then $L(d) \subseteq S$ (closed under scaling). If $S = L(d)$, S is a line. Done.

Step 4 – Case C: S also has a vector $w \notin L(d)$. Then d, w are not multiples of each other, so $\det[d\ w] \neq 0$. Therefore every $(x, y) \in \mathbb{R}^2$ solves $(x, y) = a\,d + b\,w$ uniquely. *Example:* $d = (1, 0)$, $w = (1, 1)$: $(x, y) = (x - y)(1, 0) + y(1, 1)$. Check: $(x - y + y,\ y) = (x, y)$ ✓. So every vector of $\mathbb{R}^2$ is in S (a subspace contains all mixes): $S = \mathbb{R}^2$.

One-line summary: A subspace of $\mathbb{R}^2$ is $\{\mathbf{0}\}$, a line through $\mathbf{0}$, or all of $\mathbb{R}^2$.

Question 1.6 (Exercise 3.1.18, item 3)

Let P, Q be two subspaces of V. Show that $P \cap Q$ is a subspace of V. Also show that $P \cup Q$ need not be a subspace. When is $P \cup Q$ a subspace?

Solution.

Step 1 – $P \cap Q$ is a subspace. $\mathbf{0} \in P$ and $\mathbf{0} \in Q$, so $\mathbf{0} \in P \cap Q$ (non-empty). Let $u, v \in P \cap Q$ and α, β scalars. Since $u, v \in P$ and P is a subspace, $\alpha u + \beta v \in P$. Same for Q. So $\alpha u + \beta v \in P \cap Q$. ✓

Step 2 – $P \cup Q$ need not be. $P = \{(x, 0)\}$, $Q = \{(0, y)\}$ in $\mathbb{R}^2$. $(1, 0) \in P$, $(0, 1) \in Q$, but $(1, 0) + (0, 1) = (1, 1) \notin P \cup Q$.

Step 3 – the exact condition. **Claim:** $P \cup Q$ is a subspace $\iff$ $P \subseteq Q$ or $Q \subseteq P$.

Step 4 – ($\Leftarrow$). If $P \subseteq Q$ then $P \cup Q = Q$, a subspace. Same if $Q \subseteq P$.

Step 5 – ($\Rightarrow$) by contradiction. Suppose $P \cup Q$ is a subspace but $P \not\subseteq Q$ and $Q \not\subseteq P$. Pick $p \in P \setminus Q$ and $q \in Q \setminus P$. Then $p + q \in P \cup Q$, so $p + q \in P$ or $p + q \in Q$. If $p + q \in P$ then $q = (p + q) - p \in P$ – contradiction with $q \notin P$. If $p + q \in Q$ then $p = (p + q) - q \in Q$ – contradiction with $p \notin Q$. So $P \subseteq Q$ or $Q \subseteq P$.

One-line summary: $P \cap Q$ always a subspace; $P \cup Q$ is a subspace only when one contains the other.

Question 1.7 (Exercise 3.1.18, item 4)

Let P, Q be subspaces of V and $P + Q = \{u + v : u \in P, v \in Q\}$. Show that $P + Q$ is a subspace and that $L(P \cup Q) = P + Q$.

Solution.

Step 1 – non-empty. $\mathbf{0} = \mathbf{0} + \mathbf{0} \in P + Q$.

Step 2 – mix two members. Let $x = u_1 + v_1$ and $y = u_2 + v_2$ with $u_i \in P$, $v_i \in Q$. Then $\alpha x + \beta y = (\alpha u_1 + \beta u_2) + (\alpha v_1 + \beta v_2)$. The first bracket is in P, the second in Q. So $\alpha x + \beta y \in P + Q$. ✓

Step 3 – $L(P \cup Q) \subseteq P + Q$. Each $u \in P$ equals $u + \mathbf{0} \in P + Q$; each $v \in Q$ equals $\mathbf{0} + v \in P + Q$. So $P \cup Q \subseteq P + Q$. $P + Q$ is a subspace, so by Theorem 3.1.14 (L is the smallest subspace containing the set), $L(P \cup Q) \subseteq P + Q$.

Step 4 – $P + Q \subseteq L(P \cup Q)$. $u + v = 1 \cdot u + 1 \cdot v$ is a mix of two vectors from $P \cup Q$.

One-line summary: $P + Q$ is a subspace, and it equals the span of $P \cup Q$.

Question 1.8 (Exercise 3.1.18, item 5)

Let $S = \{x_1, x_2, x_3, x_4\}$ with $x_1 = (1,0,0,0)$, $x_2 = (1,1,0,0)$, $x_3 = (1,2,0,0)$, $x_4 = (1,1,1,0)$. Determine all x_i such that $L(S) = L(S \setminus \{x_i\})$.

Solution.

Step 1 – the key fact. $L(S) = L(S \setminus \{x\})$ exactly when $x \in L(S \setminus \{x\})$. (If x is a mix of the others, any mix using x can be rewritten without x.)

Step 2 – x_4 is needed. x_4 has third coordinate 1, but every mix of x_1, x_2, x_3 has third coordinate 0. So $x_4 \notin L(x_1, x_2, x_3)$, and removing x_4 shrinks the span.

Step 3 – x_1, x_2, x_3 can each be removed. $x_1 = 2x_2 - x_3$ (check: $(2,2,0,0) - (1,2,0,0) = (1,0,0,0)$ ✓). $x_2 = \frac{1}{2}(x_1 + x_3)$ (check: $\frac{1}{2}(2,2,0,0) = (1,1,0,0)$ ✓). $x_3 = 2x_2 - x_1$ (check: $(2,2,0,0) - (1,0,0,0) = (1,2,0,0)$ ✓).

One-line summary: x_1, x_2, x_3 (each one separately); not x_4.

Question 1.9 (Exercise 3.1.18, item 6)

Let $W_1 = \{f \in C[-1,1] : f(0.2) = 0\}$ and $W_2 = \{f \in C[-1,1] : f(\frac{1}{4}) \text{ exists}\}$. Are W_1, W_2 subspaces of $C[-1,1]$?

Solution.

Step 1 – W_1 is a subspace. Same argument as Question 1.1(c) with the point 0.2: the zero function is in W_1; if $f(0.2) = g(0.2) = 0$ then $(af + bg)(0.2) = 0$.

Step 2 – W_2 is everything. Every $f \in C[-1,1]$ is a function defined at *every* point of $[-1,1]$, and $\frac{1}{4} \in [-1,1]$. So $f(\frac{1}{4})$ always exists: $W_2 = C[-1,1]$, which is a (trivial) subspace of itself.

One-line summary: Both are subspaces (W_2 is the whole space).

Question 1.10 (Exercise 3.1.18, item 7)

Let $V = \{(x,y) : x, y \in \mathbb{R}\}$ over $\mathbb{R}$ with $(x,y) \oplus (x_1,y_1) = (x + x_1, 0)$ and $\alpha \odot (x,y) = (\alpha x, 0)$. Show V is not a vector space.

Solution. It is enough to show one rule fails.

Step 1 – test rule 2b: $1 \odot u = u$. Take $u = (2,5)$. $1 \odot (2,5) = (1 \cdot 2,\ 0) = (2,0) \neq (2,5)$. Rule 2b fails.

Step 2 – (also) rule 1c fails. A zero vector $e = (e_1, e_2)$ would need $(2,5) \oplus e = (2,5)$. But $(2,5) \oplus e = (2 + e_1,\ 0)$, whose second coordinate is $0 \neq 5$. No such e exists.

One-line summary: The second coordinate is always killed to 0, so rules 2b and 1c fail.

Question 1.11 (Exercise 3.1.18, item 8)

Prove that (a) $sl_n = \{A \in M_n(\mathbb{R}) : \text{trace}(A) = 0\}$, (b) $\text{Sym}_n = \{A : A = A^t\}$, (c) $\text{Skew}_n = \{A : A + A^t = 0\}$ are subspaces of $M_n(\mathbb{R})$.

Solution. In each part $\mathbf{0}$ (the zero matrix) is in the set, and we take A, B in the set and $a, b \in \mathbb{R}$.

Step 1 – (a) trace zero. $\text{trace}(aA + bB) = \sum_i (aA_{ii} + bB_{ii}) = a\sum_i A_{ii} + b\sum_i B_{ii} = a \cdot 0 + b \cdot 0 = 0$. ✓ (Example: $\begin{pmatrix} 1 & 5 \\ 3 & -1 \end{pmatrix}$ has trace 0.)

Step 2 – (b) symmetric. Use $(X+Y)^t = X^t + Y^t$ and $(cX)^t = cX^t$: $(aA+bB)^t = aA^t + bB^t = aA + bB$. So $aA + bB$ is symmetric. ✓ (Example: $\begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}$.)

Step 3 – (c) skew-symmetric. $(aA + bB) + (aA + bB)^t = a(A + A^t) + b(B + B^t) = a\mathbf{0} + b\mathbf{0} = \mathbf{0}$. ✓

(Example: $\begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix}$.)

One-line summary: All three are closed under mixing and contain the zero matrix.

Question 1.12 (Exercise 3.1.18, item 9)

Let $V = \mathbb{R}$. Define $x \oplus y = x - y$ and $\alpha \odot x = -\alpha x$. Which vector space axioms are not satisfied?

Solution. Test each rule with numbers.

Step 1 – 1a (commutative) fails. $3 \oplus 1 = 2$ but $1 \oplus 3 = -2$.

Step 2 – 1b (associative) fails. $(1 \oplus 1) \oplus 1 = 0 \oplus 1 = -1$ but $1 \oplus (1 \oplus 1) = 1 \oplus 0 = 1$.

Step 3 – 1c (zero) holds. $x \oplus 0 = x - 0 = x$; the zero vector is 0.

Step 4 – 1d (opposite) holds. $x \oplus x = x - x = 0$: each x is its own opposite.

Step 5 – 2a fails. $1 \odot (1 \odot 1) = 1 \odot (-1) = 1$, but $(1 \cdot 1) \odot 1 = -1$.

Step 6 – 2b fails. $1 \odot x = -x \neq x$ (e.g. $x = 1$).

Step 7 – 3a holds. $\alpha \odot (x \oplus y) = -\alpha(x - y) = -\alpha x + \alpha y$ and $(\alpha \odot x) \oplus (\alpha \odot y) = -\alpha x - (-\alpha y) = -\alpha x + \alpha y$. Equal.

Step 8 – 3b fails. $(1 + 1) \odot 1 = -2$ but $(1 \odot 1) \oplus (1 \odot 1) = (-1) \oplus (-1) = -1 - (-1) = 0$.

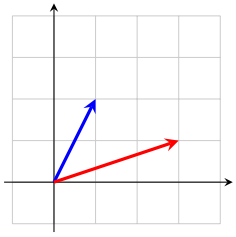
One-line summary: Fail: 1a, 1b, 2a, 2b, 3b. Hold: 1c, 1d, 3a.

3.2 Linear Independence

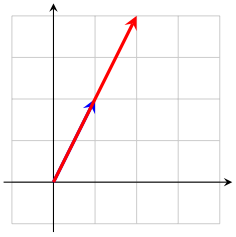
3.2.1 Is a list of vectors wasteful?

Picture it

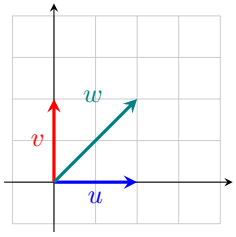
Suppose you want to describe every point of the plane using a few direction vectors. Take $u = (1, 0)$, $v = (0, 1)$ and $w = (1, 1)$. The third one is *wasteful*: anything w can do, u and v already do, because $w = u + v$. We say $\{u, v, w\}$ is **linearly dependent**. But $\{u, v\}$ has no waste: neither is a mix of the other. We say $\{u, v\}$ is **linearly independent**.
So: *independent = every vector brings a genuinely new direction; dependent = some vector is a mix of the others.*



independent: $(1, 2), (3, 1)$



dependent: $(1, 2), (2, 4)$ parallel



dependent: $w = u + v$

Definition – Linear independence and dependence (Definition 3.2.1)

Let $S = \{u_1, \ldots, u_m\}$ be a non-empty subset of V. If there are scalars $\alpha_1, \ldots, \alpha_m$, **not all zero**, with

$$\alpha_1 u_1 + \alpha_2 u_2 + \cdots + \alpha_m u_m = \mathbf{0}, \tag{3.2.1}$$

then S is **linearly dependent**. Otherwise S is **linearly independent**, i.e. equation (3.2.1) has *only* the solution $\alpha_1 = \cdots = \alpha_m = 0$.

Plain English – break every phrase down

- **Equation (3.2.1) always has the "boring" solution** $\alpha_1 = \cdots = \alpha_m = 0$ (that gives $0u_1 + \cdots + 0u_m = \mathbf{0}$). It is called the **trivial solution**. So the *only* question is: *does (3.2.1) have any other solution?*
- **"Not all zero"** means *at least one* $\alpha_i \neq 0$. Some of the α's may be 0. Dependent does *not* mean all coefficients are non-zero.
- **Why this matches "wasteful":** if $\alpha_i \neq 0$, solve (3.2.1) for u_i: $u_i = -\frac{1}{\alpha_i} \sum_{j \neq i} \alpha_j u_j$, a mix of the others. Conversely, if $u_i = \sum_{j \neq i} c_j u_j$ then $1 \cdot u_i - \sum_{j \neq i} c_j u_j = \mathbf{0}$ is a relation with a non-zero coefficient (1 on u_i).
- **Independent:** the only way to mix the vectors to get $\mathbf{0}$ is to take none of them.

Recipe 3 – test independence (the method you will use 90% of the time)

1. Write $\alpha_1 v_1 + \cdots + \alpha_k v_k = \mathbf{0}$. This is a *homogeneous* system whose coefficient matrix has $v_1, \ldots, v_k$ as its **columns**.
2. Row-reduce that matrix.
3. **Every column is a pivot column** (rank $= k$) $\Rightarrow$ only the trivial solution $\Rightarrow$ **independent**.
   Some column is free (rank $< k$) $\Rightarrow$ a non-trivial solution exists $\Rightarrow$ **dependent**. Setting the free variable to 1 gives the coefficients of an explicit relation.

Shortcuts. (a) k vectors in $\mathbb{F}^n$ with $k > n$: always dependent (more unknowns than equations, Key Fact K2). (b) n vectors in $\mathbb{F}^n$: independent $\iff$ $\det \neq 0$ (Key Fact K1). (c) Two vectors: independent $\iff$ neither is a multiple of the other. (d) A set containing $\mathbf{0}$, or two equal vectors: dependent.

Example 3.2.2 of your notes – solved step by step

(1) Show $S = \{(1,2,1),(2,1,4),(3,3,5)\} \subset \mathbb{R}^3$ is dependent. **(2)** Show $S = \{(1,1,1),(1,1,0),(1,0,1)\}$ is independent.

Solution (1).

Step 1 – set up. $\alpha_1(1,2,1)+\alpha_2(2,1,4)+\alpha_3(3,3,5)=(0,0,0)$. The coefficient matrix has these three vectors as columns: $\begin{pmatrix}1&2&3\\2&1&3\\1&4&5\end{pmatrix}$.

Step 2 – row-reduce. $R_2 \to R_2 - 2R_1$, $R_3 \to R_3 - R_1$ give $\begin{pmatrix}1&2&3\\0&-3&-3\\0&2&2\end{pmatrix}$; scale and clear: $\begin{pmatrix}1&0&1\\0&1&1\\0&0&0\end{pmatrix}$.

Step 3 – read. Column 3 is free (no pivot), so there is a non-trivial solution. Take $\alpha_3 = t$: $\alpha_1 = -t$, $\alpha_2 = -t$.

Step 4 – pick one. $t=-1$ gives $(\alpha_1,\alpha_2,\alpha_3)=(1,1,-1)$. **Check:** $1(1,2,1)+1(2,1,4)-1(3,3,5) = (1+2-3,\ 2+1-3,\ 1+4-5)=(0,0,0)$ ✓. Not all scalars are zero, so S is **dependent**. (This is the relation printed in the notes; here we *found* it by solving, not by guessing.)

Solution (2).

Step 1 – set up. $\alpha(1,1,1)+\beta(1,1,0)+\gamma(1,0,1)=(0,0,0)$ gives three equations (one per coordinate): $\alpha+\beta+\gamma=0$ (first coordinates), $\alpha+\beta=0$ (second), $\alpha+\gamma=0$ (third).

Step 2 – subtract. (first) − (second): $\gamma=0$. (first) − (third): $\beta=0$. Then $\alpha=0$ from the second equation.

Step 3 – conclude. Only the trivial solution, so S is **independent**. (Cross-check with the determinant: $\det\begin{pmatrix}1&1&1\\1&1&0\\1&0&1\end{pmatrix} = 1(1)-1(1)+1(-1) = -1 \neq 0$ ✓.)

Extra Example 3.2-A – more tests, with all the shortcuts

(1) Two vectors. $\{(1,2),(2,4)\}$: $2(1,2)-1(2,4)=\mathbf{0}$, so dependent (parallel). $\{(1,2),(3,1)\}$: $\det\begin{pmatrix}1&3\\2&1\end{pmatrix} = 1-6=-5\neq 0$, so independent.

(2) Three vectors in $\mathbb{R}^3$ with a hidden relation. $\{(1,2,3),(4,5,6),(7,8,9)\}$. $\det\begin{pmatrix}1&4&7\\2&5&8\\3&6&9\end{pmatrix}=0$, so dependent. The relation: $1(1,2,3)-2(4,5,6)+1(7,8,9)=(1-8+7,\ 2-10+8,\ 3-12+9)=(0,0,0)$ ✓.

(3) Too many vectors. $\{(1,0,0),(0,1,0),(0,0,1),(1,2,3)\}$: 4 vectors in $\mathbb{R}^3$, so dependent by shortcut (a). The relation is visible: $(1,2,3)=1e_1+2e_2+3e_3$, i.e. $1e_1+2e_2+3e_3-1\cdot(1,2,3)=\mathbf{0}$.

(4) Three vectors in $\mathbb{R}^4$. $v_1 = (1,0,2,1)$, $v_2 = (0,1,1,2)$, $v_3 = (1,1,0,0)$. Row reduction of the 4×3 matrix $[v_1\, v_2\, v_3]$ has 3 pivots (rank 3), so independent. (Here a determinant is unavailable, since the matrix is not square. Use rank.)
(5) Polynomials. $\{1+x,\ x+x^2,\ 1+x^2\} \subset P_2(\mathbb{R})$. Write each as a coefficient vector in the basis $1, x, x^2$: $(1,1,0), (0,1,1), (1,0,1)$. $\det \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} = 2 \neq 0$. Independent.
By contrast $\{1+x, 1-x, 1\} \subset P_1(\mathbb{R})$ is dependent: $1 = \frac{1}{2}(1+x) + \frac{1}{2}(1-x)$ (three vectors in a space with 2 coordinates).
(6) Functions. $\{\sin x, \cos x\}$ in $C(\mathbb{R})$: if $a \sin x + b \cos x = 0$ for *all* x, put $x = 0$: $b = 0$; put $x = \frac{\pi}{2}$: $a = 0$. Independent.
(7) The field matters. In $\mathbb{C}(\mathbb{C})$, $\{1, i\}$ is dependent: $i \cdot 1 + (-1) \cdot i = 0$ with scalars $i, -1 \neq 0$. In $\mathbb{C}(\mathbb{R})$ it is independent: $a \cdot 1 + b \cdot i = 0$ with a, b real forces $a = b = 0$. **Same vectors, different answer.**

3.2.2 Basic rules about independent sets (Proposition 3.2.3)

Proposition 3.2.3

Let V be a vector space. (1) The zero vector cannot belong to a linearly independent set. (2) Every subset of a linearly independent set is linearly independent. (3) Every set containing a linearly dependent set is linearly dependent.

Plain English

(1) **0** is always a "wasted" vector. (2) If a list has no waste, deleting vectors keeps it free of waste. (3) If a list already has waste, adding more vectors cannot cure it.

Proof in small steps – All three parts (the notes proved only part 1)

(1) Let $S = \{\mathbf{0} = u_1, u_2, \ldots, u_n\}$. For any $\gamma \neq 0$: $\gamma u_1 + 0u_2 + \cdots + 0u_n = \gamma \mathbf{0} = \mathbf{0}$. This is a relation with $\gamma \neq 0$, so S is dependent.
(2) Let S be independent and $T \subseteq S$, $T = \{t_1, \ldots, t_k\}$. Suppose $\sum_i c_i t_i = \mathbf{0}$. Give the vectors of S that are *not* in T the coefficient 0: this is a relation among the vectors of S. Since S is independent, all coefficients are 0, in particular all $c_i = 0$. So T is independent.
(3) Let S be dependent, with a relation $\sum \alpha_i u_i = \mathbf{0}$, not all $\alpha_i = 0$, and let $T \supseteq S$. Use the same coefficients on the vectors of S and coefficient 0 on the extra vectors of T. It is a relation on T with some non-zero coefficient. So T is dependent. ■

3.2.3 Growing a list: Theorem 3.2.4 and its corollaries

Theorem 3.2.4, Corollary 3.2.5, Corollary 3.2.6

3.2.4. Let $\{v_1, \ldots, v_p\}$ be independent and suppose $\{v_1, \ldots, v_p, v_{p+1}\}$ is dependent. Then $v_{p+1} \in L(v_1, \ldots, v_p)$.
3.2.5. If $\{u_1, \ldots, u_n\}$ is dependent, there is a smallest k $(2 \leq k \leq n)$ with $L(u_1, \ldots, u_k) = L(u_1, \ldots, u_{k-1})$.
3.2.6. If $\{v_1, \ldots, v_p\}$ is independent and $v \notin L(v_1, \ldots, v_p)$, then $\{v_1, \ldots, v_p, v\}$ is independent.
(In some printings of your PDF the slash in "$\notin$" in 3.2.6 is missing; the correct statement has $\notin$.)

Plain English

3.2.4: If a list was clean and adding one new vector spoils it, the new vector must be a mix of the old ones. **3.2.6** is the same fact seen from the other side and is the main tool to *grow* an independent list: *add a vector that is outside the span and the list stays independent.* **3.2.5:** in a dependent list, scanning left to right, the first vector that is a mix of the earlier ones can be thrown away without shrinking the span.

Proof in small steps – Theorem 3.2.4 (with the numbers $v_1 = (1,0)$, $v_2 = (0,1)$, $v_3 = (3,4)$ alongside)

Step 1 – use dependence of the longer list. There are $\alpha_1, \dots, \alpha_{p+1}$, not all zero, with $\alpha_1 v_1 + \cdots + \alpha_p v_p + \alpha_{p+1} v_{p+1} = \mathbf{0}$ (3.2.2). *Example:* $3v_1 + 4v_2 - 1 \cdot v_3 = \mathbf{0}$, so $(\alpha_1, \alpha_2, \alpha_3) = (3, 4, -1)$.

Step 2 – claim $\alpha_{p+1} \neq 0$. Suppose $\alpha_{p+1} = 0$. Then (3.2.2) becomes $\alpha_1 v_1 + \cdots + \alpha_p v_p = \mathbf{0}$ with not all of $\alpha_1, \dots, \alpha_p$ zero (they are not all zero, and α_{p+1} is). That says $\{v_1, \dots, v_p\}$ is dependent, contradicting the hypothesis. So $\alpha_{p+1} \neq 0$. *Example:* $\alpha_3 = -1 \neq 0$.

Step 3 – solve for v_{p+1}. Divide by α_{p+1}: $v_{p+1} = -\frac{\alpha_1}{\alpha_{p+1}} v_1 - \cdots - \frac{\alpha_p}{\alpha_{p+1}} v_p$. The quotients lie in $\mathbb{F}$ because $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$. *Example:* $v_3 = -\frac{3}{-1} v_1 - \frac{4}{-1} v_2 = 3v_1 + 4v_2 = (3,4)$. ✓ ■

Proof in small steps – Corollaries 3.2.5 and 3.2.6

3.2.6. It is the *contrapositive* of 3.2.4: if $\{v_1, \dots, v_p, v\}$ were dependent, then $v \in L(v_1, \dots, v_p)$. But $v \notin L$, so it is independent. ■

3.2.5. Add $u_1, u_2, \dots$ one at a time. A single vector $u_1 \neq \mathbf{0}$ is independent. At some first stage k the list $u_1, \dots, u_k$ becomes dependent (it must, since the full list is dependent). The list $u_1, \dots, u_{k-1}$ is independent, so by 3.2.4 $u_k \in L(u_1, \dots, u_{k-1})$, hence adding it does not enlarge the span: $L(u_1, \dots, u_k) = L(u_1, \dots, u_{k-1})$. ■

Extra Example 3.2-B – growing an independent list with Corollary 3.2.6

Start with $\{(1,0,0)\}$ (independent: it is non-zero). $(0,1,0) \notin L\{(1,0,0)\}$ (every member has second coordinate 0), so $\{(1,0,0),(0,1,0)\}$ is independent. $(0,0,1) \notin L\{(1,0,0),(0,1,0)\}$ (every member has third coordinate 0), so $\{e_1, e_2, e_3\}$ is independent. We have rebuilt the standard basis of $\mathbb{R}^3$ by repeatedly stepping *outside* the span.
Wrong use: adding $(2,3,0)$, which *is* in $L\{e_1, e_2\}$, would give a dependent list $(2e_1 + 3e_2 - (2,3,0) = \mathbf{0})$.

3.2.4 Questions for Section 3.2 (Exercise 3.2.7 of your notes)

Question 2.1 (Exercise 3.2.7, item 1)

Consider $\mathbb{R}^2$ and let $u_1 = (1,0)$. Find all choices of u_2 such that $\{u_1, u_2\}$ is independent. Do there exist u_2, u_3 with $\{u_1, u_2, u_3\}$ independent in $\mathbb{R}^2$?
Solution.

Step 1 – write $u_2 = (a,b)$ and set up. $c_1(1,0) + c_2(a,b) = (0,0)$ gives $c_1 + c_2 a = 0$ and $c_2 b = 0$.

Step 2 – case $b \neq 0$. Then $c_2 = 0$, then $c_1 = 0$. Only the trivial solution: independent.

Step 3 – case $b = 0$. Then $u_2 = (a,0) = a\,u_1$ is a multiple of u_1: e.g. $a \cdot u_1 - u_2 = \mathbf{0}$ (coefficient $-1 \neq 0$). Dependent.

Step 4 – answer for two vectors. $u_2 = (a,b)$ with $b \neq 0$ (any a). Equivalently $\det \begin{pmatrix} 1 & a \\ 0 & b \end{pmatrix} = b \neq 0$.

Step 5 – three vectors: no. $c_1u_1+c_2u_2+c_3u_3=\mathbf{0}$ is 2 equations in 3 unknowns, so it has a non-zero solution (Key Fact K2). *Example:* $u_2=(0,1)$, $u_3=(5,7)$: $5u_1+7u_2-u_3=\mathbf{0}$.

One-line summary: $u_2=(a,b)$ with $b\neq 0$; no u_3 can be added.

Question 2.2 (Exercise 3.2.7, item 2)

If none of the elements on the principal diagonal of a lower triangular matrix is zero, show that its row vectors are independent in $\mathbb{R}^n$. The same is true for column vectors.

Solution. Example to keep in mind: $L=\begin{pmatrix}1&0&0\\2&3&0\\4&5&6\end{pmatrix}$.

Step 1 – rows. Let $r_i=(l_{i1},\dots,l_{ii},0,\dots,0)$ and suppose $c_1r_1+\cdots+c_nr_n=\mathbf{0}$.

Step 2 – last coordinate. Only r_n can have a non-zero last entry (rows above have 0 there). So $c_nl_{nn}=0$, and $l_{nn}\neq 0$ gives $c_n=0$.

Step 3 – second-last coordinate. Now only r_{n-1} contributes ($c_n=0$): $c_{n-1}l_{n-1,n-1}=0\Rightarrow c_{n-1}=0$. Continue upward until $c_1=0$. So the rows are independent. *(In the example: $c_3\cdot 6=0$, then $c_2\cdot 3=0$, then $c_1\cdot 1=0$.)*

Step 4 – columns. The columns of L are the rows of L^t, which is upper triangular with the same non-zero diagonal. Suppose $d_1C_1+\cdots+d_nC_n=\mathbf{0}$. In the *first* coordinate only C_1 is non-zero (columns C_j, $j>1$, have 0 in row 1), so $d_1l_{11}=0\Rightarrow d_1=0$. Then the second coordinate gives $d_2=0$, and so on downward.

Step 5 – shortcut. $\det L=l_{11}\cdots l_{nn}\neq 0$ (e.g. $1\cdot 3\cdot 6=18$), so by Key Fact K1 both rows and columns are independent.

One-line summary: Back-substitution from the corner shows only the trivial relation exists.

Question 2.3 (Exercise 3.2.7, item 3)

Let $S=\{(1,1,1,1),(1,-1,1,2),(1,1,-1,1)\}\subset\mathbb{R}^4$. Determine whether $(1,1,2,1)\in L(S)$.

Solution.

Step 1 – set up. Look for α,β,γ with $\alpha(1,1,1,1)+\beta(1,-1,1,2)+\gamma(1,1,-1,1)=(1,1,2,1)$:
$\alpha+\beta+\gamma=1,\quad \alpha-\beta+\gamma=1,\quad \alpha+\beta-\gamma=2,\quad \alpha+2\beta+\gamma=1.$

Step 2 – eliminate. Equation 1 minus equation 2: $2\beta=0$, so $\beta=0$. Then equations 1 and 3 become $\alpha+\gamma=1$ and $\alpha-\gamma=2$: add them, $2\alpha=3$, $\alpha=\frac{3}{2}$; then $\gamma=1-\frac{3}{2}=-\frac{1}{2}$.

Step 3 – check the unused (fourth) equation. $\alpha+2\beta+\gamma=\frac{3}{2}+0-\frac{1}{2}=1$ ✓. The system is consistent.

Step 4 – verify. $\frac{3}{2}(1,1,1,1)-\frac{1}{2}(1,1,-1,1)=(\frac{3}{2}-\frac{1}{2},\ \frac{3}{2}-\frac{1}{2},\ \frac{3}{2}+\frac{1}{2},\ \frac{3}{2}-\frac{1}{2})=(1,1,2,1)$ ✓.

One-line summary: Yes: $(1,1,2,1)=\frac{3}{2}v_1+0\,v_2-\frac{1}{2}v_3$.

Question 2.4 (Exercise 3.2.7, item 4)

Show that $S=\{(1,2,3),(-2,1,1),(8,6,10)\}$ is linearly dependent in $\mathbb{R}^3$.

Solution.

Step 1 – determinant test. $\det\begin{pmatrix}1&-2&8\\2&1&6\\3&1&10\end{pmatrix}=1(10-6)+2(20-18)+8(2-3)=4+4-8=0.$

So dependent (Key Fact K1).

Step 2 – find the relation (do not guess). Solve $a(1,2,3)+b(-2,1,1)+c(8,6,10)=\mathbf{0}$. Row-

reducing the matrix above gives $\begin{pmatrix} 1 & 0 & 4 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{pmatrix}$. Columns 1, 2 are pivot columns; column 3 is free, so put $c = t$. The rows say $a + 4t = 0$ and $b - 2t = 0$, i.e. $(a, b, c) = t(-4, 2, 1)$.

Step 3 – pick $t = -1$. $(a, b, c) = (4, -2, -1)$. **Check:** $4(1, 2, 3) - 2(-2, 1, 1) - 1(8, 6, 10) = (4 + 4 - 8,\ 8 - 2 - 6,\ 12 - 2 - 10) = (0, 0, 0)$ ✓. Not all scalars are zero, so S is dependent.

One-line summary: $\det = 0$, *and* $(8, 6, 10) = 4(1, 2, 3) - 2(-2, 1, 1)$.

Question 2.5 (Exercise 3.2.7, item 5)

Show that $S = \{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$ is independent in $\mathbb{R}^3$. In general, if $\{f_1, f_2, f_3\}$ is independent, then so is $\{f_1,\ f_1 + f_2,\ f_1 + f_2 + f_3\}$.

Solution.

Step 1 – the numerical case. The matrix with these columns is upper triangular with diagonal $1, 1, 1$, so $\det = 1 \neq 0$. Independent.

Step 2 – the general case: assume a relation. Suppose $af_1 + b(f_1 + f_2) + c(f_1 + f_2 + f_3) = \mathbf{0}$.

Step 3 – collect like terms. $(a + b + c)f_1 + (b + c)f_2 + c\,f_3 = \mathbf{0}$.

Step 4 – use independence of f_1, f_2, f_3. All three coefficients must be 0: $a + b + c = 0$, $b + c = 0$, $c = 0$.

Step 5 – back-substitute. $c = 0 \Rightarrow b = 0 \Rightarrow a = 0$. Only the trivial relation, so $\{f_1, f_1 + f_2, f_1 + f_2 + f_3\}$ is independent. (The numerical case is $f_1 = e_1$, $f_2 = e_2$, $f_3 = e_3$.)

One-line summary: Collect terms on f_1, f_2, f_3 and use their independence.

Question 2.6 (Exercise 3.2.7, item 6)

In $\mathbb{R}^3$ give an example of three vectors u, v, w such that $\{u, v, w\}$ is dependent but any two of them form an independent set.

Solution.

Step 1 – choose. $u = (1, 0, 0)$, $v = (0, 1, 0)$, $w = (1, 1, 0)$.

Step 2 – the three are dependent. $w = u + v$, i.e. $u + v - w = \mathbf{0}$.

Step 3 – each pair is independent. $\{u, v\}$: neither is a multiple of the other. $\{u, w\}$: $w = (1, 1, 0)$ is not a multiple of $(1, 0, 0)$ (second coordinate). $\{v, w\}$: w is not a multiple of $(0, 1, 0)$ (first coordinate). Two non-zero vectors are independent exactly when neither is a multiple of the other.

One-line summary: Three vectors lying in one plane, no two of them parallel.

Question 2.7 (Exercise 3.2.7, item 7)

What is the maximum number of linearly independent vectors in $\mathbb{R}^3$?

Solution.

Step 1 – 3 is possible. e_1, e_2, e_3 are independent (Extra Example 3.2-B).

Step 2 – 4 is impossible. Any 4 vectors in $\mathbb{R}^3$ are dependent (Question 2.8: 3 equations, 4 unknowns).

One-line summary: The maximum is 3 *(this number is* $\dim \mathbb{R}^3$*, Section 3.3).*

Question 2.8 (Exercise 3.2.7, item 8)

Show that any set of k vectors in $\mathbb{R}^3$ is dependent if $k \geq 4$.

Solution.

Step 1 – set up. Let $v_1, \dots, v_k \in \mathbb{R}^3$. Consider $\alpha_1 v_1 + \cdots + \alpha_k v_k = \mathbf{0}$.

Step 2 – count. Coordinate by coordinate this is 3 homogeneous equations in the k unknowns $\alpha_1, \dots, \alpha_k$, with $k \geq 4 > 3$.

Step 3 – use Key Fact K2. A homogeneous system with more unknowns than equations has a non-zero solution (at least one free variable). So some $\alpha_i \neq 0$ and the set is dependent. *Example:* $(1,0,0), (0,1,0), (0,0,1), (2,3,4)$ has the relation $2e_1 + 3e_2 + 4e_3 - (2,3,4) = \mathbf{0}$.

One-line summary: More unknowns than equations forces a non-trivial solution.

Question 2.9 (Exercise 3.2.7, item 9)

Is $\{(1,0),(i,0)\}$ a linearly independent subset of $\mathbb{C}^2(\mathbb{R})$?
Solution.

Step 1 – remember the scalars. In $\mathbb{C}^2(\mathbb{R})$ the scalars a, b are *real*.

Step 2 – set up. $a(1,0) + b(i,0) = (0,0)$ gives the first coordinate $a + bi = 0$ (the second coordinate is $0 = 0$).

Step 3 – real and imaginary parts. Since a, b are real, $a + bi = 0$ forces $a = 0$ and $b = 0$.

Step 4 – conclude. Only the trivial solution: **independent**. (Over $\mathbb{C}(\mathbb{C})$ it would be *dependent*: $i(1,0) - 1 \cdot (i,0) = \mathbf{0}$ with scalars $i, -1$.)

One-line summary: Yes, independent over $\mathbb{R}$.

Question 2.10 (Exercise 3.2.7, item 10)

Under what conditions on α are the vectors $(1+\alpha,\ 1-\alpha)$ and $(\alpha-1,\ 1+\alpha)$ in $\mathbb{C}^2(\mathbb{R})$ independent?
Solution. The scalars used in the relation are real: $a, b \in \mathbb{R}$. (Take $\alpha \in \mathbb{R}$; a remark at the end covers complex α.)

Step 1 – set up. $a(1+\alpha, 1-\alpha) + b(\alpha-1, 1+\alpha) = (0,0)$:
$a(1+\alpha) + b(\alpha-1) = 0$ (E1), $a(1-\alpha) + b(1+\alpha) = 0$ (E2).

Step 2 – add and subtract. (E1)+(E2): $2a + 2b\alpha = 0$, i.e. $a = -b\alpha$. (E1)−(E2): $2a\alpha - 2b = 0$, i.e. $b = a\alpha$.

Step 3 – combine. Put $b = a\alpha$ into $a = -b\alpha$: $a = -a\alpha^2$, so $a(1+\alpha^2) = 0$.

Step 4 – real α. $1 + \alpha^2 \geq 1 > 0$, so $a = 0$; then $b = a\alpha = 0$. Only the trivial relation: **independent for every real α.**

Step 5 – cross-check. The determinant is $\det \begin{pmatrix} 1+\alpha & \alpha-1 \\ 1-\alpha & 1+\alpha \end{pmatrix} = (1+\alpha)^2 + (\alpha-1)^2 = 2(1+\alpha^2) \neq 0$.
✓ (E.g. $\alpha = 2$: $(3,-1), (1,3)$, $\det = 10 = 2 \cdot 5$.)

Step 6 – remark for complex α. If α is allowed to be complex, $a(1+\alpha^2) = 0$ with $a \neq 0$ forces $\alpha = \pm i$, and then $b = a\alpha = \pm ia$ is *not real*. So even then the only real relation is trivial: still independent. (Over $\mathbb{C}$ scalars they would be dependent exactly at $\alpha = \pm i$.)

One-line summary: Independent for every real α ($\det = 2(1+\alpha^2) \neq 0$).

Question 2.11 (Exercise 3.2.7, item 11)

Let $u, v \in V$ and M a subspace of V. Let K be the subspace spanned by M and u, and H the subspace spanned by M and v. Show that if $v \in K$ and $v \notin M$, then $u \in H$.
Solution. Example: $V = \mathbb{R}^2$, $M = L\{(1,0)\}$, $u = (0,1)$, $v = (1,1)$.

Step 1 – describe K. A mix of vectors from M and u is (a mix of vectors of M) $+ a\,u$, and a mix of vectors of M lies in M. So $K = \{m + au : m \in M,\ a \in \mathbb{F}\}$.

Step 2 – use $v \in K$. $v = m + au$ for some $m \in M$, $a \in \mathbb{F}$. *(Example:* $(1,1) = (1,0) + 1 \cdot (0,1)$*.)*

Step 3 – $a \neq 0$. If $a = 0$ then $v = m \in M$, contradicting $v \notin M$.

Step 4 – solve for u. $u = \frac{1}{a}v - \frac{1}{a}m$. This is a mix of v and $m \in M$, so $u \in H$. *(Example:* $(0,1) = 1 \cdot (1,1) - 1 \cdot (1,0)$*.)* ✓

One-line summary: Since v really uses u ($a \neq 0$), u can be recovered from v and M.

3.3 Bases and Dimension

3.3.1 What is a basis? (the "just right" list)

Picture it

We now combine the two ideas of Sections 3.1 and 3.2. Think of a list of vectors as a set of "directions" you may travel along.

- **Too few** directions: you cannot reach everything (the list does not *span*).
- **Too many** directions: some are wasteful (the list is *dependent*).
- **Just right:** you reach every point (spans), and nothing is wasted (independent). That is a **basis**.

List in $\mathbb{R}^2$	Spans $\mathbb{R}^2$?	Independent?	Basis?
$\{(1,0)\}$	no (only the x-axis)	yes	no (too few)
$\{(1,0),(0,1)\}$	yes	yes	**yes** (standard basis)
$\{(1,0),(1,1)\}$	yes	yes	**yes** (another basis!)
$\{(1,0),(0,1),(1,1)\}$	yes	no ($(1,1) = (1,0)+(0,1)$)	no (too many)
$\{(1,2),(2,4)\}$	no (only a line)	no	no

Definition – Basis (Definition 3.3.1)

A non-empty subset B of a vector space V is a **basis** of V if
(a) B is linearly independent, and
(b) $L(B) = V$, i.e. every vector of V is a linear combination of the elements of B.
A vector in B is called a **basis vector**.

Plain English

Basis = **independent** (no waste) + **spanning** (reaches everything). To prove that a set B is a basis you must prove *both* halves. A vector space has *many* different bases (see the table), but all of them have the same size (next pages).

Remark 3.3.2 – unique representation

If $\{v_1,\dots,v_p\}$ is a basis of $V(\mathbb{F})$, then every $v \in V$ is a *unique* linear combination $v = \alpha_1 v_1 + \cdots + \alpha_p v_p$. Conversely, if every vector has exactly one such expression, the set is a basis. By convention $L(\varnothing) = \{\mathbf{0}\}$, so $\varnothing$ is a basis of the zero space $\{\mathbf{0}\}$.

Proof in small steps – Why the representation is unique

Step 1 – suppose two representations. $v = \alpha_1 v_1 + \cdots + \alpha_p v_p$ and $v = \beta_1 v_1 + \cdots + \beta_p v_p$.

Step 2 – subtract. $\mathbf{0} = v - v = (\alpha_1 - \beta_1)v_1 + \cdots + (\alpha_p - \beta_p)v_p$.

Step 3 – use independence. The v_i are independent, so every coefficient is 0: $\alpha_i - \beta_i = 0$, i.e. $\alpha_i = \beta_i$. ■

Example. Basis $\{(1,0),(1,1)\}$ of $\mathbb{R}^2$ and $v = (3,2)$: $v = a(1,0) + b(1,1)$ gives $b = 2$, then $a = 1$. So $v = 1\cdot(1,0) + 2\cdot(1,1)$, and no other pair (a,b) works. (Check: $(1+2,\ 0+2) = (3,2)$ ✓.)

Why uniqueness matters

Because the representation is unique, the list of coefficients $(\alpha_1, \ldots, \alpha_p)$ is a perfect "address" of v. This is exactly the idea of *coordinates* in Section 3.4. And it gives a quick **test that something is NOT a basis**: if one vector can be written in two different ways, the set is not a basis (Example 3.3.3.3 below).

Example 3.3.3 of your notes – items 1 to 6, solved in detail

Item 1. $V = \{(x, y, 0) : x, y \in \mathbb{R}\} \subset \mathbb{R}^3$. Check that $B_1 = \{(1,0,0), (0,1,0)\}$, $B_2 = \{(1,0,0), (1,1,0)\}$, $B_3 = \{(2,0,0), (1,3,0)\}$ are all bases of V.
Method for each: solve $(x, y, 0) = a\, b_1 + c\, b_2$ for a, c (spans), and see the solution is forced (independent: it is unique, in particular for $(x, y) = (0, 0)$ it gives $a = c = 0$).

- B_1: $(x, y, 0) = x(1,0,0) + y(0,1,0)$, forced. ✓
- B_2: $a(1,0,0) + c(1,1,0) = (a + c,\, c,\, 0) = (x, y, 0) \Rightarrow c = y,\ a = x - y$. Forced. ✓ (E.g. $(5,2,0) = 3(1,0,0) + 2(1,1,0)$.)
- B_3: $a(2,0,0) + c(1,3,0) = (2a + c,\, 3c,\, 0) = (x, y, 0) \Rightarrow c = \frac{y}{3},\ a = \frac{1}{2}(x - \frac{y}{3})$. Forced. ✓

Three different bases of the same space.
Item 2 – the standard basis. $e_i = (0, \ldots, 0, 1, 0, \ldots, 0) \in \mathbb{R}^n$ with the 1 in place i. *Spans:* $(x_1, \ldots, x_n) = x_1 e_1 + \cdots + x_n e_n$. *Independent:* if $\alpha_1 e_1 + \cdots + \alpha_n e_n = \mathbf{0}$, the vector on the left is $(\alpha_1, \ldots, \alpha_n)$, so every $\alpha_i = 0$. For $n = 3$: $\{(1,0,0), (0,1,0), (0,0,1)\}$.
Item 3. $V = \{(x, y, z) : x + y - z = 0\}$ and $S = \{(1,1,2), (2,1,3), (1,2,3)\} \subset V$ (each satisfies $x + y - z = 0$: $1 + 1 - 2 = 0$, $2 + 1 - 3 = 0$, $1 + 2 - 3 = 0$). Show S is *not* a basis, and find a basis.

Step 1 – two representations of one vector. $(3,2,5) \in V$ ($3+2-5 = 0$). $(1,1,2)+(2,1,3) = (3,2,5)$ and $4(1,1,2) - (1,2,3) = (4-1,\, 4-2,\, 8-3) = (3,2,5)$. The coefficient lists $(1,1,0)$ and $(4,0,-1)$ differ.

Step 2 – conclude. By Remark 3.3.2, S is not a basis. (Indeed subtracting gives $-3(1,1,2)+(2,1,3)+(1,2,3) = \mathbf{0}$: dependent.)

Step 3 – find a real basis. $x + y - z = 0 \iff z = x + y$. So $(x, y, z) = (x, y, x + y) = x(1,0,1) + y(0,1,1)$. Thus $\{(1,0,1), (0,1,1)\}$ spans V. Independent: neither is a multiple of the other. **Basis of** V (so V has 2 vectors in a basis).
Item 4. $V = \{a + ib\}$ with $\mathbb{F} = \mathbb{C}$. Every $a + ib = (a + ib) \cdot 1$, so $\{1\}$ spans; $1 \neq 0$ so it is independent. Basis $\{1\}$.
Item 5. The *same* V with $\mathbb{F} = \mathbb{R}$. Every $a + ib = a \cdot 1 + b \cdot i$ with a, b real, so $\{1, i\}$ spans. Independent over $\mathbb{R}$: $a + bi = 0$ with a, b real forces $a = b = 0$. Basis $\{1, i\}$. (Notes: "$i \cdot (1 + 0i)$ is not defined" here, because $i \notin \mathbb{R}$ is not an allowed scalar.) **Same set, different scalars, different basis size.**
Item 6. $P(\mathbb{R})$ has the *infinite* basis $\{1, x, x^2, \ldots\}$. *Spans:* each polynomial is a *finite* sum $a_0 + a_1 x + \cdots + a_m x^m$. *Independent:* (for an infinite set this means every finite subset is independent) if $a_0 + a_1 x + \cdots + a_m x^m$ is the zero function, all $a_i = 0$, because a non-zero polynomial of degree $\leq m$ has at most m roots, yet this one vanishes at every x.

Definition – Finite dimensional (Definition 3.3.4)

V is **finite dimensional** if it has a basis with finitely many elements; otherwise it is **infinite dimensional**.

In Example 3.3.3 only $P(\mathbb{R})$ (item 6) is infinite dimensional; all the others are finite dimensional.

Remark 3.3.5 – how to *build* a basis by growing a list

Step 1. Pick any non-zero $v_1 \in V$. $\{v_1\}$ is independent.
Step 2. If $L(v_1) = V$, stop: $\{v_1\}$ is a basis. Otherwise pick $v_2 \in V$ with $v_2 \notin L(v_1)$. By Corollary 3.2.6, $\{v_1, v_2\}$ is independent.
Step i**.** If $L(v_1, \ldots, v_i) = V$ stop. Otherwise pick $v_{i+1} \notin L(v_1, \ldots, v_i)$; the list $\{v_1, \ldots, v_{i+1}\}$ is independent (Cor. 3.2.6).

The process stops because V is finite dimensional. **Idea in one line:** keep adding a vector from outside the current span, and stop when nothing is left outside.

Extra Example 3.3-A – Remark 3.3.5 in action on $V = \{x + y - z = 0\}$

Pick $v_1 = (1, 0, 1) \in V$ $(1 + 0 - 1 = 0)$. $L(v_1) \neq V$ because $(0, 1, 1) \in V$ $(0 + 1 - 1 = 0)$ but $(0, 1, 1)$ is not a multiple of $(1, 0, 1)$. So take $v_2 = (0, 1, 1)$; by Cor. 3.2.6 $\{v_1, v_2\}$ is independent. Is $L(v_1, v_2) = V$? Every $(x, y, z) \in V$ has $z = x + y$ and equals $xv_1 + yv_2$. Yes, so we stop: $\{(1, 0, 1), (0, 1, 1)\}$ is a basis. (Any vector $(x, y, z) \in V$ not in $L(v_1, v_2)$ would have been the next choice, but there is none.)

3.3.2 "Too many vectors" and dimension

Picture it

In the plane $\mathbb{R}^2$ you can have at most 2 independent directions: any third vector is a mix of two good ones. In $\mathbb{R}^3$ at most 3. This "maximum number of independent directions" is the **dimension**. The next theorem says it is the same number for every basis, which is what makes "dimension" a well-defined idea.

Theorem 3.3.7

Let $\{v_1, \ldots, v_n\}$ be a basis of V. If $\{w_1, \ldots, w_m\}$ is a set of vectors of V with $m > n$, then $\{w_1, \ldots, w_m\}$ is linearly dependent.

Plain English, with numbers first

In a space with a basis of n vectors, any list of *more than* n vectors has waste. *Example ($n = 2$, $m = 3$):* $w_1 = (1, 2)$, $w_2 = (3, 4)$, $w_3 = (5, 6)$ in $\mathbb{R}^2$. We claim a relation exists: $1 \cdot w_1 - 2 \cdot w_2 + 1 \cdot w_3 = (1 - 6 + 5,\ 2 - 8 + 6) = (0, 0)$. ✓ The proof shows *why* such a relation must exist.

Proof in small steps – Theorem 3.3.7 (running example: basis $v_1 = (1, 0)$, $v_2 = (0, 1)$ and the three w's above)

Step 1 – what we need. A solution $(\alpha_1, \ldots, \alpha_m) \neq \mathbf{0}$ of $\alpha_1 w_1 + \cdots + \alpha_m w_m = \mathbf{0}$ (3.3.1).

Step 2 – write each w_j in the basis. Since $\{v_i\}$ spans, $w_j = a_{1j}v_1 + a_{2j}v_2 + \cdots + a_{nj}v_n$ for $j = 1, \ldots, m$. *(Here $w_1 = 1v_1 + 2v_2$, $w_2 = 3v_1 + 4v_2$, $w_3 = 5v_1 + 6v_2$.)*

Step 3 – substitute into (3.3.1) and collect the v_i.

$$\alpha_1 w_1 + \cdots + \alpha_m w_m = \Big(\sum_{j=1}^{m} a_{1j}\alpha_j\Big)v_1 + \cdots + \Big(\sum_{j=1}^{m} a_{nj}\alpha_j\Big)v_n.$$

Step 4 – make every bracket zero. If we choose α so that $\sum_j a_{ij}\alpha_j = 0$ for every $i = 1, \ldots, n$, the right side is $\mathbf{0}$. That is the homogeneous system $A\alpha = \mathbf{0}$, where $A = (a_{ij})$ is $n \times m$. *(Example:* $A = \begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{pmatrix}$.*)*

Step 5 – count. $A\alpha = \mathbf{0}$ has n equations in $m > n$ unknowns, so it has a non-zero solution (Key Fact K2). *(Example: $\alpha = (1, -2, 1)$.)*

Step 6 – conclude. For that non-zero α, (3.3.1) holds. So $\{w_1, \ldots, w_m\}$ is dependent. ■

Corollary 3.3.10 and Definition 3.3.11

3.3.10. In a finite dimensional V, any two bases have the same number of vectors.
3.3.11. This common number is the **dimension** $\dim(V)$.

Proof in small steps – Corollary 3.3.10

Let $\{u_1, \ldots, u_n\}$ and $\{v_1, \ldots, v_m\}$ be bases and suppose $m > n$. Using the first as "the basis", Theorem 3.3.7 says $\{v_1, \ldots, v_m\}$ is dependent. But it is a basis, hence independent. Contradiction. Similarly $n > m$ is impossible. So $m = n$. ■

Space	A basis	Dimension
$\{\mathbf{0}\}$	$\varnothing$	0
$\mathbb{R}^n(\mathbb{R})$	$e_1, \ldots, e_n$	n
$\mathbb{C}^n(\mathbb{C})$	$e_1, \ldots, e_n$	n
$\mathbb{C}^n(\mathbb{R})$	$e_1, ie_1, \ldots, e_n, ie_n$	$2n$
$P_n(\mathbb{R})$	$1, x, \ldots, x^n$	$n+1$
$M_{m\times n}(\mathbb{R})$	the matrices E_{ij} (single 1)	mn
$P(\mathbb{R})$, $C[-1,1]$	infinite	infinite

Example 3.3.12 and Remark 3.3.13 of your notes – the field changes the dimension

(1) $\mathbb{C}^2(\mathbb{C})$**.** $(a+ib,\ c+id) = (a+ib)(1,0)+(c+id)(0,1)$ (complex scalars $a+ib$, $c+id$). So $\{(1,0),(0,1)\}$ spans. Independent (standard). $\dim = \mathbf{2}$.
(2) $\mathbb{C}^2(\mathbb{R})$**.** Only real scalars. $(a+ib,\ c+id) = a(1,0)+b(i,0)+c(0,1)+d(0,i)$ with a,b,c,d real. So $\{(1,0),(i,0),(0,1),(0,i)\}$ spans. Independent: if $a(1,0)+b(i,0)+c(0,1)+d(0,i) = (0,0)$ with real a,b,c,d, then $a+ib = 0$ and $c+id = 0$ give $a = b = c = d = 0$. $\dim = \mathbf{4}$.
Remark 3.3.13. The dimension of a vector space can change when the field of scalars changes. Same set $\mathbb{C}^2$; $\dim_{\mathbb{C}} = 2$ but $\dim_{\mathbb{R}} = 4$. Always ask: *which scalars?*

Example 3.3.14 of your notes – the space of linear functions $\mathbb{R}^n \to \mathbb{R}$

The set-up in plain words. V is the set of all functions $f : \mathbb{R}^n \to \mathbb{R}$ with $f(x+y) = f(x)+f(y)$ and $f(\alpha x) = \alpha f(x)$ ("linear functions"). Add functions pointwise: $(f \oplus g)(x) = f(x)+g(x)$; scale by t via $(t \odot f)(x) = f(tx)$. Because $f(tx) = tf(x)$, this is the same as the ordinary multiple $(tf)(x)$.
A concrete member ($n = 2$). $f(x_1, x_2) = 3x_1 - 2x_2$. Check $f(x+y) = 3(x_1+y_1) - 2(x_2+y_2) = f(x)+f(y)$ ✓. The functions $e_i(x_1, \ldots, x_n) = x_i$ are also in V. Here $f = 3e_1 + (-2)e_2$, and note $3 = f(1,0)$, $-2 = f(0,1)$.
Claim: $\{e_1, \ldots, e_n\}$ is a basis, so $\dim V = n$. Write $\varepsilon_1, \ldots, \varepsilon_n$ for the standard basis vectors of $\mathbb{R}^n$ (to avoid confusion with the functions e_i).

Step 1 – spans. Let $f \in V$ and $x = (x_1, \ldots, x_n) = \sum x_i\varepsilon_i$. Then $f(x) = \sum f(x_i\varepsilon_i)$ (additivity, repeatedly) $= \sum x_i f(\varepsilon_i)$ (homogeneity) $= \sum f(\varepsilon_i)\, e_i(x)$. So $f = f(\varepsilon_1)e_1 \oplus \cdots \oplus f(\varepsilon_n)e_n$.

Step 2 – independent. If $c_1e_1 \oplus \cdots \oplus c_ne_n$ is the zero function, evaluate at $x = \varepsilon_j$: $e_i(\varepsilon_j) = 1$ if $i = j$, else 0, so we get $c_j = 0$. This holds for every j.

Step 3 – conclude. $\dim V = n$. (The coordinates of f are just the numbers $f(\varepsilon_1), \ldots, f(\varepsilon_n)$.)

3.3.3 Extending and shrinking: Theorem 3.3.15 and Corollary 3.3.16

Theorem 3.3.15 and Corollary 3.3.16

3.3.15. Every linearly independent subset S of a finite dimensional V can be extended to a basis of V. (Equivalently: if $\dim V = n$ and $\{v_1, \ldots, v_r\}$ is independent, there are $v_{r+1}, \ldots, v_n$ with $\{v_1, \ldots, v_n\}$ a basis.)
3.3.16. Let $\dim V = n$. Then any n independent vectors form a basis; and any $m > n$ vectors are dependent.

Plain English (three rules you will use constantly)

1. **Extend:** take your independent list and keep adding vectors from *outside* its span until the span is everything.
2. **Shortcut for checking a basis:** in an n-dimensional space you need only check that n vectors are *independent* (spanning comes free), or only that they *span*.
3. **Counting rules:** more than n vectors $\Rightarrow$ dependent. Fewer than n vectors $\Rightarrow$ cannot span.

Proof in small steps – 3.3.15 and 3.3.16

3.3.15. Start with S independent. If $L(S) = V$, done. Otherwise pick $v \notin L(S)$; $S \cup \{v\}$ is independent (Cor. 3.2.6). Repeat. Independent lists in V have at most $n = \dim V$ vectors (Thm 3.3.7), so the process must stop, and it stops exactly when the span is V. ■
3.3.16. *First part:* extend the n independent vectors to a basis (3.3.15). A basis has exactly n vectors (Cor. 3.3.10), so nothing was added: the n vectors were already a basis. *Second part:* Theorem 3.3.7. ■ *Fewer than n cannot span:* if $m < n$ vectors spanned V, throwing away redundant ones (Cor. 3.2.5) would leave a basis with $\leq m < n$ vectors, contradicting Cor. 3.3.10.

Extra Example 3.3-B – using the shortcut

(1) $\{(1,0,0),(1,1,0),(1,1,1)\}$ is a basis of $\mathbb{R}^3$: three vectors, $\det = 1 \neq 0$ (so independent), and $\dim \mathbb{R}^3 = 3$. No separate spanning check needed.
(2) $\{(1,2),(2,4),(0,1)\}$ cannot be a basis of $\mathbb{R}^2$: $3 > 2$ vectors, so dependent.
(3) $\{(1,0,0,0),(0,1,0,0),(0,0,1,0)\}$ is not a basis of $\mathbb{R}^4$: only $3 < 4$ vectors, so it cannot span (it misses $(0,0,0,1)$).
(4) Extend $\{(1,2,0)\}$ to a basis of $\mathbb{R}^3$: add $(0,1,0)$ ($\notin L\{(1,2,0)\}$, since multiples of $(1,2,0)$ have first and second coordinates in ratio $1:2$), then add $(0,0,1)$ (not in the span of the first two, whose third coordinates are 0). Result: $\{(1,2,0),(0,1,0),(0,0,1)\}$ with $\det = 1 \neq 0$. Another valid extension: $(1,0,0),(0,0,1)$.

3.3.4 Three ways to get a basis from a spanning set

The problem

You are given vectors $S = \{s_1, \ldots, s_k\}$ (possibly wasteful) and asked for a basis of $L(S)$. Remark 3.3.8 of your notes gives one method; here are three. **Method C is the safest.**

Recipe 4 – basis of $L(S)$ for $S \subset \mathbb{R}^n$

Method A (your notes' Remark 3.3.8). Put the vectors as the *rows* of A. Use row operations of the type $R_i \to cR_i$ and "add a multiple of one row to another" (no swaps), until as many rows as possible are zero. The vectors of S whose rows are *non-zero* form a basis. *Safety check:* confirm each

discarded vector really is a mix of the kept ones.
Method B (rows of the RREF). Put the vectors as rows, row-reduce fully. The *non-zero rows of the RREF* are a basis of $L(S)$ (Theorem 3.1.17). These are new vectors, not members of S.
Method C (pivot columns). Put the vectors as *columns* of a matrix. Row-reduce. The columns of the *original* matrix that are **pivot columns** form a basis of $L(S)$. *Why:* row operations do not change which columns are mixes of which (a relation $Ax = \mathbf{0}$ has the same solutions after row operations). In the RREF the pivot columns are independent and every other column is a mix of them; the same is therefore true for the original columns.

Example 3.3.9 of your notes – all three methods

$S = \{(1,1,1,1),(1,1,-1,1),(1,1,0,1),(1,-1,1,1)\} \subset \mathbb{R}^4$. Find a basis of $L(S)$. Call the vectors s_1, s_2, s_3, s_4.

Method A (the notes). Rows: $\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & -1 & 1 & 1 \end{pmatrix}$.

Step 1 – clear the first column. Subtract row 1 from rows 2, 3, 4: $\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & -2 & 0 & 0 \end{pmatrix}$.

Step 2 – make a row zero. $R_2 \to R_2 - 2R_3$ gives row $2 = (0,0,0,0)$: $\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & -2 & 0 & 0 \end{pmatrix}$.

Step 3 – read. Non-zero rows: $1, 3, 4$. So $\{s_1, s_3, s_4\} = \{(1,1,1,1),(1,1,0,1),(1,-1,1,1)\}$ is a basis of $L(S)$, and $\dim L(S) = 3$.

Step 4 – safety check. The discarded $s_2 = (1,1,-1,1)$ should be a mix of the kept ones: $2s_3 - s_1 = (2,2,0,2) - (1,1,1,1) = (1,1,-1,1) = s_2$. ✓

Step 5 – the alternative in the notes. Instead of step 2, use $R_3 \to R_3 - \frac{1}{2}R_2$: row 3 becomes zero, giving the basis $\{s_1, s_2, s_4\}$. Check: $s_3 = \frac{1}{2}(s_1 + s_2) = \frac{1}{2}(2,2,0,2) = (1,1,0,1)$. ✓ Both are correct; bases are not unique.

Method B. Row-reduce fully; the RREF of the matrix above is $\begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$. A basis of $L(S)$: $\{(1,0,0,1),(0,1,0,0),(0,0,1,0)\}$. (Different vectors, same span, still 3 of them.)

Method C. Columns: $[s_1\, s_2\, s_3\, s_4]$. Its RREF is $\begin{pmatrix} 1 & 0 & \frac{1}{2} & 0 \\ 0 & 1 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$ with pivot columns $1, 2, 4$. Basis: $\{s_1, s_2, s_4\}$ (the first, second, fourth *original* vectors). The free column 3 says $s_3 = \frac{1}{2}s_1 + \frac{1}{2}s_2$ – exactly the relation found in step 5. This also *shows* the dependency.

3.3.5 Example 3.3.17 explained very slowly (basis of $V \cap W$ and how to extend it)

What is being asked, in plain words

We are given two subspaces V, W of $\mathbb{R}^5$, each described by *equations*. We must find
- a basis of V and a basis of W *such that both contain a basis of the overlap* $V \cap W$.

Plan: (1) find a basis of the overlap; (2) add vectors to it, staying inside W, until it is a basis of W; (3) add vectors to it, staying inside V, until it is a basis of V. Step (2) and (3) are exactly Theorem 3.3.15 (extend an independent list to a basis).

Recipe 5 – the "minor test" for independence (used in every extension step)

Let $u_1, \dots, u_k \in \mathbb{R}^n$. Write them as the k rows of a $k \times n$ matrix. **If some $k \times k$ block of columns has non-zero determinant, then $u_1, \dots, u_k$ are independent.**
Why: suppose $c_1u_1 + \cdots + c_ku_k = \mathbf{0}$. A vector that is zero is zero in *every* coordinate, in particular in those k chosen coordinates. That gives k equations in the k unknowns $c_1, \dots, c_k$ whose matrix is the $k \times k$ block (transposed). Its determinant is $\neq 0$, so the only solution is $c_1 = \cdots = c_k = 0$ (Key Fact K1).

Example 3.3.17 of your notes, in small steps

$V = \{(v,w,x,y,z) \in \mathbb{R}^5 : \ v + x - 3y + z = 0\}$ and $W = \{(v,w,x,y,z) \in \mathbb{R}^5 : \ w - x - z = 0, \ v = y\}$.

Step 1 – what does a vector of $V \cap W$ satisfy?. It must be in V *and* in W, so it satisfies *all three* equations together: (E1) $v + x - 3y + z = 0$, (E2) $w - x - z = 0$, (E3) $v = y$.

Step 2 – solve the three equations. By (E3) replace v by y. Then (E1) becomes $y + x - 3y + z = 0$, i.e. $z = 2y - x$. (E2) says $w = x + z = x + (2y - x) = 2y$. The unknowns x, y are free (they can be anything), and

$$(v,w,x,y,z) = (y,\ 2y,\ x,\ y,\ 2y - x) = y\,(1,2,0,1,2) + x\,(0,0,1,0,-1).$$

Step 3 – read off a basis of $V \cap W$. $b_1 = (1,2,0,1,2)$, $b_2 = (0,0,1,0,-1)$. They span $V \cap W$ (the line above) and are independent (b_1 has a 1 in place 4 where b_2 has 0; b_2 has a 1 in place 3 where b_1 has 0). So $\dim(V \cap W) = 2$.
Check that they lie in both: b_1: $1+0-3+2 = 0$ ✓, $2-0-2 = 0$ ✓, $v = y = 1$ ✓. b_2: $0+1-0-1 = 0$ ✓, $0-1+1 = 0$ ✓, $v = y = 0$ ✓.

Step 4 – count the dimensions. V: five unknowns, one equation, so $5 - 1 = 4$ free variables: $\dim V = 4$. W: five unknowns, two independent equations, so $5 - 2 = 3$ free variables: $\dim W = 3$. Hence we must *add* 1 *vector* to $\{b_1, b_2\}$ to get a basis of W, and *add* 2 *vectors* to get a basis of V.

Step 5 – extend inside W. A general vector of W has $v = y$ and $w = x + z$, so it looks like $(y,\ x+z,\ x,\ y,\ z)$ with x, y, z free. We need one vector of W outside $L(b_1, b_2)$. Try $x = 1$, $y = 1$, $z = 0$: $u = (1,1,1,1,0) \in W$ (check: $w - x - z = 1 - 1 - 0 = 0$ ✓, $v = y$ ✓).
Why is $\{b_1, b_2, u\}$ *independent?* Use the minor test on the first three coordinates (v,w,x): the rows are $b_1 \to (1,2,0)$, $b_2 \to (0,0,1)$, $u \to (1,1,1)$, and $\det \begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix} = 1\cdot(0-1) - 2\cdot(0-1) + 0 = -1 + 2 = 1 \neq 0$.
Why is it a basis of W*?* Three independent vectors in a 3-dimensional space (Cor. 3.3.16). **Basis of W: $\{(1,2,0,1,2),\ (0,0,1,0,-1),\ (1,1,1,1,0)\}$.**

Step 6 – extend inside V. A general vector of V has $z = 3y - v - x$: $(v,\ w,\ x,\ y,\ 3y - v - x)$ with v, w, x, y free. Choose two: $(v,w,x,y) = (0,1,0,0)$ gives $p = (0,1,0,0,0)$ (check $0 + 0 - 0 + 0 = 0$ ✓); $(v,w,x,y) = (0,1,1,1)$ gives $z = 3 - 0 - 1 = 2$, so $q = (0,1,1,1,2)$ (check $0 + 1 - 3 + 2 = 0$ ✓).

Independence of $\{b_1, b_2, p, q\}$*:* minor test on the coordinates (v, w, x, y): rows $b_1 \to (1,2,0,1)$, $b_2 \to (0,0,1,0)$, $p \to (0,1,0,0)$, $q \to (0,1,1,1)$, and the 4×4 determinant is $-1 \neq 0$. Four independent vectors in the 4-dimensional V form a basis. **Basis of V:** $\{(1,2,0,1,2), (0,0,1,0,-1), (0,1,0,0,0), (0,1,1,1,2)\}$.

Step 7 – sanity check with Grassmann (Theorem 3.3.18 below). $\dim(V+W) = \dim V + \dim W - \dim(V \cap W) = 4 + 3 - 2 = 5$, so $V + W = \mathbb{R}^5$. In fact $\{b_1, b_2, u, p, q\}$ (the overlap basis, then the extra vector of W, then the two extras of V) is a basis of $\mathbb{R}^5$; its 5×5 determinant is $1 \neq 0$.

The "no guessing" way to extend: pivot columns (always works)

If you do not want to guess u, p, q, use Method C of Recipe 4. **For W:** write down a spanning set of W (from $(y, x+z, x, y, z)$: put $(x,y,z) = (1,0,0), (0,1,0), (0,0,1)$ to get $w_1 = (0,1,1,0,0)$, $w_2 = (1,0,0,1,0)$, $w_3 = (0,1,0,0,1)$). Make the columns $[\, b_1 \; b_2 \; w_1 \; w_2 \; w_3 \,]$ and row-reduce:

$$\begin{pmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 2 & -1 \\ 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \text{pivot columns } 1, 2, 3.$$

So $\{b_1, b_2, w_1\}$ is a basis of W that contains $\{b_1, b_2\}$. (The first columns b_1, b_2 are independent, so they are automatically pivot columns. That is why this method keeps them.) **For V:** the same idea with the spanning vectors $(1,0,0,0,-1)$, $(0,1,0,0,0)$, $(0,0,1,0,-1)$, $(0,0,0,1,3)$ (put each of v, w, x, y equal to 1 in turn, others 0) gives the pivot columns $b_1, b_2, (1,0,0,0,-1), (0,1,0,0,0)$. Both answers differ from the notes' answer, and both are right: *bases are not unique.*

3.3.6 Grassmann's dimension formula (Theorem 3.3.18)

Picture it: counting with sets

For finite sets, $|A \cup B| = |A| + |B| - |A \cap B|$ (the overlap is counted twice, so subtract it once). Subspaces behave the same way, with "dimension" instead of "number of elements". Two planes in $\mathbb{R}^3$: dim = 2 each. They add up to 4, but the space only has dimension 3, so the overlap must have dimension $4 - 3 = 1$: **two different planes through the origin always meet in a line.**

Theorem 3.3.18 (Grassmann)

Let $V(\mathbb{F})$ be finite dimensional and M, N subspaces. Then

$$\dim(M) + \dim(N) = \dim(M+N) + \dim(M \cap N).$$

Recall $M + N = \{u + v : u \in M,\ v \in N\}$ (Extra Example 3.1-D).

Proof in small steps – Grassmann's formula (running example: M = the xy-plane, N = the yz-plane in $\mathbb{R}^3$)

Step 1 – basis of the overlap. Let $\{u_1, \ldots, u_k\}$ be a basis of $M \cap N$ ($k = 0$ allowed). *Example:* $M \cap N$ is the y-axis, $u_1 = e_2$, $k = 1$.

Step 2 – extend on each side. Extend to a basis of M: $\{u_1, \ldots, u_k, m_1, \ldots, m_a\}$, so $\dim M = k + a$. Extend to a basis of N: $\{u_1, \ldots, u_k, n_1, \ldots, n_b\}$, so $\dim N = k + b$ (Theorem 3.3.15). *Example:* $m_1 = e_1$

$(a=1)$, $n_1 = e_3$ $(b=1)$.

Step 3 – the claim. $T = \{u_1, \ldots, u_k, m_1, \ldots, m_a, n_1, \ldots, n_b\}$ is a basis of $M+N$. If so, $\dim(M+N) = k+a+b$ and then $\dim M + \dim N = (k+a)+(k+b) = (k+a+b)+k = \dim(M+N)+\dim(M\cap N)$. *Example:* $T = \{e_2, e_1, e_3\}$, a basis of $\mathbb{R}^3 = M+N$; and $2+2=3+1$ ✓.

Step 4 – T spans $M+N$. A vector of $M+N$ is $m+n$. Here m is a mix of the u's and m's, and n is a mix of the u's and n's. Adding the two mixes gives a mix of T.

Step 5 – T is independent. Suppose $\sum \alpha_i u_i + \sum \beta_j m_j + \sum \gamma_l n_l = \mathbf{0}$. Put $w = \sum \gamma_l n_l$. Then $w = -\sum \alpha_i u_i - \sum \beta_j m_j$.

Step 6 – where does w live?. The first expression shows $w \in N$ (a mix of the n_l). The second shows $w \in M$ (a mix of the u_i, m_j). So $w \in M \cap N$, hence $w = \sum \delta_i u_i$ for some scalars δ_i.

Step 7 – kill the γ's and δ's. From $w = \sum \gamma_l n_l = \sum \delta_i u_i$ we get $\sum \delta_i u_i - \sum \gamma_l n_l = \mathbf{0}$, a relation between vectors of the basis of N. So all $\delta_i = 0$ and all $\gamma_l = 0$. Then $w = \mathbf{0}$.

Step 8 – kill the α's and β's. Now $\sum \alpha_i u_i + \sum \beta_j m_j = -w = \mathbf{0}$, a relation between vectors of the basis of M. So all $\alpha_i = 0$ and $\beta_j = 0$. Hence T is independent, so a basis of $M+N$. ■

What Grassmann gives you (read this before the questions)

1. **Find one dimension from the other three:** $\dim(M\cap N) = \dim M + \dim N - \dim(M+N)$, and $\dim(M+N) = \dim M + \dim N - \dim(M\cap N)$.
2. **Subspaces of a big space must overlap:** since $M+N \subseteq V$, we have $\dim(M+N) \le \dim V$, therefore $\dim(M\cap N) \ge \dim M + \dim N - \dim V$. If the right-hand side is positive, the subspaces share a non-zero vector. (Question 3.10: $4+4-7 = 1 > 0$.)
3. **Direct sum:** if $M\cap N = \{\mathbf{0}\}$ then $\dim(M+N) = \dim M + \dim N$. We write $M \oplus N$ and call it a **direct sum**. If moreover $M \oplus N = V$, every $v \in V$ splits *uniquely* as $v = m+n$, $m \in M$, $n \in N$ (proved in Question 3.13). If $M\cap N \neq \{\mathbf{0}\}$, the splitting is **not** unique (Question 3.14).

Recipe 6 – dimensions of $M\cap N$ and $M+N$

When M and N are given by equations: $M\cap N$ is the solution set of *all the equations together* (as in Example 3.3.17). Solve, read off a basis (free variables). Then $\dim(M+N)$ comes from Grassmann. **When $M = L(S)$ and $N = L(T)$:** $M+N = L(S\cup T)$. Put all the vectors of S and T as the columns (or rows) of one matrix; its rank is $\dim(M+N)$, and the pivot columns give a basis. Then $\dim(M\cap N) = \dim M + \dim N - \dim(M+N)$ from Grassmann. To actually *find* a vector of $M\cap N$, solve $\sum a_i s_i = \sum c_j t_j$.

Extra Example 3.3-C – Grassmann used both ways

(1) Equations. $M = \{x+y+z=0\}$ and $N = \{x-y=0\}$ in $\mathbb{R}^3$. $\dim M = 3-1 = 2$, $\dim N = 3-1 = 2$. Intersection: $x = y$ and $2x+z = 0$, so $(x,y,z) = x(1,1,-2)$, $\dim(M\cap N) = 1$. By Grassmann $\dim(M+N) = 2+2-1 = 3$, so $M+N = \mathbb{R}^3$. (Consistent with the picture "two planes meet in a line".)
(2) Spanning sets. $M = L\{(1,1,0),(0,1,1)\}$, $N = L\{(1,0,1),(1,2,1)\}$ in $\mathbb{R}^3$. Both have dimension 2 (each pair is not proportional). Put the four vectors as columns and row-reduce: $\begin{pmatrix}1&0&1&1\\1&1&0&2\\0&1&1&1\end{pmatrix} \to \begin{pmatrix}1&0&0&1\\0&1&0&1\\0&0&1&0\end{pmatrix}$, rank 3, so $\dim(M+N) = 3$. Grassmann: $\dim(M\cap N) = 2+2-3 = 1$. *Find it:* column 4 of the RREF says $(1,2,1) = 1\cdot(1,1,0) + 1\cdot(0,1,1)$, so $(1,2,1)$ is in both: $M\cap N = L\{(1,2,1)\}$. ✓

3.3.7 Row rank equals column rank (Proposition 3.3.20), and the rank–nullity count

Why this is surprising and useful

For the $m \times n$ matrix A the rows live in $\mathbb{R}^n$ and the columns in $\mathbb{R}^m$ – different spaces, even different sizes. Yet the *number* of independent rows always equals the number of independent columns. So "rank" is one well-defined number, and you may compute it from rows or from columns, whichever is easier.

Proposition 3.3.20

For every real $m \times n$ matrix A: row rank(A) = column rank(A). (Here row rank = $\dim \mathrm{Row}(A)$ and column rank = $\dim \mathrm{Col}(A)$.)

Proof in small steps – Proposition 3.3.20 (running example $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 1 & 2 & 2 \end{pmatrix}$)

Step 1 – choose a row basis. Let r = row rank and let $u_1, \ldots, u_r \in \mathbb{R}^n$ be a basis of $\mathrm{Row}(A)$. *Example:* $r = 2$; the non-zero RREF rows $u_1 = (1, 2, 0)$, $u_2 = (0, 0, 1)$ are a basis of $\mathrm{Row}(A)$.

Step 2 – every row is a mix of the u's. Each row $R_i = \alpha_{i1}u_1 + \cdots + \alpha_{ir}u_r$. *Example:* $R_1 = (1, 2, 3) = 1\,u_1 + 3\,u_2$, $R_2 = (2, 4, 5) = 2u_1 + 5u_2$, $R_3 = (1, 2, 2) = 1u_1 + 2u_2$.

Step 3 – write this as a matrix product. Let $C = (\alpha_{ij})$ $(m \times r)$ and let U be the $r \times n$ matrix whose rows are $u_1, \ldots, u_r$. Then $A = CU$. *Example:* $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 1 & 2 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 2 & 5 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ ✓ (multiply to check).

Step 4 – read $A = CU$ column by column. The k-th column of A equals $u_{1k} \cdot$ (column 1 of C) $+ \cdots + u_{rk} \cdot$ (column r of C). So *every column of A is a mix of the r columns of C*. *Example:* column 3 of A is $(3, 5, 2) = 0 \cdot (1, 2, 1) + 1 \cdot (3, 5, 2)$ (columns of C).

Step 5 – count. $\mathrm{Col}(A) \subseteq L$(the r columns of C), and r vectors span a space of dimension $\leq r$. So column rank $\leq r$ = row rank.

Step 6 – use the transpose. Apply Steps 1–5 to A^t (its rows are the columns of A and vice versa): row rank$(A^t) \geq$ column rank(A^t), i.e. column rank$(A) \geq$ row rank(A). Combine with Step 5: equal. ■

Recipe 7 – everything about a matrix A $(m \times n)$ in one go

Row-reduce A to RREF D. Let r = number of non-zero rows (pivots).
Row space $\mathrm{Row}(A)$: the r non-zero rows of D. $\dim = r$.
Column space $\mathrm{Col}(A)$ (range): the r columns of the *original* A in the pivot positions. $\dim = r$ (Prop. 3.3.20).
Null space $N(A) = \{x : Ax = \mathbf{0}\}$: solve $Dx = \mathbf{0}$; each of the $n - r$ free variables gives one basis vector (set that free variable $= 1$, the other free ones $= 0$). $\dim = n - r$.
Count: $\dim \mathrm{Col}(A) + \dim N(A) = r + (n - r) = n =$ number of columns.
Matrix P with $PA = D$: row-reduce the augmented matrix $[A \mid I_m]$; when the left half becomes D, the right half is P.

Extra Example 3.3-D – Recipe 7 on $A = \begin{pmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & 3 & 1 \\ 1 & 2 & 2 & 1 \end{pmatrix}$

Step 1 – row-reduce. $R_2 \to R_2 - 2R_1$, $R_3 \to R_3 - R_1$ gives $\begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix}$. Then $R_1 \to R_1 - R_2$ and $R_3 \to R_3 - R_2$ give the RREF $D = \begin{pmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$. Pivots in columns 1 and 3, so $r = 2$.

Step 2 – row space. $\{(1,2,0,-1),\ (0,0,1,1)\}$, dim $= 2$.

Step 3 – column space. Columns 1 and 3 of the *original* A: $\{(1,2,1),(1,3,2)\}$, dim $= 2$. (Not the columns of D!)

Step 4 – null space. $Dx = \mathbf{0}$: $x_1 + 2x_2 - x_4 = 0$, $x_3 + x_4 = 0$. Free variables $x_2 = s$, $x_4 = t$; then $x_1 = -2s + t$, $x_3 = -t$. So $x = s(-2,1,0,0) + t(1,0,-1,1)$; basis $\{(-2,1,0,0),(1,0,-1,1)\}$, dim $= n - r = 4 - 2 = 2$.
Check: $A(-2,1,0,0)^t = (-2{+}2,\ -4{+}4,\ -2{+}2) = (0,0,0)$ ✓; $A(1,0,-1,1)^t = (1{-}1,\ 2{-}3{+}1,\ 1{-}2{+}1) = (0,0,0)$ ✓.

Step 5 – count. 2 (column space) $+2$ (null space) $= 4 = n$. ✓ Row rank 2 = column rank 2. ✓

3.3.8 Questions for Section 3.3 (Exercises 3.3.6 and 3.3.19 of your notes)

Before you start – the toolkit for every question below

1. **"Is this a basis?"** Show *independent* and *spanning* – or, if you already know $\dim V = n$, just show that you have n vectors and that they are independent (Cor. 3.3.16).
2. **"Find a basis of a space given by equations."** Solve the equations, write the solution with free variables, split into one vector per free variable. dim = number of free variables = $n-$ number of independent equations.
3. **Sums and intersections:** intersection = all equations together; then Grassmann: $\dim(M+N) = \dim M + \dim N - \dim(M \cap N)$.
4. **Always ask: which field?** $\mathbb{C}^n(\mathbb{C})$ has dimension n; $\mathbb{C}^n(\mathbb{R})$ has dimension $2n$.

Question 3.1 (Exercise 3.3.6, item 1)

Let $S = \{v_1, v_2, \ldots, v_p\}$ be a subset of a vector space $V(\mathbb{F})$. Suppose $L(S) = V$ but S is not a linearly independent set. Then prove that each vector in V can be expressed in more than one way as a linear combination of vectors from S.

Solution.

Step 1 – use dependence. S is dependent, so there are scalars $c_1, \ldots, c_p$, *not all zero*, with $c_1v_1 + \cdots + c_pv_p = \mathbf{0}$ (R).

Step 2 – take any vector $v \in V$. Since $L(S) = V$, $v = a_1v_1 + \cdots + a_pv_p$ for some scalars a_i (first way).

Step 3 – add the zero vector. Add (R) to this: $v = v + \mathbf{0} = (a_1 + c_1)v_1 + \cdots + (a_p + c_p)v_p$ (second way).

Step 4 – the two ways are different. Some $c_j \neq 0$, so $a_j + c_j \neq a_j$: the coefficient lists (a_i) and $(a_i + c_i)$ differ. So v has at least two representations. As v was arbitrary, this holds for *every* v (stronger

than "some"). ■

Step 5 – example. $S = \{(1,0),(0,1),(1,1)\}$ spans $\mathbb{R}^2$; (R): $1(1,0)+1(0,1)-1(1,1)=\mathbf{0}$. For $v=(3,2)$: $3(1,0)+2(0,1)+0(1,1)$ and $4(1,0)+3(0,1)-1(1,1)=(4-1,\,3-1)=(3,2)$ ✓.

Question 3.2 (Exercise 3.3.6, item 2)

Show that the set $\{(1,0,1),(1,i,0),(1,1,1-i)\}$ is a basis of $\mathbb{C}^3(\mathbb{C})$.

Solution.

Step 1 – what is enough. $\dim \mathbb{C}^3(\mathbb{C}) = 3$ (standard basis e_1, e_2, e_3, complex scalars). We have 3 vectors, so it is enough to show they are *independent* (Cor. 3.3.16).

Step 2 – set up. Let $a,b,c \in \mathbb{C}$ with $a(1,0,1)+b(1,i,0)+c(1,1,1-i)=(0,0,0)$. Coordinate by coordinate: (1) $a+b+c=0$, (2) $ib+c=0$, (3) $a+(1-i)c=0$.

Step 3 – solve. From (2): $c=-ib$. Put in (3): $a=-(1-i)c=(1-i)(ib)=ib-i^2b=ib+b=(1+i)b$.

Step 4 – use (1). $a+b+c=(1+i)b+b-ib=2b$. So $2b=0$, $b=0$. Then $c=-ib=0$ and $a=(1+i)\cdot 0=0$.

Step 5 – conclude. Only the trivial solution: independent. Hence a basis of $\mathbb{C}^3(\mathbb{C})$. ■

Step 6 – cross-check with a determinant. $\det\begin{pmatrix}1&0&1\\1&i&0\\1&1&1-i\end{pmatrix} = 1\cdot\big(i(1-i)-0\big)-0+1\cdot(1\cdot 1-i\cdot 1) = (i+1)+(1-i)=2\neq 0$ ✓ (Key Fact K1 holds over $\mathbb{C}$ too).

Question 3.3 (Exercise 3.3.6, item 3)

Let A be a matrix of rank r. Then show that the r non-zero rows in the row-reduced echelon form of A are linearly independent and they form a basis of the row space of A.

Solution. Let D be the RREF of A and $d_1,\dots,d_r$ its non-zero rows (rank $=r$ means exactly r non-zero rows).

Step 1 – D and A have the same row space. D comes from A by row operations, and row operations do not change the row space (Lemma 3.1.16). So $\mathrm{Row}(A)=\mathrm{Row}(D)$.

Step 2 – they span. The zero rows of D add nothing to a span. So $\mathrm{Row}(D)=L(d_1,\dots,d_r)$.

Step 3 – they are independent. In RREF every non-zero row has a leading 1 in a pivot column, and that pivot column has 0 in every other row. Say d_i has its leading 1 in column p_i. If $c_1d_1+\cdots+c_rd_r=\mathbf{0}$, look at column p_i of this equation: $c_i\cdot 1+\sum_{j\neq i}c_j\cdot 0=0$, so $c_i=0$. This works for every i.

Step 4 – conclude. $\{d_1,\dots,d_r\}$ is independent and spans $\mathrm{Row}(A)$: a basis, and $\dim \mathrm{Row}(A)=r$. ■

Step 5 – example. $A=\begin{pmatrix}1&2&3\\2&4&5\end{pmatrix}\to D=\begin{pmatrix}1&2&0\\0&0&1\end{pmatrix}$ (pivot columns 1, 3). $c_1(1,2,0)+c_2(0,0,1)=(c_1,2c_1,c_2)=\mathbf{0}$ forces $c_1=c_2=0$ ✓.

Question 3.4 (Exercise 3.3.19, item 1)

Find a basis of the vector space $P_n(\mathbb{R})$. Also, find $\dim(P_n(\mathbb{R}))$. What can you say about the dimension of $P(\mathbb{R})$?

Solution.

Step 1 – guess. $B=\{1,x,x^2,\dots,x^n\}$. (For $n=2$: $\{1,x,x^2\}$.)

Step 2 – spans. Every $p\in P_n(\mathbb{R})$ is $a_0+a_1x+\cdots+a_nx^n=a_0\cdot 1+a_1\cdot x+\cdots+a_n\cdot x^n$. *Example:* $3-x+2x^2=3\cdot 1+(-1)x+2x^2$.

Step 3 – independent. Suppose $a_0+a_1x+\cdots+a_nx^n=0$ for *every* real x. A non-zero polynomial

of degree $\leq n$ has at most n roots, but this one has infinitely many roots (all real numbers). So it must be the zero polynomial: $a_0 = \cdots = a_n = 0$.

Step 4 – dimension. B has $n+1$ elements, so $\dim P_n(\mathbb{R}) = n+1$.

Step 5 – $P(\mathbb{R})$. Suppose $P(\mathbb{R})$ had a finite basis with m elements. Then every set of $m+1$ vectors would be dependent (Thm 3.3.7). But $\{1, x, \ldots, x^m\}$ has $m+1$ elements and is independent (Step 3). Contradiction. So $P(\mathbb{R})$ is *infinite dimensional*, with infinite basis $\{1, x, x^2, \ldots\}$.

Question 3.5 (Exercise 3.3.19, item 2)

Consider the real vector space $C([0, 2\pi])$ of all real valued continuous functions. For each n consider the vector e_n defined by $e_n(x) = \sin(nx)$. Prove that the collection $\{e_n : 1 \leq n < \infty\}$ is a linearly independent set.

Solution. An infinite set is independent if every *finite* subset is independent.

Step 1 – suppose not. There are finitely many $k_1 < k_2 < \cdots < k_\ell$ and real $\alpha_1, \ldots, \alpha_\ell$, not all 0, with $\alpha_1 \sin(k_1 x) + \cdots + \alpha_\ell \sin(k_\ell x) = 0$ for all $x \in [0, 2\pi]$.

Step 2 – the key integral. For positive integers m, k: $\sin(mx)\sin(kx) = \frac{1}{2}[\cos((m-k)x) - \cos((m+k)x)]$. If $m \neq k$: $\int_0^{2\pi} \sin(mx)\sin(kx)\,dx = \frac{1}{2}\left[\frac{\sin((m-k)x)}{m-k} - \frac{\sin((m+k)x)}{m+k}\right]_0^{2\pi} = 0$ (the sine of a multiple of 2π is 0). If $m = k$: $\int_0^{2\pi} \sin^2(mx)dx = \frac{1}{2}\int_0^{2\pi}(1 - \cos 2mx)\,dx = \frac{1}{2} \cdot 2\pi = \pi$.

Step 3 – multiply and integrate. Fix any j with $1 \leq j \leq \ell$. Multiply the relation of Step 1 by $\sin(k_j x)$ and integrate over $[0, 2\pi]$: $\sum_i \alpha_i \int_0^{2\pi} \sin(k_j x)\sin(k_i x)\,dx = 0$.

Step 4 – collapse. By Step 2 only the term $i = j$ survives and equals $\alpha_j \pi$. So $\alpha_j \pi = 0$, i.e. $\alpha_j = 0$.

Step 5 – contradiction. j was arbitrary, so all $\alpha_j = 0$ – contradicting "not all 0". Hence $\{e_n\}$ is independent. ■

Step 6 – tiny example. $a \sin x + b \sin 2x = 0$. Multiply by $\sin x$ and integrate: $a\pi + b \cdot 0 = 0$, so $a = 0$. Multiply by $\sin 2x$: $b = 0$.

Question 3.6 (Exercise 3.3.19, item 3)

Show that the set $\{(1,0,0), (1,1,0), (1,1,1)\}$ is a basis of $\mathbb{C}^3(\mathbb{C})$. Is it a basis of $\mathbb{C}^3(\mathbb{R})$ also?

Solution.

Step 1 – $\mathbb{C}^3(\mathbb{C})$. $\dim \mathbb{C}^3(\mathbb{C}) = 3$ and we have 3 vectors. Independent over $\mathbb{C}$: the matrix with these rows is triangular with $\det = 1 \cdot 1 \cdot 1 = 1 \neq 0$ (Key Fact K1). So by Cor. 3.3.16 it is a basis of $\mathbb{C}^3(\mathbb{C})$.

Step 2 – $\mathbb{C}^3(\mathbb{R})$: count. $\dim \mathbb{C}^3(\mathbb{R}) = 6$ (basis e_j, ie_j for $j = 1, 2, 3$). A basis of $\mathbb{C}^3(\mathbb{R})$ must have 6 vectors. We have only 3, so the set *cannot span*: **not a basis of** $\mathbb{C}^3(\mathbb{R})$.

Step 3 – a vector it misses. Is $(i, 0, 0)$ in the real span? $a(1,0,0) + b(1,1,0) + c(1,1,1) = (a+b+c,\ b+c,\ c)$ with real a, b, c. Equal to $(i, 0, 0)$ needs $c = 0$, then $b = 0$, then $a = i$, *not real*. So $(i, 0, 0) \notin$ the real span. (The set is still independent over $\mathbb{R}$, but too small.)

One-line summary: Basis of $\mathbb{C}^3(\mathbb{C})$: yes. Basis of $\mathbb{C}^3(\mathbb{R})$: no – it spans only a 3-dimensional part of a 6-dimensional space.

Question 3.7 (Exercise 3.3.19, item 4)

Let $W = \{(x, y, z, w) \in \mathbb{R}^4 : \ x + y - z + w = 0\}$ be a subspace of $\mathbb{R}^4$. Find a basis and the dimension of W.

Solution.

Step 1 – solve the equation. $x + y - z + w = 0 \iff z = x + y + w$. The variables x, y, w are free.

Step 2 – split by free variable. $(x, y, z, w) = (x, y, x + y + w, w) = x(1, 0, 1, 0) + y(0, 1, 1, 0) +$

$w(0,0,1,1)$.

Step 3 – spanning. So $\{(1,0,1,0),(0,1,1,0),(0,0,1,1)\}$ spans W. (Each lies in W: $1+0-1+0=0$, $0+1-1+0=0$, $0+0-1+1=0$ ✓.)

Step 4 – independent. In coordinates (x,y,w) the three vectors are $(1,0,0),(0,1,0),(0,0,1)$ – the identity matrix, $\det = 1 \neq 0$ (minor test). So independent.

Step 5 – conclude. Basis: $\{(1,0,1,0),(0,1,1,0),(0,0,1,1)\}$, $\dim W = 3 = 4-1$ (four unknowns, one equation).

Question 3.8 (Exercise 3.3.19, item 5)

Let $V = \{(x,y,z,w) \in \mathbb{R}^4 : x+y-z+w=0,\ x+y+z+w=0\}$ and $W = \{(x,y,z,w) \in \mathbb{R}^4 : x-y-z+w=0,\ x+2y-w=0\}$ be two subspaces of $\mathbb{R}^4$. Find bases and dimensions of V, W, $V \cap W$ and $V+W$.

Solution.

Step 1 – basis of V. Subtract the two equations: $2z=0$, so $z=0$. Then $x+y+w=0$, i.e. $w=-x-y$. $(x,y,z,w)=(x,y,0,-x-y)=x(1,0,0,-1)+y(0,1,0,-1)$. Check in V: $1+0-0-1=0$ ✓ both equations; $0+1-0-1=0$ ✓. Independent (look at coordinates x,y: identity). **Basis of V:** $\{(1,0,0,-1),(0,1,0,-1)\}$, $\dim V = 2$.

Step 2 – basis of W. From the second equation $w=x+2y$. First equation: $x-y-z+(x+2y)=0$, so $z=2x+y$. $(x,y,z,w)=(x,y,2x+y,x+2y)=x(1,0,2,1)+y(0,1,1,2)$. Check: $1-0-2+1=0$ ✓, $1+0-1=0$ ✓; $0-1-1+2=0$ ✓, $0+2-2=0$ ✓. **Basis of W:** $\{(1,0,2,1),(0,1,1,2)\}$, $\dim W = 2$.

Step 3 – $V \cap W$: all four equations together. From V: $z=0$ and $w=-x-y$. Put into the first equation of W: $x-y-0+(-x-y)=-2y=0$, so $y=0$. Then $w=-x$. Second equation of W: $x+2\cdot 0-(-x)=2x=0$, so $x=0$, hence $w=0$. Only the zero vector: $V \cap W = \{\mathbf{0}\}$, **basis** $\varnothing$, **dimension** 0.

Step 4 – $V+W$ by Grassmann. $\dim(V+W) = 2+2-0=4$. So $V+W=\mathbb{R}^4$.

Step 5 – a basis of $V+W$. Combine the bases: $\{(1,0,0,-1),(0,1,0,-1),(1,0,2,1),(0,1,1,2)\}$ spans $V+W$. Check independence: $\det \begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 2 \end{pmatrix} = 4 \neq 0$ ✓. Four independent vectors in $\mathbb{R}^4$: a basis, $\dim = 4$.

One-line summary: $\dim V = 2$, $\dim W = 2$, $V \cap W = \{\mathbf{0}\}$ *(dim 0)*, $V+W = \mathbb{R}^4$ *(dim 4)*.

Question 3.9 (Exercise 3.3.19, item 6)

Let V be the set of all real symmetric $n \times n$ matrices. Find its basis and dimension. What if V is the complex vector space of all $n \times n$ Hermitian matrices?

Solution. Let E_{ij} be the matrix with 1 in position (i,j) and 0 elsewhere.

Step 1 – symmetric, small case $n=2$. $A = \begin{pmatrix} a & b \\ b & c \end{pmatrix} = aE_{11} + cE_{22} + b(E_{12}+E_{21})$. Three free numbers a,b,c: dimension 3.

Step 2 – symmetric, general n. A symmetric matrix satisfies $a_{ji}=a_{ij}$, so it is determined by its entries on and above the diagonal. $A = \sum_i a_{ii}E_{ii} + \sum_{i<j} a_{ij}(E_{ij}+E_{ji})$.

Step 3 – basis. $\{E_{ii} : 1 \le i \le n\} \cup \{E_{ij}+E_{ji} : i<j\}$. Independent: each member has a position where it alone is non-zero (E_{ii} at (i,i); $E_{ij}+E_{ji}$ at (i,j) with $i<j$).

Step 4 – count. $n + \dfrac{n(n-1)}{2} = \dfrac{n(n+1)}{2}$. $\dim = \frac{n(n+1)}{2}$ ($n=2$: 3 ✓; $n=3$: 6).

Step 5 – Hermitian ($A = \overline{A}^t$). Then $a_{ji} = \overline{a_{ij}}$ and the diagonal entries are real ($a_{ii} = \overline{a_{ii}}$). Careful: if A is Hermitian, iA is *not* (since $\overline{iA}^t = -iA$). So Hermitian matrices are **not closed under complex scalars**: they form a *real* vector space (the notes' "complex" should be read as "complex entries").

Step 6 – basis over $\mathbb{R}$. Free real numbers: the n diagonal entries, and for each $i < j$ the complex number $a_{ij} = p + iq$ (two real numbers p, q). Basis: E_{ii} (n matrices), $E_{ij} + E_{ji}$ and $i(E_{ij} - E_{ji})$ for $i < j$ ($2 \cdot \frac{n(n-1)}{2}$ matrices). *Example* $n = 2$: $\begin{pmatrix} a & p+iq \\ p-iq & c \end{pmatrix} = aE_{11} + cE_{22} + p\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + q\begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}$.

Step 7 – count. $n + n(n-1) = n^2$. **Hermitian: real dimension n^2** ($n = 2$: 4).

Question 3.10 (Exercise 3.3.19, item 7)

If M and N are 4-dimensional subspaces of a vector space V of dimension 7, then show that M and N have at least one non-zero vector in common.

Solution.

Step 1 – Grassmann. $\dim(M \cap N) = \dim M + \dim N - \dim(M + N) = 8 - \dim(M + N)$.

Step 2 – bound the sum. $M + N$ is a subspace of V, so $\dim(M + N) \leq \dim V = 7$.

Step 3 – conclude. $\dim(M \cap N) \geq 8 - 7 = 1$. A subspace of dimension ≥ 1 contains a non-zero vector (its basis is non-empty). So $M \cap N$ contains a non-zero vector. ■

Step 4 – example. In $\mathbb{R}^7$: $M = L(e_1, \ldots, e_4)$, $N = L(e_4, \ldots, e_7)$ meet in $L(e_4)$ – dimension exactly 1.

Question 3.11 (Exercise 3.3.19, item 8)

Let $P = L\{(1,0,0), (1,1,0)\}$ and $Q = L\{(1,1,1)\}$ be vector subspaces of $\mathbb{R}^3$. Show that $P + Q = \mathbb{R}^3$ and $P \cap Q = \{\mathbf{0}\}$. If $u \in \mathbb{R}^3$, determine u_P, u_Q such that $u = u_P + u_Q$ where $u_P \in P$, $u_Q \in Q$. Is it necessary that u_P and u_Q are unique?

Solution.

Step 1 – $P + Q = \mathbb{R}^3$. $P + Q = L\{(1,0,0), (1,1,0), (1,1,1)\}$. Three vectors, $\det \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} = 1 \neq 0$, so independent, so a basis of $\mathbb{R}^3$ (dim 3).

Step 2 – $P \cap Q = \{\mathbf{0}\}$. $P = L\{(1,0,0), (1,1,0)\}$ is the xy-plane (every vector $(s,t,0) = (s-t)(1,0,0) + t(1,1,0)$ has third coordinate 0). A vector of Q is $c(1,1,1)$, with third coordinate c. Being in P forces $c = 0$. So only $\mathbf{0}$. (Grassmann agrees: $2 + 1 = 3 + \dim(P \cap Q)$ gives 0.)

Step 3 – find the decomposition. For $u = (x,y,z)$ we need $u_Q = c(1,1,1)$ and $u - u_Q \in P$, i.e. third coordinate of $u - u_Q$ is 0: $z - c = 0$, so $c = z$.

Step 4 – answer. $u_Q = (z,z,z)$, $u_P = u - u_Q = (x - z,\ y - z,\ 0) = (x-y)(1,0,0) + (y-z)(1,1,0)$.

Step 5 – example. $u = (5,3,2)$: $u_Q = (2,2,2)$, $u_P = (3,1,0) = 2(1,0,0) + 1(1,1,0)$. Check: $(3,1,0) + (2,2,2) = (5,3,2)$ ✓.

Step 6 – unique?. Yes. In Step 3 the value $c = z$ was *forced*. In general: if $u = p + q = p' + q'$ then $p - p' = q' - q \in P \cap Q = \{\mathbf{0}\}$, so $p = p'$, $q = q'$ (this is Question 3.13).

Question 3.12 (Exercise 3.3.19, item 9)

Let W_1 be a k-dimensional subspace of an n-dimensional vector space $V(\mathbb{F})$ where $k \geq 1$. Prove that there exists an $(n-k)$-dimensional subspace W_2 of V such that $W_1 \cap W_2 = \{\mathbf{0}\}$ and $W_1 + W_2 = V$.

Solution.

Step 1 – basis of W_1. Let $\{w_1, \ldots, w_k\}$ be a basis of W_1. It is independent in V.

Step 2 – extend. By Theorem 3.3.15 there are $w_{k+1}, \ldots, w_n$ such that $\{w_1, \ldots, w_n\}$ is a basis of V.

Step 3 – define W_2. $W_2 = L(w_{k+1}, \ldots, w_n)$. These $n-k$ vectors are independent (a subset of a basis), so $\dim W_2 = n-k$.

Step 4 – $W_1 + W_2 = V$. $W_1 + W_2 = L(w_1, \ldots, w_k, w_{k+1}, \ldots, w_n) = V$.

Step 5 – $W_1 \cap W_2 = \{\mathbf{0}\}$. Grassmann: $\dim(W_1 \cap W_2) = k + (n-k) - \dim V = n - n = 0$. (Directly: $v = \sum_{i \le k} a_i w_i = \sum_{j>k} b_j w_j$ gives $\sum a_i w_i - \sum b_j w_j = \mathbf{0}$, so all $a_i, b_j = 0$ by independence of the full basis.) ■

Step 6 – example. $V = \mathbb{R}^3$, $W_1 = L\{(1,1,0)\}$ ($k=1$). Extend by $(0,1,0), (0,0,1)$ ($\det \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = 1 \neq 0$). $W_2 = L\{(0,1,0),(0,0,1)\}$ has $\dim 2 = 3 - 1$ ✓. (W_2 is not unique: other extensions give other complements.)

Question 3.13 (Exercise 3.3.19, item 10)

Let P and Q be subspaces of $\mathbb{R}^n$ such that $P + Q = \mathbb{R}^n$ and $P \cap Q = \{\mathbf{0}\}$. Then show that each $u \in \mathbb{R}^n$ can be uniquely expressed as $u = u_P + u_Q$ where $u_P \in P$ and $u_Q \in Q$.

Solution.

Step 1 – existence. $P + Q = \{p + q : p \in P, q \in Q\} = \mathbb{R}^n$, so every u equals some $p + q$.

Step 2 – two decompositions. Suppose $u = p + q = p' + q'$ with $p, p' \in P$ and $q, q' \in Q$.

Step 3 – rearrange. $p - p' = q' - q$.

Step 4 – where does it live?. The left side is in P (P is a subspace, closed under subtraction). The right side is in Q. So the common vector lies in $P \cap Q = \{\mathbf{0}\}$.

Step 5 – conclude. $p - p' = \mathbf{0}$ and $q' - q = \mathbf{0}$: $p = p'$, $q = q'$. Unique. ■

Step 6 – example. $P = L\{(1,0)\}$, $Q = L\{(0,1)\}$ in $\mathbb{R}^2$: $(3,2) = (3,0) + (0,2)$ and no other split works.

Question 3.14 (Exercise 3.3.19, item 11)

Let $P = L\{(1,-1,0),(1,1,0)\}$ and $Q = L\{(1,1,1),(1,2,1)\}$ be vector subspaces of $\mathbb{R}^3$. Show that $P + Q = \mathbb{R}^3$ and $P \cap Q \neq \{\mathbf{0}\}$. Show that there exists a vector $u \in \mathbb{R}^3$ such that u cannot be written uniquely in the form $u = u_P + u_Q$ where $u_P \in P$ and $u_Q \in Q$.

Solution.

Step 1 – $P + Q = \mathbb{R}^3$. $P + Q = L$ of the four vectors, which contains $(1,-1,0), (1,1,0), (1,1,1)$. $\det \begin{pmatrix} 1 & -1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} = 1 \cdot (1) - (-1) \cdot (1) + 0 = 2 \neq 0$, so these three are a basis of $\mathbb{R}^3$. Hence $P + Q = \mathbb{R}^3$.

Step 2 – dimensions. $\dim P = 2$ and $\dim Q = 2$ (each pair is not proportional). So $\dim P + \dim Q = 4 > 3 = \dim \mathbb{R}^3$.

Step 3 – Grassmann. $\dim(P \cap Q) = 4 - 3 = 1 \neq 0$. So $P \cap Q \neq \{\mathbf{0}\}$.

Step 4 – find a common vector. $(1,2,1) - (1,1,1) = (0,1,0) \in Q$. Also $(0,1,0) = \frac{1}{2}(1,1,0) - \frac{1}{2}(1,-1,0) \in P$. So $P \cap Q = L\{(0,1,0)\}$.

Step 5 – a non-unique u. Take $u = (1,1,1)$. *First way:* $u_P = (0,0,0)$, $u_Q = (1,1,1)$. *Second way:* $u_P = (0,1,0) \in P$ and $u_Q = (1,0,1) = 2(1,1,1) - (1,2,1) \in Q$; check $(0,1,0) + (1,0,1) = (1,1,1)$ ✓. Two different decompositions. ■

Step 6 – why. If $u = p + q$ then also $u = (p + w) + (q - w)$ for any $w \in P \cap Q$. A non-zero overlap

destroys uniqueness (compare Question 3.13).

Question 3.15 (Exercise 3.3.19, item 12)

Recall the vector space $P_4(\mathbb{R})$. Is the set $W = \{p(x) \in P_4(\mathbb{R}) : \ p(-1) = p(1) = 0\}$ a subspace of $P_4(\mathbb{R})$? If yes, find its dimension.

Solution.

Step 1 – subspace. The zero polynomial has 0 at ± 1. If $p(\pm 1) = 0$, $q(\pm 1) = 0$ and $a, b \in \mathbb{R}$, then $(ap + bq)(1) = a \cdot 0 + b \cdot 0 = 0$ and similarly at -1. So W is a subspace.

Step 2 – what does $p \in W$ look like?. A polynomial with roots 1 and -1 is divisible by $(x-1)(x+1) = x^2 - 1$ (Factor Theorem). Since $\deg p \leq 4$: $p(x) = (x^2 - 1)(a + bx + cx^2)$ with $a, b, c \in \mathbb{R}$.

Step 3 – expand. $p = a(x^2 - 1) + b(x^3 - x) + c(x^4 - x^2)$. So $\{x^2 - 1, \ x^3 - x, \ x^4 - x^2\}$ spans W.

Step 4 – independent. If $a(x^2 - 1) + b(x^3 - x) + c(x^4 - x^2) = 0$: the coefficient of x^4 is $c = 0$; of x^3 is $b = 0$; then $a(x^2 - 1) = 0$ gives $a = 0$.

Step 5 – conclude. Basis $\{x^2 - 1, \ x^3 - x, \ x^4 - x^2\}$, $\dim W = 3$.

Step 6 – cross-check by counting. $\dim P_4(\mathbb{R}) = 5$. With $p = a_0 + \cdots + a_4 x^4$: $p(1) = 0$ and $p(-1) = 0$ give (add and subtract) $a_0 + a_2 + a_4 = 0$ and $a_1 + a_3 = 0$: two independent equations, so $5 - 2 = 3$ ✓. (Example: $x^4 - 1 = (x^2 - 1) + (x^4 - x^2) \in W$.)

Question 3.16 (Exercise 3.3.19, item 13)

Let V be the set of all 2×2 matrices with complex entries and $a_{11} + a_{22} = 0$. Show that V is a real vector space. Find its basis. Also let $W = \{A \in V : \ a_{21} = -a_{12}\}$. Show that W is a vector subspace of V, and find its dimension.

Solution.

Step 1 – V is a real vector space. The set $M_2(\mathbb{C})$ of all complex 2×2 matrices is a vector space over $\mathbb{R}$ (add matrices, multiply by real numbers; all rules hold entry by entry). V contains the zero matrix and, for $A, B \in V$ and real α, β, $(\alpha A + \beta B)_{11} + (\alpha A + \beta B)_{22} = \alpha(a_{11} + a_{22}) + \beta(b_{11} + b_{22}) = 0$. So V is a subspace of $M_2(\mathbb{C})(\mathbb{R})$, hence a real vector space.

Step 2 – what does $A \in V$ look like?. $A = \begin{pmatrix} a & b \\ c & -a \end{pmatrix}$ with $a, b, c \in \mathbb{C}$. Write $a = a_1 + ia_2$, $b = b_1 + ib_2$, $c = c_1 + ic_2$ ($a_1, a_2, b_1, b_2, c_1, c_2$ real): *six real numbers.*

Step 3 – split. $A = a_1 \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} + a_2 \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} + b_1 \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + b_2 \begin{pmatrix} 0 & i \\ 0 & 0 \end{pmatrix} + c_1 \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 & 0 \\ i & 0 \end{pmatrix}$.

Step 4 – independent over $\mathbb{R}$. If this is the zero matrix: entry $(1,1)$ gives $a_1 + ia_2 = 0$, so $a_1 = a_2 = 0$; entry $(1,2)$: $b_1 + ib_2 = 0$; entry $(2,1)$: $c_1 + ic_2 = 0$. All six real coefficients vanish.

Step 5 – basis of V. These six matrices: $\dim_{\mathbb{R}} V = 6$. (Over $\mathbb{C}$ the dimension would be 3; here only real scalars are allowed.)

Step 6 – W is a subspace. W: $a_{11} + a_{22} = 0$ and $a_{21} - (-a_{12}) = a_{21} + a_{12} = 0$. Both conditions are linear, and $0 \in W$. If $A, B \in W$ and $\alpha, \beta \in \mathbb{R}$, then $\alpha A + \beta B$ satisfies both conditions. So W is a subspace of V.

Step 7 – basis and dimension of W. $A = \begin{pmatrix} a & b \\ -b & -a \end{pmatrix}$, $a, b \in \mathbb{C}$: four real numbers. $A = a_1 \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} + a_2 \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} + b_1 \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + b_2 \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}$, independent as in Step 4. $\dim_{\mathbb{R}} W = 4$.

Question 3.17 (Exercise 3.3.19, item 14)

Let $A = \begin{pmatrix} 1 & 2 & 1 & 3 & 2 \\ 0 & 2 & 2 & 2 & 4 \\ 2 & -2 & 4 & 0 & 8 \\ 4 & 2 & 5 & 6 & 10 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 4 & 0 & 6 \\ -1 & 0 & -2 & 5 \\ -3 & -5 & 1 & -4 \\ -1 & -1 & 1 & 2 \end{pmatrix}$. For both A and B find: (a) their row-reduced echelon forms; (b) matrices P_1 and P_2 such that P_1A and P_2B are in row-reduced form; (c) a basis each for the row spaces of A and B; (d) a basis each for the range spaces (column spaces) of A and B; (e) bases of the null spaces of A and B; (f) the dimensions of all the vector subspaces so obtained.

Solution for A. Do the row operations on $[A \mid I_4]$: the left half becomes the RREF, the right half becomes P_1. Below I show the left half (the matrix A changing) and, after each step, the right half (the identity changing).

Step 1 – clear column 1. $R_3 \to R_3 - 2R_1$, $R_4 \to R_4 - 4R_1$:

$$A \to \begin{pmatrix} 1 & 2 & 1 & 3 & 2 \\ 0 & 2 & 2 & 2 & 4 \\ 0 & -6 & 2 & -6 & 4 \\ 0 & -6 & 1 & -6 & 2 \end{pmatrix}, \qquad I \to \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -2 & 0 & 1 & 0 \\ -4 & 0 & 0 & 1 \end{pmatrix}.$$

Step 2 – pivot in column 2. $R_2 \to \frac{1}{2}R_2$, then $R_1 \to R_1 - 2R_2$:

$$A \to \begin{pmatrix} 1 & 0 & -1 & 1 & -2 \\ 0 & 1 & 1 & 1 & 2 \\ 0 & -6 & 2 & -6 & 4 \\ 0 & -6 & 1 & -6 & 2 \end{pmatrix}, \qquad I \to \begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 \\ -2 & 0 & 1 & 0 \\ -4 & 0 & 0 & 1 \end{pmatrix}.$$

Step 3 – clear column 2. $R_3 \to R_3 + 6R_2$, $R_4 \to R_4 + 6R_2$:

$$A \to \begin{pmatrix} 1 & 0 & -1 & 1 & -2 \\ 0 & 1 & 1 & 1 & 2 \\ 0 & 0 & 8 & 0 & 16 \\ 0 & 0 & 7 & 0 & 14 \end{pmatrix}, \qquad I \to \begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 \\ -2 & 3 & 1 & 0 \\ -4 & 3 & 0 & 1 \end{pmatrix}.$$

Step 4 – pivot in column 3. $R_3 \to \frac{1}{8}R_3$; then $R_1 \to R_1 + R_3$, $R_2 \to R_2 - R_3$, $R_4 \to R_4 - 7R_3$:

$$A \to D_A = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \qquad I \to P_1 = \frac{1}{8}\begin{pmatrix} 6 & -5 & 1 & 0 \\ 2 & 1 & -1 & 0 \\ -2 & 3 & 1 & 0 \\ -18 & 3 & -7 & 8 \end{pmatrix}.$$

(a) RREF of A is D_A (rank 3, pivot columns $1, 2, 3$). **(b)** P_1 above, and $P_1A = D_A$. (Check row 1 of P_1A: $\frac{1}{8}(6R_1 - 5R_2 + R_3) = \frac{1}{8}\big((6,12,6,18,12) - (0,10,10,10,20) + (2,-2,4,0,8)\big) = \frac{1}{8}(8,0,0,8,0) = (1,0,0,1,0)$ ✓. P_1 is not unique, because the zero row of D_A leaves freedom in the last row.)

Step 5 – (c) row space. The non-zero rows of D_A: $\{(1,0,0,1,0), (0,1,0,1,0), (0,0,1,0,2)\}$. dim = **3.**

Step 6 – (d) column space. The pivot columns $1, 2, 3$ of the *original* A: $\{(1,0,2,4), (2,2,-2,2), (1,2,4,5)\}$. dim = **3.** (Why enough: column 4 = col 1 + col 2 = $(3,2,0,6)$ ✓, column 5 = 2· col 3 = $(2,4,8,10)$ ✓.)

Step 7 – (e) null space. $D_A x = \mathbf{0}$: $x_1 + x_4 = 0$, $x_2 + x_4 = 0$, $x_3 + 2x_5 = 0$. Free: $x_4 = s$, $x_5 = t$. $x = (-s, -s, -2t, s, t) = s(-1, -1, 0, 1, 0) + t(0, 0, -2, 0, 1)$. **Basis** $\{(-1, -1, 0, 1, 0), (0, 0, -2, 0, 1)\}$, $\dim = 2 = 5 - 3$. (Check: $A(-1, -1, 0, 1, 0)^t = (-1 - 2 + 3,\ -2 + 2,\ -2 + 2,\ -4 - 2 + 6) = \mathbf{0}$ ✓.)

Step 8 – (f) dimensions for A. Row space 3, column space 3 (equal, as Prop. 3.3.20 says), null space 2; $3 + 2 = 5 =$ number of columns ✓.

Solution for B.

Step 1 – pivot in column 1. $R_1 \to \frac{1}{2}R_1$; then $R_2 \to R_2 + R_1$, $R_3 \to R_3 + 3R_1$, $R_4 \to R_4 + R_1$:

$$B \to \begin{pmatrix} 1 & 2 & 0 & 3 \\ 0 & 2 & -2 & 8 \\ 0 & 1 & 1 & 5 \\ 0 & 1 & 1 & 5 \end{pmatrix}, \qquad I \to \begin{pmatrix} \frac{1}{2} & 0 & 0 & 0 \\ \frac{1}{2} & 1 & 0 & 0 \\ \frac{3}{2} & 0 & 1 & 0 \\ \frac{1}{2} & 0 & 0 & 1 \end{pmatrix}.$$

Step 2 – pivot in column 2. $R_2 \to \frac{1}{2}R_2$; then $R_1 \to R_1 - 2R_2$, $R_3 \to R_3 - R_2$, $R_4 \to R_4 - R_2$:

$$B \to \begin{pmatrix} 1 & 0 & 2 & -5 \\ 0 & 1 & -1 & 4 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 2 & 1 \end{pmatrix}, \qquad I \to \begin{pmatrix} 0 & -1 & 0 & 0 \\ \frac{1}{4} & \frac{1}{2} & 0 & 0 \\ \frac{5}{4} & -\frac{1}{2} & 1 & 0 \\ \frac{1}{4} & -\frac{1}{2} & 0 & 1 \end{pmatrix}.$$

Step 3 – pivot in column 3. $R_3 \to \frac{1}{2}R_3$; then $R_4 \to R_4 - 2R_3$, $R_1 \to R_1 - 2R_3$, $R_2 \to R_2 + R_3$:

$$B \to D_B = \begin{pmatrix} 1 & 0 & 0 & -6 \\ 0 & 1 & 0 & \frac{9}{2} \\ 0 & 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad I \to P_2 = \frac{1}{8}\begin{pmatrix} -10 & -4 & -8 & 0 \\ 7 & 2 & 4 & 0 \\ 5 & -2 & 4 & 0 \\ -8 & 0 & -8 & 8 \end{pmatrix}.$$

(a) RREF D_B (rank 3, pivots $1, 2, 3$). **(b)** $P_2 B = D_B$. (Check row 1: $\frac{1}{8}(-10R_1 - 4R_2 - 8R_3) = \frac{1}{8}\big((-20, -40, 0, -60) + (4, 0, 8, -20) + (24, 40, -8, 32)\big) = \frac{1}{8}(8, 0, 0, -48) = (1, 0, 0, -6)$ ✓.)

Step 4 – (c) row space. $\{(1, 0, 0, -6), (0, 1, 0, \frac{9}{2}), (0, 0, 1, \frac{1}{2})\}$. $\dim = 3$.

Step 5 – (d) column space. Pivot columns $1, 2, 3$ of the original B: $\{(2, -1, -3, -1), (4, 0, -5, -1), (0, -2, 1, 1)\}$. $\dim = 3$. (Column $4 = -6\,c_1 + \frac{9}{2}c_2 + \frac{1}{2}c_3 = (6, 5, -4, 2)$ ✓.)

Step 6 – (e) null space. $x_1 - 6x_4 = 0$, $x_2 + \frac{9}{2}x_4 = 0$, $x_3 + \frac{1}{2}x_4 = 0$; free $x_4 = t$: $x = t(6, -\frac{9}{2}, -\frac{1}{2}, 1)$. Take $t = 2$ to clear fractions: **basis** $\{(12, -9, -1, 2)\}$, $\dim = 1 = 4 - 3$. (Check: $B(12, -9, -1, 2)^t = (24 - 36 + 12,\ -12 + 2 + 10,\ -36 + 45 - 1 - 8,\ -12 + 9 - 1 + 4) = \mathbf{0}$ ✓.)

Step 7 – (f) dimensions for B. Row 3, column 3, null 1; $3 + 1 = 4$ columns ✓.

Question 3.18 (Exercise 3.3.19, item 15)

Let $M(n, \mathbb{R})$ denote the space of all $n \times n$ real matrices. For the sets given below, check that they are subspaces of $M(n, \mathbb{R})$ and also find their dimension: (a) $sl(n, \mathbb{R}) = \{A \in M(n, \mathbb{R}) :\ \mathrm{tr}(A) = 0\}$, where $\mathrm{tr}(A)$ is the trace of A; (b) $S(n, \mathbb{R}) = \{A \in M(n, \mathbb{R}) :\ A = A^t\}$; (c) $A(n, \mathbb{R}) = \{A \in M(n, \mathbb{R}) :\ A + A^t = 0\}$.

Solution. $\dim M(n, \mathbb{R}) = n^2$ (basis E_{ij}).

Step 1 – (a) subspace. $\mathrm{tr}(\alpha A + \beta B) = \alpha\,\mathrm{tr}\,A + \beta\,\mathrm{tr}\,B = 0$ and $\mathrm{tr}(0) = 0$.

Step 2 – (a) dimension. One equation $a_{11} + \cdots + a_{nn} = 0$ on n^2 unknowns: $n^2 - 1$ free. Basis: E_{ij} for

$i \neq j$ ($n^2 - n$ matrices) and $E_{ii} - E_{nn}$ for $1 \leq i \leq n-1$ ($n-1$ matrices). Count: $n^2 - n + n - 1 = n^2 - 1$. $\dim sl(n, \mathbb{R}) = n^2 - 1$. ($n = 2$: $\{E_{12}, E_{21}, E_{11} - E_{22}\}$, dim 3.)

Step 3 – (b) subspace. If $A = A^t$, $B = B^t$ then $(\alpha A + \beta B)^t = \alpha A^t + \beta B^t = \alpha A + \beta B$. And $0^t = 0$.

Step 4 – (b) dimension. As in Question 3.9: $\{E_{ii}\} \cup \{E_{ij} + E_{ji} : i < j\}$, $\dim S(n, \mathbb{R}) = \dfrac{n(n+1)}{2}$.

Step 5 – (c) subspace. If $A^t = -A$ and $B^t = -B$ then $(\alpha A + \beta B)^t = -(\alpha A + \beta B)$. And $0 \in A(n, \mathbb{R})$.

Step 6 – (c) dimension. $A + A^t = 0$ means $a_{ji} = -a_{ij}$. For $i = j$: $2a_{ii} = 0$, so diagonal entries are 0. Free: the entries above the diagonal. Basis: $\{E_{ij} - E_{ji} : i < j\}$. $\dim A(n, \mathbb{R}) = \dfrac{n(n-1)}{2}$. ($n = 3$: $\begin{pmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{pmatrix}$, 3 free numbers.)

Step 7 – a beautiful check. $\dim S + \dim A = \frac{n(n+1)}{2} + \frac{n(n-1)}{2} = n^2 = \dim M(n, \mathbb{R})$. Indeed $S \cap A = \{\mathbf{0}\}$ (a matrix both symmetric and skew has $a_{ij} = a_{ji} = -a_{ij}$, so 0), and every matrix splits uniquely as $A = \frac{A + A^t}{2} + \frac{A - A^t}{2}$ (symmetric + skew-symmetric): $M(n, \mathbb{R}) = S \oplus A$, exactly the situation of Question 3.13.

3.4 Ordered Bases and Coordinates

3.4.1 Why the *order* of a basis matters

Picture it: an address needs an order

The point $(3, 2)$ and the point $(2, 3)$ are different, although they use the same two numbers. A coordinate list is an *address*, and an address only makes sense if we agree which number refers to which direction. So far a basis was a *set* $\{u_1, \ldots, u_n\}$, where order is meaningless. To use a basis as a coordinate system, we fix an order: "first direction, second direction, ...".

Definition – Ordered basis (Definition 3.4.1)

An **ordered basis** of an n-dimensional $V(\mathbb{F})$ is a basis $\{u_1, \ldots, u_n\}$ together with a fixed order (a first vector, a second vector, ...). We write it with round brackets: $(u_1, u_2, \ldots, u_n)$.

Plain English

Same vectors, different order, **different ordered basis**. For example (u_1, u_2, u_3), (u_2, u_3, u_1) and (u_3, u_2, u_1) are three different ordered bases, even though they contain the same three vectors. Square or curly brackets { } = set (no order); round brackets ( ) = ordered list.

Example 3.4.2 of your notes – $P_2(\mathbb{R})$ with two different orders

$\{1 - x,\ 1 + x,\ x^2\}$ is a basis of $P_2(\mathbb{R})$. For any $a_0 + a_1x + a_2x^2$:

$$a_0 + a_1x + a_2x^2 = \frac{a_0 - a_1}{2}(1 - x) + \frac{a_0 + a_1}{2}(1 + x) + a_2x^2.$$

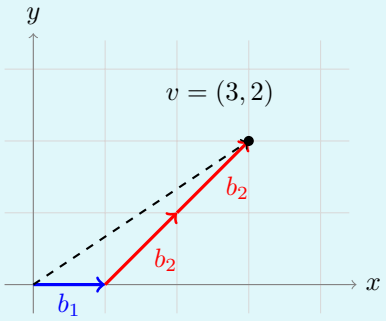
Check: $\frac{a_0-a_1}{2}(1 - x) + \frac{a_0+a_1}{2}(1 + x) = \frac{(a_0-a_1)+(a_0+a_1)}{2} + x\,\frac{-(a_0-a_1)+(a_0+a_1)}{2} = a_0 + a_1x$ ✓.
Concrete: $p = 3 + x + 2x^2$. Then $\frac{3-1}{2} = 1$ and $\frac{3+1}{2} = 2$, so $p = 1 \cdot (1 - x) + 2 \cdot (1 + x) + 2 \cdot x^2$ (check: $(1 - x) + (2 + 2x) + 2x^2 = 3 + x + 2x^2$ ✓).

- Order $(1 - x,\ 1 + x,\ x^2)$: first component $\frac{a_0-a_1}{2}$, second $\frac{a_0+a_1}{2}$, third a_2. For p: $(1, 2, 2)$.
- Order $(1 + x,\ 1 - x,\ x^2)$: first component $\frac{a_0+a_1}{2}$, second $\frac{a_0-a_1}{2}$, third a_2. For p: $(2, 1, 2)$.

Same polynomial, same vectors in the basis, but the list of numbers changed because the order changed.

A picture of a "skew" coordinate system in $\mathbb{R}^2$

Take the ordered basis $B = (b_1, b_2)$ with $b_1 = (1, 0)$ and $b_2 = (1, 1)$, and the point $v = (3, 2)$. To *reach* v using b_1, b_2 as the only allowed steps: go 1 step along b_1, then 2 steps along b_2. So $v = 1 \cdot b_1 + 2 \cdot b_2$ and the coordinates are $(1, 2)$.



$$v = 1 \cdot b_1 + 2 \cdot b_2$$
$$[v]_B = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

3.4.2 Coordinates of a vector (Definition 3.4.3)

Definition – Coordinates (Definition 3.4.3)

Let $B = (v_1, \dots, v_n)$ be an ordered basis of $V(\mathbb{F})$ and $v \in V$. If $v = \beta_1 v_1 + \cdots + \beta_n v_n$, the list $(\beta_1, \dots, \beta_n)$ is the **coordinate vector** of v with respect to B. It is written as the column

$$[v]_B = (\beta_1, \dots, \beta_n)^t = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_n \end{pmatrix}.$$

Three facts that are all you need

1. **Existence:** because B spans, every v has coordinates.
2. **Uniqueness:** because B is independent, they are unique (Remark 3.3.2). That is why "$[v]_B$" is a meaningful name.
3. **They depend on the ordered basis.** Change B, or just its order, and $[v]_B$ changes. If B' is B with two vectors swapped, $[v]_{B'}$ is $[v]_B$ with the same two entries swapped.

Recipe 8 – how to *find* $[v]_B$ (never guess; always solve)

Step 1. Write $v = \beta_1 v_1 + \cdots + \beta_n v_n$ with unknown β_i.
Step 2. Compare coordinates: this is a linear system $\beta_1 v_1 + \cdots + \beta_n v_n = v$, i.e. the matrix with columns $v_1, \dots, v_n$ times β equals v.
Step 3. Solve it (row-reduce $[\, v_1 \ \cdots \ v_n \mid v \,]$). The unique solution is $[v]_B$.
Step 4. Check: rebuild $\beta_1 v_1 + \cdots + \beta_n v_n$ and compare with v.

Example 3.4.4 of your notes – one vector, three coordinate systems

$V = \mathbb{R}^3$, $B_1 = \big((1,0,0),(0,1,0),(0,0,1)\big)$, $B_2 = \big((1,0,0),(1,1,0),(1,1,1)\big)$, $B_3 = \big((1,1,1),(1,1,0),(1,0,0)\big)$, and $u = (1,-1,1)$.

Step 1 – $[u]_{B_1}$. $(1,-1,1) = 1\cdot(1,0,0) + (-1)\cdot(0,1,0) + 1\cdot(0,0,1)$. So $[u]_{B_1} = (1,-1,1)^t$ (for the standard basis the coordinates are the entries themselves).

Step 2 – $[u]_{B_2}$: solve. $a(1,0,0) + b(1,1,0) + c(1,1,1) = (a+b+c,\ b+c,\ c) = (1,-1,1)$. Third coordinate: $c = 1$. Second: $b + c = -1$, so $b = -2$. First: $a + b + c = 1$, so $a = 1 + 2 - 1 = 2$. $[u]_{B_2} = (2,-2,1)^t$. *Check:* $2(1,0,0) - 2(1,1,0) + (1,1,1) = (2-2+1,\ -2+1,\ 1) = (1,-1,1)$ ✓.

Step 3 – $[u]_{B_3}$: solve. $a(1,1,1) + b(1,1,0) + c(1,0,0) = (a+b+c,\ a+b,\ a) = (1,-1,1)$. Third: $a = 1$. Second: $a + b = -1$, so $b = -2$. First: $a + b + c = 1$, so $c = 2$. $[u]_{B_3} = (1,-2,2)^t$. *Check:* $(1,1,1) - 2(1,1,0) + 2(1,0,0) = (1-2+2,\ 1-2,\ 1) = (1,-1,1)$ ✓.

Step 4 – what to notice. B_3 is B_2 written backwards, and $[u]_{B_3} = (1,-2,2)$ is $[u]_{B_2} = (2,-2,1)$ written backwards. One vector, three different coordinate lists.

Extra Example 3.4-A – coordinates in a polynomial space

$B = (1,\ 1+x,\ 1+x+x^2)$ is an ordered basis of $P_2(\mathbb{R})$. Find $[p]_B$ for $p = 4 + 3x + 2x^2$.
Solve $\beta_1 \cdot 1 + \beta_2(1+x) + \beta_3(1+x+x^2) = 4 + 3x + 2x^2$. Compare coefficients. x^2: $\beta_3 = 2$. x: $\beta_2 + \beta_3 = 3$, so $\beta_2 = 1$. Constant: $\beta_1 + \beta_2 + \beta_3 = 4$, so $\beta_1 = 1$. So $[p]_B = (1,1,2)^t$. *Check:* $1 + (1+x) + 2(1+x+x^2) = 4 + 3x + 2x^2$ ✓.

Extra Example 3.4-B – coordinates in a matrix space

In $M_2(\mathbb{R})$ take $B = (E_{11},\ E_{22},\ E_{12}+E_{21},\ E_{12}-E_{21})$. Find $[A]_B$ for $A = \begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix}$.

$A = \beta_1 E_{11} + \beta_2 E_{22} + \beta_3(E_{12}+E_{21}) + \beta_4(E_{12}-E_{21})$. Entry $(1,1)$: $\beta_1 = 2$. Entry $(2,2)$: $\beta_2 = 3$. Entry $(1,2)$: $\beta_3 + \beta_4 = 5$. Entry $(2,1)$: $\beta_3 - \beta_4 = 1$. Add: $2\beta_3 = 6$, $\beta_3 = 3$; then $\beta_4 = 2$. $[A]_B = (2,3,3,2)^t$. *Check:* entry $(1,2)$: $3+2=5$ ✓; entry $(2,1)$: $3-2=1$ ✓.

3.4.3 Change of basis (Theorem 3.4.5)

The question

You know the coordinates of v in one ordered basis B_2. Someone else uses another ordered basis B_1. Is there a *machine* that turns your coordinates $[v]_{B_2}$ into theirs $[v]_{B_1}$ – the same machine for every v? Yes, a single matrix A, with $[v]_{B_1} = A\,[v]_{B_2}$. Its columns are simply the *vectors of B_2, written in the coordinates of B_1*.

First, a tiny example ($\mathbb{R}^2$): build the machine and test it

$B_1 = (u_1, u_2) = \big((1,0),(1,1)\big)$ and $B_2 = (v_1, v_2) = \big((1,2),(0,1)\big)$.

Step 1 – write v_1 in B_1. $(1,2) = a(1,0) + b(1,1) = (a+b,\ b)$: $b = 2$, $a = -1$. So $[v_1]_{B_1} = (-1,2)^t$.

Step 2 – write v_2 in B_1. $(0,1) = (a+b, b)$: $b = 1$, $a = -1$. So $[v_2]_{B_1} = (-1,1)^t$.

Step 3 – the matrix. Columns $= [v_1]_{B_1}, [v_2]_{B_1}$: $A = \begin{pmatrix} -1 & -1 \\ 2 & 1 \end{pmatrix}$.

Step 4 – test it. Take v with $[v]_{B_2} = (3,4)^t$, i.e. $v = 3v_1 + 4v_2 = 3(1,2) + 4(0,1) = (3,10)$. Now $A[v]_{B_2} = \begin{pmatrix} -1 & -1 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} -7 \\ 10 \end{pmatrix}$. And directly: $(3,10) = a(1,0) + b(1,1)$ gives $b = 10$, $a = -7$, so $[v]_{B_1} = (-7,10)^t$ ✓. The machine works.

Theorem 3.4.5

Let V be n-dimensional with ordered bases $B_1 = (u_1, \ldots, u_n)$ and $B_2 = (v_1, \ldots, v_n)$. Let $A = \big[\,[v_1]_{B_1},\ [v_2]_{B_1},\ \ldots,\ [v_n]_{B_1}\,\big]$ (the i-th column is the coordinate vector of v_i with respect to B_1). Then for every $v \in V$:

$$[v]_{B_1} = A\,[v]_{B_2}.$$

Proof in small steps – Theorem 3.4.5 (the tiny example is in brackets)

Step 1 – write each v_i in B_1. Since B_1 is a basis, $v_i = a_{1i}u_1 + a_{2i}u_2 + \cdots + a_{ni}u_n = \sum_{j=1}^{n} a_{ji}u_j$. So $[v_i]_{B_1} = (a_{1i}, \ldots, a_{ni})^t$ is the i-th column of $A = (a_{ji})$. [$v_1 = -1u_1 + 2u_2$, $v_2 = -1u_1 + 1u_2$.]

Step 2 – take any v. Let $[v]_{B_2} = (\alpha_1, \ldots, \alpha_n)^t$, i.e. $v = \sum_i \alpha_i v_i$. [$\alpha = (3,4)$.]

Step 3 – substitute Step 1. $v = \sum_i \alpha_i \Big(\sum_j a_{ji}u_j\Big) = \sum_j \Big(\sum_i a_{ji}\alpha_i\Big)u_j$. [$v = 3(-u_1 + 2u_2) + 4(-u_1 + u_2) = (-3-4)u_1 + (6+4)u_2$.]

Step 4 – read the B_1-coordinates. By uniqueness of coordinates, the coefficient of u_j is the j-th entry of $[v]_{B_1}$: $[v]_{B_1} = \Big(\sum_i a_{1i}\alpha_i,\ \ldots,\ \sum_i a_{ni}\alpha_i\Big)^t$. [$(-7,10)$.]

Step 5 – recognise a matrix product. The j-th entry is the j-th row of A times the column α. So $[v]_{B_1} = A\,[v]_{B_2}$. ■

Direction – the #1 place where marks are lost

A turns B_2-coordinates into B_1-coordinates. Columns of $A =$ the *new/second* basis vectors written in the *first* basis. Memory trick: in $[v]_{B_1} = A[v]_{B_2}$ the "B_2" next to A is what A *eats*; the "B_1" on the left is what it *produces*. To go the other way use the inverse: $[v]_{B_2} = A^{-1}[v]_{B_1}$. A is always invertible: it converts coordinates in a reversible way (and its columns, the coordinate vectors of the independent set B_2, are independent).

Recipe 9 – build the change-of-basis matrix in one go

Put the vectors of B_1 as the columns of a matrix P_1 and the vectors of B_2 as the columns of P_2 (in the standard coordinates of $\mathbb{R}^n$). Row-reduce $[\,P_1 \mid P_2\,]$. When the left half becomes I, the right half is A (with $[v]_{B_1} = A[v]_{B_2}$). *Why:* solving $P_1 x =$ (each column of P_2) finds the B_1-coordinates of each B_2-vector all at once. In short, $A = P_1^{-1}P_2$.

Tiny example: $[P_1|P_2] = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 2 & 1 \end{pmatrix} \to \begin{pmatrix} 1 & 0 & -1 & -1 \\ 0 & 1 & 2 & 1 \end{pmatrix}$; the right half is the matrix A found above ✓.

Example 3.4.6 of your notes – all four items, slowly

$B_1 = \big((1,0,0),(1,1,0),(1,1,1)\big)$ and $B_2 = \big((1,1,1),(1,-1,1),(1,1,0)\big)$ are ordered bases of $\mathbb{R}^3$.

Step 1 – formula for $[(x,y,z)]_{B_1}$. Solve $a(1,0,0)+b(1,1,0)+c(1,1,1) = (a+b+c,\ b+c,\ c) = (x,y,z)$. So $c = z$, $b = y - z$, $a = x - y$:

$$[(x,y,z)]_{B_1} = (x-y,\ y-z,\ z)^t.$$

Step 2 – formula for $[(x,y,z)]_{B_2}$. Solve $a(1,1,1)+b(1,-1,1)+c(1,1,0) = (x,y,z)$, i.e. $a+b+c = x$, $a-b+c = y$, $a+b = z$. Subtract the first two: $2b = x - y$, so $b = \frac{x-y}{2}$. Third: $a = z - b = \frac{y-x}{2} + z$. First: $c = x - a - b = x - z$:

$$[(x,y,z)]_{B_2} = \left(\tfrac{y-x}{2} + z,\ \tfrac{x-y}{2},\ x - z\right)^t.$$

Check with $v = (1,-1,1)$: $\left(\frac{-2}{2}+1,\ \frac{2}{2},\ 0\right) = (0,1,0)$, and indeed $v = (1,-1,1)$ is the second vector of B_2 ✓.

Step 3 – the matrix A (Theorem 3.4.5). Columns are $[v_i]_{B_1}$, using the formula of Step 1: $[(1,1,1)]_{B_1} = (0,0,1)^t$, $[(1,-1,1)]_{B_1} = (2,-2,1)^t$, $[(1,1,0)]_{B_1} = (0,1,0)^t$. So

$$A = \begin{pmatrix} 0 & 2 & 0 \\ 0 & -2 & 1 \\ 1 & 1 & 0 \end{pmatrix}.$$

Step 4 – verify $[(x,y,z)]_{B_1} = A[(x,y,z)]_{B_2}$. With $c_1 = \frac{y-x}{2} + z$, $c_2 = \frac{x-y}{2}$, $c_3 = x - z$: first row $2c_2 = x - y$ ✓; second row $-2c_2 + c_3 = -(x-y) + x - z = y - z$ ✓; third row $c_1 + c_2 = \frac{y-x}{2} + z + \frac{x-y}{2} = z$ ✓.

Step 5 – the inverse. $\det A = 2 \neq 0$, so A is invertible and $[(x,y,z)]_{B_2} = A^{-1}[(x,y,z)]_{B_1}$ with $A^{-1} = \begin{pmatrix} -\frac{1}{2} & 0 & 1 \\ \frac{1}{2} & 0 & 0 \\ 1 & 1 & 0 \end{pmatrix}$. *Check on* $(x-y,\ y-z,\ z)$: first entry $-\frac{1}{2}(x-y) + z = \frac{y-x}{2} + z$ ✓; second $\frac{1}{2}(x-y)$ ✓; third $(x-y) + (y-z) = x - z$ ✓.

The whole section in five lines

1. Fix an order of a basis: an *ordered basis* B.
2. $[v]_B$ = the unique numbers with $v = \beta_1 v_1 + \cdots + \beta_n v_n$. Find them by *solving*, not guessing.
3. Different ordered bases give different coordinate lists for the same vector.
4. $[v]_{B_1} = A[v]_{B_2}$, where column i of A is $[v_i]_{B_1}$ (v_i the i-th vector of B_2).
5. A is invertible, and A^{-1} goes back: $[v]_{B_2} = A^{-1}[v]_{B_1}$.

3.4.4 Questions for Section 3.4 (Exercise 3.4.7 of your notes)

Question 4.1 (Exercise 3.4.7, item 1)

Determine the coordinates of the vectors $(1,2,1)$ and $(4,-2,2)$ with respect to the ordered basis $B = \big((2,1,0),(2,1,1),(2,2,1)\big)$ of $\mathbb{R}^3$.

Solution.

Step 1 – set up for a general vector (p,q,r). $a(2,1,0)+b(2,1,1)+c(2,2,1) = (2a+2b+2c,\ a+b+2c,\ b+c) = (p,q,r)$.

Step 2 – solve once for all. Equation 1 divided by 2: $a+b+c=\frac{p}{2}$. Subtract this from equation 2: $c = q - \frac{p}{2}$. Equation 3: $b = r - c = r - q + \frac{p}{2}$. Then $a = \frac{p}{2} - b - c = \frac{p}{2} - r$. So

$$[(p,q,r)]_B = \left(\tfrac{p}{2} - r,\quad r - q + \tfrac{p}{2},\quad q - \tfrac{p}{2}\right)^t.$$

(The solution is unique for every (p,q,r), which also shows B really is a basis; $\det = -2 \neq 0$.)

Step 3 – $(1,2,1)$. $p=1, q=2, r=1$: $a = \frac{1}{2} - 1 = -\frac{1}{2}$, $b = 1-2+\frac{1}{2} = -\frac{1}{2}$, $c = 2 - \frac{1}{2} = \frac{3}{2}$. $[(1,2,1)]_B = (-\frac{1}{2}, -\frac{1}{2}, \frac{3}{2})^t$. *Check:* $-\frac{1}{2}(2,1,0) - \frac{1}{2}(2,1,1) + \frac{3}{2}(2,2,1) = (-1-1+3,\ -\frac{1}{2}-\frac{1}{2}+3,\ -\frac{1}{2}+\frac{3}{2}) = (1,2,1)$ ✓.

Step 4 – $(4,-2,2)$. $p=4, q=-2, r=2$: $a = 2-2 = 0$, $b = 2+2+2 = 6$, $c = -2-2 = -4$. $[(4,-2,2)]_B = (0,6,-4)^t$. *Check:* $6(2,1,1) - 4(2,2,1) = (12-8,\ 6-8,\ 6-4) = (4,-2,2)$ ✓.

Question 4.2 (Exercise 3.4.7, item 2)

Consider the vector space $P_3(\mathbb{R})$. (a) Show that $B_1 = (1-x,\ 1+x^2,\ 1-x^3,\ 3+x^2-x^3)$ and $B_2 = (1,\ 1-x,\ 1+x^2,\ 1-x^3)$ are bases of $P_3(\mathbb{R})$. (b) Find the coordinates of the vector $u = 1+x+x^2+x^3$ with respect to the ordered bases B_1 and B_2. (c) Find the matrix A such that $[u]_{B_2} = A[u]_{B_1}$. (d) Let $v = a_0 + a_1 x + a_2 x^2 + a_3 x^3$. Then verify that $[v]_{B_1} = \begin{pmatrix} -a_1 \\ -a_0 - a_1 + 2a_2 - a_3 \\ -a_0 - a_1 + a_2 - 2a_3 \\ a_0 + a_1 - a_2 + a_3 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix} [v]_{B_2}$.

Solution. $\dim P_3(\mathbb{R}) = 4$ (basis $1, x, x^2, x^3$).

Step 1 – (a) solve for a general v in B_2. $c_0 \cdot 1 + c_1(1-x) + c_2(1+x^2) + c_3(1-x^3) = a_0 + a_1 x + a_2 x^2 + a_3 x^3$. Compare coefficients. x: $-c_1 = a_1$. x^2: $c_2 = a_2$. x^3: $-c_3 = a_3$. Constant: $c_0 + c_1 + c_2 + c_3 = a_0$. So $c_1 = -a_1$, $c_2 = a_2$, $c_3 = -a_3$, $c_0 = a_0 + a_1 - a_2 + a_3$. **Unique solution for every** v (also for $v = 0$ it gives all $c_i = 0$), so B_2 spans and is independent: a basis. And

$$[v]_{B_2} = (a_0 + a_1 - a_2 + a_3,\ -a_1,\ a_2,\ -a_3)^t.$$

Step 2 – (a) the same for B_1. $d_1(1-x) + d_2(1+x^2) + d_3(1-x^3) + d_4(3+x^2-x^3) = v$. x: $-d_1 = a_1$. x^2: $d_2 + d_4 = a_2$. x^3: $-d_3 - d_4 = a_3$. Constant: $d_1 + d_2 + d_3 + 3d_4 = a_0$. Then $d_1 = -a_1$, $d_2 = a_2 - d_4$, $d_3 = -a_3 - d_4$, and the constant equation becomes $-a_1 + a_2 - d_4 - a_3 - d_4 + 3d_4 = a_0$, so $d_4 = a_0 + a_1 - a_2 + a_3$. Hence $d_2 = -a_0 - a_1 + 2a_2 - a_3$ and $d_3 = -a_0 - a_1 + a_2 - 2a_3$. Again a unique

solution for every v: B_1 is a basis, and

$$[v]_{B_1} = \big(-a_1,\ -a_0 - a_1 + 2a_2 - a_3,\ -a_0 - a_1 + a_2 - 2a_3,\ a_0 + a_1 - a_2 + a_3\big)^t.$$

This is exactly the first vector in (d). ✓

Step 3 – (b) plug in $u = 1 + x + x^2 + x^3$. $a_0 = a_1 = a_2 = a_3 = 1$. $[u]_{B_1} = (-1,\ -1-1+2-1,\ -1-1+1-2,\ 1+1-1+1)^t = (-1,-1,-3,2)^t$. $[u]_{B_2} = (1+1-1+1,\ -1,\ 1,\ -1)^t = (2,-1,1,-1)^t$.
Check B_1: $-(1-x)-(1+x^2)-3(1-x^3)+2(3+x^2-x^3) = (-1-1-3+6)+x+(-1+2)x^2+(3-2)x^3 = 1+x+x^2+x^3$ ✓.
Check B_2: $2-(1-x)+(1+x^2)-(1-x^3) = (2-1+1-1)+x+x^2+x^3$ ✓.

Step 4 – (c) columns of A (Theorem 3.4.5 with the roles: A eats B_1-coordinates and gives B_2-coordinates). Column i is the B_2-coordinates of the i-th vector of B_1. Use the formula for $[v]_{B_2}$ with each B_1 vector: $1-x$ ($a = (1,-1,0,0)$): $(1-1-0+0,\ 1,\ 0,\ 0) = (0,1,0,0)$. $1+x^2$ ($a = (1,0,1,0)$): $(1+0-1+0,\ 0,\ 1,\ 0) = (0,0,1,0)$. $1-x^3$ ($a = (1,0,0,-1)$): $(1+0-0-1,\ 0,\ 0,\ 1) = (0,0,0,1)$. $3+x^2-x^3$ ($a = (3,0,1,-1)$): $(3+0-1-1,\ 0,\ 1,\ 1) = (1,0,1,1)$.

$$A = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix}.$$

Test: $A(-1,-1,-3,2)^t = (2,\ -1,\ -1+2,\ -3+2)^t = (2,-1,1,-1)^t = [u]_{B_2}$ ✓.

Step 5 – (d) the other direction. The matrix $Q = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix}$ should satisfy $[v]_{B_1} = Q[v]_{B_2}$.

Put $[v]_{B_2} = (c_0,c_1,c_2,c_3) = (a_0+a_1-a_2+a_3,\ -a_1,\ a_2,\ -a_3)$. Row 1: $c_1 = -a_1$ ✓. Row 2: $-c_0+c_2 = -a_0-a_1+a_2-a_3+a_2 = -a_0-a_1+2a_2-a_3$ ✓. Row 3: $-c_0+c_3 = -a_0-a_1+a_2-a_3-a_3 = -a_0-a_1+a_2-2a_3$ ✓. Row 4: $c_0 = a_0+a_1-a_2+a_3$ ✓. All four rows match the formula for $[v]_{B_1}$ in Step 2. ■

Step 6 – consistency. Q is the matrix from B_2-coordinates to B_1-coordinates, so $Q = A^{-1}$; indeed $QA = I_4$ (multiply to check).

Part M Your mid-sem paper: the Chapter 3 questions, solved

What this part is

This is your **NIT Silchar Mid Semester (PG/PhD), Linear Optimization, CS 5103** paper. Six of its parts come straight from Chapter 3: **1(c), 2(a), 2(b), 2(c), 2(d) and 4(b)**. Their marks add up to $2+1+2+2+1+3=11$ out of 30. Every one of them is solved below in the same style as the exercises, and each can be done with a recipe from this PDF. (The remaining parts – 1(a) inverse of AB, 1(b) reduced row echelon form, 3 linear system with parameters, 4(a) determinant and inverse – belong to Chapters 1 and 2.)

Paper	What it asks	Answer	Learn it from
1(c) [2]	Is $W=\{2x-3y+z=0\}$ a subspace of $\mathbb{R}^3$?	Yes	Section 3.1: subspace test; Questions 1.1, 1.2
2(a) [1]	Are $(1,2,3),(0,1,2),(2,3,4)$ independent?	Dependent: $2v_1-v_2-v_3=\mathbf{0}$	Section 3.2: Recipe 3; Questions 2.1–2.4
2(b) [2]	Basis of the null space of $\begin{pmatrix}1&2&-1\\2&4&1\end{pmatrix}$	$\{(-2,1,0)\}$	Section 3.1 (row, column, null space); Recipe 7; Question 3.17(e)
2(c) [2]	Coordinates of $v\in\mathbb{R}^2$ in the ordered basis $\big((1,1),(1,-1)\big)$	$[v]_B=\left(\frac{p+q}{2},\frac{p-q}{2}\right)$	Section 3.4: Recipe 8; Question 4.1
2(d) [1]	Define span and basis; standard basis of $\mathbb{R}^3$	definitions	Sections 3.1 and 3.3
4(b) [3]	Basis and dim of $L(v_1,\dots,v_4)\subset\mathbb{R}^4$; extend to a basis of $\mathbb{R}^4$	$\dim=2$, then add e_3,e_4	Section 3.3: Recipe 4, Remark 3.3.5, Example 3.3.17, Question 3.17

Two digits in your photo are smudged – read this

In the photo, the first coordinate of v in 2(c) is blurred (it looks like 5), and the last entry of v_4 in 4(b) has an unreadable sign (it looks like -2). The solutions below are done for the most likely readings: $v=(5,7)$ and $v_4=(1,0,1,-2)$. For 2(c) I also give a formula that works for *any* $v=(p,q)$. For 4(b) I add a one-paragraph note for the other reading $v_4=(1,0,1,2)$. Please compare with your question paper.

Mid-sem Question 1(c) [2 marks]

Let $W=\{(x,y,z)\in\mathbb{R}^3:\ 2x-3y+z=0\}$. Determine whether W is a subspace of $\mathbb{R}^3$ by verifying the closure properties.

Solution.

Step 1 – what must be checked. W is a subspace if it is non-empty (contains $\mathbf{0}$), closed under addition and closed under scalar multiplication.

Step 2 – zero vector. $(0,0,0)$: $2\cdot0-3\cdot0+0=0$, so $(0,0,0)\in W$. W is non-empty.

Step 3 – closed under addition. Let $u=(x_1,y_1,z_1)$, $v=(x_2,y_2,z_2)\in W$. So $2x_1-3y_1+z_1=0$ and $2x_2-3y_2+z_2=0$. Then $u+v=(x_1+x_2,\ y_1+y_2,\ z_1+z_2)$ and $2(x_1+x_2)-3(y_1+y_2)+(z_1+z_2)=(2x_1-3y_1+z_1)+(2x_2-3y_2+z_2)=0+0=0$. So $u+v\in W$.

Step 4 – closed under scalar multiplication. For $\alpha\in\mathbb{R}$: $\alpha u=(\alpha x_1,\alpha y_1,\alpha z_1)$ and $2\alpha x_1-3\alpha y_1+\alpha z_1=\alpha(2x_1-3y_1+z_1)=\alpha\cdot0=0$. So $\alpha u\in W$.

Step 5 – conclusion. All three conditions hold. W **is a subspace of** $\mathbb{R}^3$ (a plane through the origin). ■

Step 6 – numerical check (optional, one line). $u = (1,1,1)$: $2-3+1=0$ ✓; $v=(0,1,3)$: $0-3+3=0$ ✓; $u+v=(1,2,4)$: $2-6+4=0$ ✓; $5u=(5,5,5)$: $10-15+5=0$ ✓.

One-line summary: A homogeneous linear equation (right side 0) always gives a subspace, whatever the coefficients. (If the right side were non-zero, even the zero vector would fail.) If the exam also asked for a basis: $z=3y-2x$ gives $(x,y,3y-2x)=x(1,0,-2)+y(0,1,3)$, so basis $\{(1,0,-2),(0,1,3)\}$ and $\dim W = 2$.

Mid-sem Question 2(a) [1 mark]

Determine whether the set of vectors $v_1=(1,2,3)$, $v_2=(0,1,2)$ and $v_3=(2,3,4)$ in $\mathbb{R}^3$ is linearly independent or linearly dependent.

Solution.

Step 1 – set up. Let $av_1+bv_2+cv_3=\mathbf{0}$, i.e. $a(1,2,3)+b(0,1,2)+c(2,3,4)=(0,0,0)$.

Step 2 – three equations. x: $a+2c=0$. y: $2a+b+3c=0$. z: $3a+2b+4c=0$.

Step 3 – solve. From the first, $a=-2c$. In the second, $b=-2a-3c=4c-3c=c$. Third: $3(-2c)+2c+4c=-6c+6c=0$ ✓– it holds for *every* c. So c is free.

Step 4 – a non-zero solution. Take $c=-1$: $a=2$, $b=-1$. Then $2v_1-v_2-v_3=(2,4,6)-(0,1,2)-(2,3,4)=(0,0,0)$ ✓.

Step 5 – conclude. A non-trivial relation exists, so the set is **linearly dependent**. ($v_3=2v_1-v_2$.)

One-line summary: Quick check you can also write: $\det\begin{pmatrix}1&2&3\\0&1&2\\2&3&4\end{pmatrix}=1(4-6)-2(0-4)+3(0-2)=-2+8-6=0$, *so dependent (Key Fact K1).*

Mid-sem Question 2(b) [2 marks]

Find a basis for the null space of the matrix $A=\begin{pmatrix}1&2&-1\\2&4&1\end{pmatrix}$.

Solution.

Step 1 – what the null space is. $N(A)=\{x\in\mathbb{R}^3 : Ax=\mathbf{0}\}$.

Step 2 – row-reduce A. $R_2\to R_2-2R_1$: $\begin{pmatrix}1&2&-1\\0&0&3\end{pmatrix}$. $R_2\to\frac{1}{3}R_2$: $\begin{pmatrix}1&2&-1\\0&0&1\end{pmatrix}$. $R_1\to R_1+R_2$: $\begin{pmatrix}1&2&0\\0&0&1\end{pmatrix}$ (RREF; pivot columns 1 and 3).

Step 3 – write the equations. $x_1+2x_2=0$ and $x_3=0$. The free variable is x_2 (the non-pivot column); put $x_2=t$.

Step 4 – solve. $x_1=-2t$, $x_3=0$: $x=(-2t,\ t,\ 0)=t\,(-2,1,0)$.

Step 5 – basis. $\{(-2,1,0)\}$. **It is a basis**: it spans $N(A)$ (every solution is a multiple of it) and it is independent (it is non-zero). $\dim N(A)=1=3-\operatorname{rank}A=3-2$.

Step 6 – check. $A(-2,1,0)^t=(1\cdot(-2)+2\cdot1+(-1)\cdot0,\ 2\cdot(-2)+4\cdot1+1\cdot0)=(0,0)$ ✓.

Mid-sem Question 2(c) [2 marks]

Find the coordinate vector $[v]_B$ of $v = (5, 7)$ in $\mathbb{R}^2$ relative to the ordered basis $B = \{b_1 = (1, 1),\ b_2 = (1, -1)\}$.

Solution.

Step 1 – what we need. Numbers a, b with $v = a\,b_1 + b\,b_2$. Then $[v]_B = (a, b)^t$.

Step 2 – is B a basis?. $\det \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = -1 - 1 = -2 \neq 0$: independent, and two independent vectors in $\mathbb{R}^2$ form a basis. (So coordinates exist and are unique.)

Step 3 – write the equation. $a(1, 1) + b(1, -1) = (a + b,\ a - b) = (5, 7)$. So $a + b = 5$ and $a - b = 7$.

Step 4 – solve. Add: $2a = 12$, so $a = 6$. Subtract: $2b = -2$, so $b = -1$.

Step 5 – answer. $[v]_B = \begin{pmatrix} 6 \\ -1 \end{pmatrix}$.

Step 6 – check. $6(1, 1) + (-1)(1, -1) = (6 - 1,\ 6 + 1) = (5, 7)$ ✓.

One-line summary: For any $v = (p, q)$: $a + b = p$, $a - b = q$, so $[v]_B = \left(\frac{p+q}{2},\ \frac{p-q}{2}\right)^t$. (E.g. if your paper had $v = (3, 7)$: $(5, -2)^t$.) Remember: order matters – *with the order (b_2, b_1) the answer would be $(-1, 6)^t$.*

Mid-sem Question 2(d) [1 mark]

Define the Span of a set of vectors and a Basis of a vector space. State the standard basis of $\mathbb{R}^3$.

Solution (write this in the exam).

Step 1 – Span. Let $S = \{v_1, \ldots, v_k\}$ be a set of vectors in a vector space $V(\mathbb{F})$. The **span** of S is the set of all linear combinations of its vectors: $L(S) = \{\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_k v_k :\ \alpha_1, \ldots, \alpha_k \in \mathbb{F}\}$. It is a subspace of V (the smallest subspace containing S).

Step 2 – Basis. A non-empty subset B of V is a **basis** of V if (i) B is *linearly independent* (i.e. $c_1 u_1 + \cdots + c_m u_m = \mathbf{0}$ for $u_i \in B$ forces all $c_i = 0$), and (ii) B *spans* V, i.e. $L(B) = V$. Every vector of V is then a *unique* linear combination of the vectors in B.

Step 3 – Standard basis of $\mathbb{R}^3$. $e_1 = (1, 0, 0)$, $e_2 = (0, 1, 0)$, $e_3 = (0, 0, 1)$. (Reason: $(x, y, z) = xe_1 + ye_2 + ze_3$, and $\alpha e_1 + \beta e_2 + \gamma e_3 = (\alpha, \beta, \gamma) = \mathbf{0}$ forces $\alpha = \beta = \gamma = 0$.) $\dim \mathbb{R}^3 = 3$.

Mid-sem Question 4(b) [3 marks]

Let W be the subspace of $\mathbb{R}^4$ spanned by the vectors $v_1 = (1, -1, 2, 3)$, $v_2 = (2, -1, 3, 1)$, $v_3 = (4, -3, 7, 7)$ and $v_4 = (1, 0, 1, -2)$. (i) Find a basis for W and determine $\dim(W)$. (ii) Extend the basis of W to form a complete basis for $\mathbb{R}^4$.

Solution.

Step 1 – part (i): put the vectors as rows. $\begin{pmatrix} 1 & -1 & 2 & 3 \\ 2 & -1 & 3 & 1 \\ 4 & -3 & 7 & 7 \\ 1 & 0 & 1 & -2 \end{pmatrix}$. Row operations do not change the span of the rows (Lemma 3.1.16), so we may reduce.

Step 2 – clear column 1. $R_2 \to R_2 - 2R_1$, $R_3 \to R_3 - 4R_1$, $R_4 \to R_4 - R_1$ give $\begin{pmatrix} 1 & -1 & 2 & 3 \\ 0 & 1 & -1 & -5 \\ 0 & 1 & -1 & -5 \\ 0 & 1 & -1 & -5 \end{pmatrix}$

(check R_2: $(2, -1, 3, 1) - (2, -2, 4, 6) = (0, 1, -1, -5)$ ✓; R_3: $(4, -3, 7, 7) - (4, -4, 8, 12) = (0, 1, -1, -5)$

✓; R_4: $(1,0,1,-2)-(1,-1,2,3)=(0,1,-1,-5)$ ✓).

Step 3 – make two rows zero. $R_3 \to R_3 - R_2$, $R_4 \to R_4 - R_2$: rows 3 and 4 become $(0,0,0,0)$. Two non-zero rows are left: **rank** $= 2$.

Step 4 – answer to (i). $\dim W = 2$. A basis (the non-zero rows): $\{(1,-1,2,3),\ (0,1,-1,-5)\}$. *A basis made of the given vectors:* $\{v_1, v_2\}$. Why this is legitimate: $v_3 = 2v_1 + v_2$ (check: $(2,-2,4,6)+(2,-1,3,1)=(4,-3,7,7)$ ✓) and $v_4 = -v_1 + v_2$ (check: $(-1,1,-2,-3)+(2,-1,3,1)=(1,0,1,-2)$ ✓), so $W = L(v_1, v_2)$; and v_1, v_2 are independent because neither is a multiple of the other.

Step 5 – part (ii): what is needed. $\dim \mathbb{R}^4 = 4$ and $\dim W = 2$, so we must add $4-2=2$ vectors such that all four are independent (Theorem 3.3.15 / Cor. 3.3.16).

Step 6 – choose and test. Try $e_3 = (0,0,1,0)$ and $e_4 = (0,0,0,1)$. Rows v_1, v_2, e_3, e_4: $\det \begin{pmatrix} 1 & -1 & 2 & 3 \\ 2 & -1 & 3 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \det \begin{pmatrix} 1 & -1 \\ 2 & -1 \end{pmatrix} \cdot \det \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = (-1+2)\cdot 1 = 1 \neq 0$. (The matrix is block upper triangular, so its determinant is the product of the two diagonal blocks. Equivalently, the 2×2 minor of v_1, v_2 in the first two coordinates is non-zero, so e_3, e_4, which are zero there, complete the list.)

Step 7 – conclude. Four independent vectors in the 4-dimensional $\mathbb{R}^4$ form a basis (Cor. 3.3.16). **Basis of $\mathbb{R}^4$:** $\{(1,-1,2,3),\ (2,-1,3,1),\ (0,0,1,0),\ (0,0,0,1)\}$.

One-line summary: $\dim W = 2$*; basis* $\{v_1, v_2\}$*; extension by* e_3, e_4. If your paper has $v_4 = (1,0,1,2)$ instead: *then* $R_4 \to R_4 - R_1$ *gives* $(0,1,-1,-1)$*, which is* not R_2'*, so after* $R_4 \to R_4 - R_2'$ *one non-zero row* $(0,0,0,4)$ *remains: rank* 3, $\dim W = 3$*, basis* $\{v_1, v_2, v_4\}$*, and one extra vector completes it, e.g.* $e_1 = (1,0,0,0)$ *(the* 4×4 *determinant is* $4 \neq 0$*).*